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Manual for the
Category B driving licence in
Easy Read

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C/ Puerto Navacerrada, 128
Industrial Estate “Las Nieves”

28935 Móstoles (Madrid)

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Topic 1. Definitions

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Vehicle categories depending on their use

Definitions related to vehicles

Vehicle

Device prepared to travel on roads, streets, roads and land of all kinds.

Types of vehicles

1. Vehicles without an engine
2. Vehicles with an engine
3. Other vehicles with an engine

Watch video

Vehicles without an engine

Vehicle drawn by animals

Cycle and bicycle

Trailer

Semi-trailer

Vehicles drawn by animals

For example, a horse-drawn carriage.

Cycle

Vehicle with two or more wheels that has pedals. It moves by the energy and effort of the person who rides it. The bicycle is a two-wheeled cycle.

Watch video

Trailer

Platform with wheels that is attached to a motor vehicle, by means of an axle, to carry loads.

Axle. Bar that goes through an object and helps it move and turn. In a trailer, the axle makes the load spread out and makes it easier to drive.

Light trailer	Heavy trailer
With load it weighs a maximum of 750 kilos.	With load it can weigh more than 750 kilos.

Semi-trailer

Trailer that is attached directly to a motor vehicle, without an axle.

Light semi-trailer	Non-light semi-trailer
With load it weighs a maximum of 750 kilos.	With load it can weigh more than 750 kilos.

Motor vehicles

Vehicles that need an engine to work. Motor vehicles are classified into:

- Motor cars They are used to carry people and goods. They are also used to move or tow other vehicles.
- Special vehicles They are used to do certain jobs or services. Some have their own engine and others are towed. For example, excavators to move earth and tractors to work in the countryside. There are different types of motor cars and special vehicles:

Motor cars

Two-wheeled motorcycles

Motorcycles with sidecar

Three-wheeled vehicles

Heavy quadricycle (or quadricycle)

Car

Pick-up

Car-derived van

Adaptable mixed vehicle

Bus / Coach

Trolleybus

Lorry

Van / Light van

Tractor unit

Articulated vehicle

Road train

Special vehicles

With an engine

Quad - ATV

Tractor

Motor cultivator

Self-propelled machine

Towed

Agricultural trailer

Towed machine

Motor cars

Motorcycles

Watch video

Two-wheeled vehicles that must meet at least one of these two characteristics related to engine capacity and speed:

- If it is petrol, have an engine with an engine capacity greater than 50 cubic centimetres or, which is the same, 0.05 litres.
- Reach a speed greater than 45 kilometres per hour.

Engine capacity. Measure used to calculate the power of an engine. The greater the engine capacity, the greater its power.

Engine capacity is measured in litres or in cubic centimetres.

Motorcycles with sidecar

They are three-wheeled motorcycles that have a seat at the side for another person to sit. They must meet the same characteristics of engine capacity and speed as two-wheeled motorcycles.

Three-wheeled vehicle

Vehicle with three symmetrical wheels that must meet the same characteristics of engine capacity and speed as two-wheeled motorcycles and sidecars.

Heavy quadricycle

Four-wheeled motor car that meets these characteristics:

- With load and passengers it weighs a maximum of 450 kilos if it carries people or 600 kilos if it carries goods.
- The maximum power the engine can have is 15 kilowatts.

Kilowatt. Unit used to measure the maximum power a device can have.

Car

Motor car used to carry people. It has at least four wheels. It can have a maximum of nine seats, including the driver's seat.

Car-derived van

Motor car with the bodywork of a car, but used to carry goods instead of people. It only has one row of seats to leave more space for goods.

Bodywork. Structure that gives shape to the motor car.

Pick up

Vehicle in which the seats for people and the load area are separated into two different spaces. With load and passengers it weighs a maximum of 3,500 kilos.

Adaptable mixed vehicle

Vehicle used to carry people or goods. The seats can be put in or taken out depending on the use you want to give the vehicle. A maximum of nine people can travel in it.

Bus or coach

Vehicle with more than nine seats used to carry people and their luggage.

Lorry

Vehicle with four wheels or more used to carry goods.

Watch video

It has two parts:

- The cab. Space with seats where the passengers and the driver travel. It can have a maximum of nine seats.
- Rest of the lorry. Space at the back. It is the place where the goods travel.

Van or light van

Vehicle with four wheels or more used to carry goods. Unlike the lorry, the cab and the rest of the lorry are together, not separate.

Tractor unit

Vehicle used to tow the semi-trailer.

Combination of vehicles

Motor car made up of a motor vehicle and a trailer or semi-trailer attached or connected to it.

Watch video

There are three types of combinations of vehicles:

1. Road train. The trailer is attached to the motor vehicle by means of an axle.
2. Articulated vehicle. The semi-trailer is connected to the motor vehicle, without an axle.
3. Euromodular configuration. Attached vehicles that have at least 6 axles on their wheels. The axles connect the wheels to each other and help them turn.

Special vehicles

Quad ATV

Vehicle with four wheels or more that is used more for driving on hills or tracks than on roads. It is driven with handlebars, like a motorcycle.

Tractor

Vehicle with four wheels or more that has a very large engine. It is used to carry or tow materials and heavy loads.

Watch video

The tractor is used mainly in field work (agricultural tractor) and in construction (works tractor).

Motor cultivator

Small vehicle that has an engine and wheels. The person who drives it walks and moves it by means of a long part, like handlebars. This long part is part of the motor cultivator. It is used mainly in gardening and agriculture to turn over the soil in vegetable gardens or small plots of land.

Self-propelled machine

Vehicle with four wheels or more used to do work in the countryside and in construction.

Towed vehicles

There are two types:

1. Agricultural trailer Vehicle to do work in the countryside. It must be towed by a tractor or by other special vehicles.
2. Towed machine Vehicle used to do work on building sites, in the countryside or in other services. It must be towed by another special vehicle.

Other types of vehicles with an engine

Some vehicles that have an engine are not considered motor vehicles because they have special characteristics.

Vehicles not considered motor vehicles

Tram

Moped

Vehicles for people with reduced mobility

Pedal bicycle with an engine

Personal mobility vehicle

Tram

It runs on rails laid in the road or in the street.

Moped

Two-, three- or four-wheeled vehicles. They have low-power engines. The maximum speed they reach is 45 kilometres per hour. A maximum of two people can travel on mopeds.

Vehicles for people with reduced mobility

They are made for use by people who have a physical disability. The maximum speed they reach is 45 kilometres per hour. It weighs a maximum of 350 kilos.

Pedal bicycles with an engine

Personal mobility vehicles

They have an electric motor that reaches a speed of between 6 and 25 kilometres per hour. For example, the electric scooter. In this type of vehicle only one person may travel.

Definitions related to people

Driver

Person who drives a vehicle. It is also the person who controls the additional controls in driving school vehicles to learn to drive.

Pedestrian

Person who walks on the pavement and the road. Pedestrians are also people who:

- Pull or push a pram, a wheelchair or any small vehicle without an engine.
- Walk and take with them a cycle or a two-wheeled moped.
- Use a wheelchair.

Vehicle owner

Person who has the vehicle registered in their name in the corresponding official register. This person will always carry in the vehicle the documentation that says they are the owner and will show it to the authorities when asked. The vehicle owner is responsible for ensuring that this vehicle is not driven by people who do not have a driving licence.

Usual driver

Person who usually drives a vehicle, even if they are not the owner. The usual driver must have the driving licence needed to drive that vehicle. The vehicle owner is the person who authorises that person to be the usual driver.

Vehicle categories depending on their use

Category	Use
M	Motor vehicles made to carry people and their luggage. For example, cars and buses.
N	Motor vehicles made to carry goods. For example, vans and lorries.
O	Trailers made to carry people and goods.

Topic 2. Documents

Contents

Documents needed to drive

Driving licence

- Types of driving licence
- Points-based driving licence
- New drivers
- When must you renew your Category B driving licence?

Vehicle registration certificate

- Which vehicles must have a vehicle registration certificate?
- What details appear on the vehicle registration certificate?

Vehicle roadworthiness inspection card (ITV)

- Which vehicles must have the vehicle roadworthiness inspection card?
- What details appear on the vehicle roadworthiness inspection card?
- When must you have the roadworthiness inspection for your vehicle (ITV)?
- Results of the roadworthiness inspection

Compulsory third-party liability insurance

- Which vehicles must have compulsory third-party liability insurance?
- What does compulsory insurance cover?
- What does compulsory insurance not cover?
- What are the consequences of not insuring the vehicle?

Responsible persons

Documents needed to drive

To drive a vehicle you need the following documents:

Documents

Driving licence

Vehicle registration certificate

Vehicle roadworthiness inspection card

Third-party liability insurance

Watch video

You must always carry these documents in the vehicle when you drive to show them to the authorities who are responsible for monitoring traffic. These documents must be the originals. You can also present a photocopy of the documents certified by an official public body. For example, the town hall, the police or a notary.

Certify. An official body certifies that the copy of the document is authentic

When you need to adapt the vehicle to drive, you must apply for a special permit and make the necessary changes to the vehicle. For example, fitting controls on the steering wheel for people who have a disability in their legs and difficulty using the car pedals. We are going to learn about the 4 documents needed that a driver must have, what they are for and how they are used.

Driving licence

Document that authorises a person to drive and ensures that they meet the requirements to do so.

Types of driving licence

You must obtain one licence or another depending on the vehicle you are going to drive:

Type of licence	Vehicles you can drive	Minimum age to drive
AM	Mopeds with two or three wheels and light quadricycles. Vehicles for people with reduced mobility.	15 years
A1	Motorcycles with a maximum engine capacity of 125 cubic centimetres and a maximum power of 11 kilowatts. Motor tricycles with a maximum power of 15 kilowatts.	16 years
A2	Motorcycles with a maximum power of 35 kilowatts.	18 years
A	Motorcycles and tricycles with any type of engine and power.	20 years

Type of licence	Vehicles you can drive	Minimum age to drive
B	Mopeds. Vehicles for people with reduced mobility. Cars that can carry up to nine passengers. Special agricultural vehicles. Special non-agricultural vehicles that reach a maximum speed of 40 kilometres per hour and do not weigh more than 3,500 kilos. Combination of vehicles when the trailer does not carry a weight greater than 750 kilos.	18 years
B+E	Combination of vehicles. The trailer can carry a load of up to 3,500 kilos.	18 years

Points-based driving licence

When you obtain the driving licence you are given 8 points. You will lose some of these points or all of them if you commit a serious or very serious offence. For example, using your mobile while driving. You can recover the points after some time by doing a driver re-education course. This course lasts 12 hours.

New drivers

A person who obtains the driving licence is called a new driver during the first year.

New drivers must display in the car a green rectangular plate with a white letter L. This plate is placed on the rear window of the car. It must be clearly visible so that other drivers know that car is being driven by a new driver.

When must you renew your Category B driving licence?

- Every 10 years if you are under 65.
- Every 5 years if you are over 65. You must apply to renew the licence before it expires. You must also apply again for the driving licence if you lose it, it is stolen, or any of your personal details change. For example, your address.

Vehicle registration certificate

Document that confirms that a vehicle has a registration number plate and is authorised to circulate.

Which vehicles must have a vehicle registration certificate?

- All motor vehicles.
- Mopeds.
- Trailers and semi-trailers that can carry more than 750 kilos.

Registration number plate. Set of letters and numbers that identify vehicles. Each vehicle has a different registration number plate which is stamped on a metal plate and placed on the vehicle.

What details appear on the vehicle registration certificate?

- First name, surname(s) and address of the vehicle owner.
- Vehicle registration number and the date it was registered.
- Number of seats the vehicle has.
- Use given to the vehicle. That is, whether it is used to carry people or goods.
- Maximum weight the vehicle can carry. The owner of a vehicle must report any change in their details to the Traffic Headquarters.

They must do it no later than 15 days after the change. They must also report if they sell or hand over the vehicle to another person. In this case, they have 10 days to report it. The person who buys or receives the vehicle must apply for the vehicle registration certificate to be renewed in their name.

Watch video

Vehicle roadworthiness inspection card (ITV)

Document that proves that a vehicle has no faults and is in good condition to operate.

Which vehicles must have the vehicle roadworthiness inspection card?

- All motor vehicles.
- All mopeds.
- All trailers and semi-trailers.

What details appear on the vehicle roadworthiness inspection card?

- Vehicle characteristics.
- Record of the inspections the vehicle has passed.

When must you have the roadworthiness inspection for your vehicle (ITV)? To know when each vehicle must be inspected, you must take into account what type of vehicle it is, how old it is and what it is used for: whether it is a car, a special vehicle...

Types of vehicle

When to have the roadworthiness inspection (ITV)

Two-wheeled mopeds First inspection at 3 years. After 3 years, inspection every 2 years.

Three-wheeled mopeds and light quadricycles

Motorcycles, three-wheeled vehicles and heavy quadricycles

Quads

First inspection at 4 years. After 4 years, inspection every 2 years.

Motor vehicles that can carry up to nine passengers	First inspection at 4 years. After 4 years, inspection every 2 years. After 10 years, inspection every year.
Motor vehicles that can carry goods with a weight of up to 3.5 tonnes	First inspection at 2 years. After 2 years, inspection every 2 years. After 6 years, inspection every year. After 10 years, inspection every 6 months.
Trailers to carry goods or people or to accommodate people Except the towed caravan	First inspection at 1 year. Up to 10 years, inspection every year. After 10 years, once every 6 months.
Towed caravan	First inspection at 6 years. After 6 years, every 2 years.

You must also have a roadworthiness inspection when the vehicle is modified, or if it has had an accident and the structure has been damaged. The vehicle owner is the person responsible for ensuring that the roadworthiness inspections are carried out.

Results of the roadworthiness inspection

When the vehicle is in good condition, the ITV technicians:

- Write on the vehicle roadworthiness inspection card that the vehicle is OK and the date of the next inspection.
- Give you a sticker. You must place it inside the vehicle, on the right-hand side of the windscreen.
- Give you a report. You must always carry it in the vehicle. When the vehicle has a fault or a serious problem:
 - You cannot continue to drive.
 - You must take it to a garage to have it repaired.
 - Once repaired, you must take it for a new inspection to check that it can now circulate.

Compulsory third-party liability insurance

People who have a motor vehicle have the obligation to take out and pay for insurance. This insurance is used to protect other people, objects or vehicles if you have an accident. This insurance is called third-party liability insurance.

Which vehicles must have compulsory third-party liability insurance?

All motor vehicles if the owner lives in Spain, except:

- Trains and trams.
- Wheelchairs.
- Motorised toys.

Other vehicles that do not need to have third-party liability insurance are trailers, semi-trailers and towed machines that cannot carry more than 750 kilos. What does compulsory insurance cover?

- Damage that your vehicle causes to an object. For example, repairing a lamppost if you crash into it.
- Damage to another person. For example, if you run over a pedestrian. However, the insurance does not cover damage to another person if it is that person who behaves badly. For example, if a pedestrian throws themselves at your car so that you run them over. It will also not cover damage to a person for reasons that are not related to driving. For example, a traffic accident caused by a hurricane or a fire.

What does compulsory insurance not cover?

- Injuries that in an accident are suffered by the driver of the insured vehicle. For example, it does not pay the medical expenses of a driver who loses a leg in an accident.
- Damage suffered by objects that are inside the car of the person who is driving.
- Damage that the accident causes to the objects of the insured person, the person who is driving, or their relatives. For example, it does not cover repairing the door of a garage that a driver breaks when entering the home of their brother.
- Damage to people who are not wearing a helmet if it is compulsory to wear it.
- Accidents and damage that occur in a stolen vehicle. When there is an accident, the drivers of all vehicles have to inform their insurers. They must do it no later than 7 days after the accident.

What are the consequences of not insuring the vehicle?

- The vehicle cannot circulate. If a traffic monitoring officer stops you, they will immobilise the vehicle at that moment.
- The vehicle owner has to pay the costs of the place where the vehicle is kept while it has no insurance.
- The vehicle owner has to pay a fine. You must always carry in the vehicle the proof of payment of the insurance. This way you can prove that the vehicle is insured. This proof will show the following information:
 - Name of the insurance company.
 - Vehicle registration number.
 - Insurance plate.

- Date when the insurance must be renewed.
- Indication of what damage the insurance covers.

Responsible persons

As a general rule, the person who commits a fault or offence as a driver or pedestrian is the one who must pay the fine or the penalty.

However, there are cases in which drivers are responsible for a fault or offence, even if they did not cause the accident. These cases are:

- Drivers and passengers who do not wear a helmet in vehicles in which it is compulsory to wear it.
- Drivers who carry minors in vehicles in which it is not allowed for children of that age to travel. When traffic faults or offences are committed by a person under 18 years old, the fine must be paid by their father, mother, or guardian. The person who is the vehicle owner is responsible for faults or offences related to:
 - The vehicle documentation.
 - Not complying with the vehicle inspections.
 - The condition the vehicle is in.

Topic 3. The condition of the driver

Contents

Physical and psychological condition of the driver

Factors that influence the condition of the driver

- Fatigue
- Drowsiness
- Alcohol
- Other drugs
- Illnesses and medicines
- Food
- Suitable clothing

Accidents due to distractions

- Smoking in the vehicle
- Use of the mobile phone
- GPS sat navs

Physical and psychological condition of the driver

You must be in good physical and psychological condition to drive. There are many physical and psychological factors that affect your safety and the safety of other people when you drive a vehicle. For example, your ability to see well, your mood, and the time you take to react to unexpected events that happen on the road.

Also, you must have enough training to handle the vehicle and understand all traffic rules.

Eyesight

Through our eyes we receive most of the information to drive well. Some of this information is: the condition of the road, traffic signs, distance to objects and the speed of other vehicles. To drive well it is necessary to see things in front of you, to the sides, and behind.

But some factors make you see less:

- Drinking alcohol.
- Taking drugs.
- Being tired.
- Driving too fast.

Mood

Changes in mood can make you lose concentration and cause a road traffic accident. For example, having a strong argument while you are driving or receiving very sad news before driving can change your mood.

Reaction time

It is the time that passes between hearing or seeing something and reacting. For example, the time that passes between when you see a red traffic light and you stop the car. In normal conditions, the reaction time of a person who is driving is between half a second and one second.

Factors that make a person take longer to react:

- Being tired (fatigue).
- Being sleepy.
- Being very old.
- Hearing or seeing badly.
- Illnesses.
- Some medicines.
- Alcohol and drugs.
- Eating a lot before driving.
- Too much heat in the car.
- Altered mood.
- Paying little attention to driving. !

Factors that influence the condition of the driver

Being tired (fatigue)

It is one of the main risks that can cause accidents on the road. Fatigue can be physical or psychological.

Physical fatigue produces a feeling of tiredness and psychological fatigue makes it harder for you to concentrate.

Watch video

What can cause fatigue?

- What the road is like.
- What the vehicle is like.
- What the driver is like. What the road is like:
- Road with heavy traffic.
- The road surface is in poor condition.
- You do not know the road.
- Difficult weather: rain, fog, snow or too much heat. What the vehicle is like:
- Too much heat inside the vehicle. The suitable temperature is 23 degrees more or less.
- You drive at night with poor lighting.
- You drive in a vehicle in poor condition or that makes too much noise.
- You are uncomfortable in the seat.

What the driver is like:

- You drive for many hours without resting or you take very short breaks.
- You drive fast for a long time.
- You drive when sleepy, after drinking alcohol, or you feel unwell.
- You do long journeys and at night when you are not used to doing them.
- You have had your driving licence for a short time.
- You drive with your body in a bad posture. What are the symptoms of fatigue? There are some signs that warn you that you are suffering fatigue and you must stop driving. These signs are:
- Difficulty concentrating on the road.
- Your eyes feel heavy and you start to see badly.
- You hear badly.
- A feeling of having numb arms.
- A feeling of pressure in your head.
- You move a lot in the seat and change posture.
- You make slower movements.
- A feeling of being more nervous and irritable.
- Difficulty making decisions about driving.
- You take longer to react.

Being sleepy (drowsiness)

Many road traffic accidents are related to driving when sleepy. It is not necessary to fall completely asleep to have an accident for this reason. The symptoms of drowsiness appear before falling fully asleep.

Drowsiness. A state in which you have a feeling of tiredness, heaviness in the body, and sleepiness.

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What can cause drowsiness?

- Sleeping fewer hours than usual.
- Changing the hours when you usually sleep.
- Sleeping badly, even if you sleep many hours.
- Driving on roads with little traffic.
- Drinking alcohol or taking medicines before driving.
- Having illnesses related to sleep.
- Driving at dawn or at midday, after eating. What are the symptoms of drowsiness? There are some signs that warn you that you are suffering drowsiness and you must stop driving. These signs are:
- Difficulty keeping your head upright, or keeping your eyes open.
- Blurred vision.
- Yawning many times.
- Losing concentration or having meaningless thoughts.
- Being distracted by anything.
- Being restless or irritable.
- Difficulty remembering the last kilometres you have travelled.
- Leaving your lane.
- Not noticing traffic signs or the place where you must leave the road.
- Driving very close to the vehicle in front of yours. Effects of drowsiness:
- Taking longer than normal to react.
- Difficulty making decisions about driving.
- Feeling that it is hard for you to make movements with your body or you make them more slowly.
- Falling asleep for a few seconds without realising.

How can you avoid fatigue and drowsiness?

- Stop the vehicle in a safe place when you feel the first symptoms and sleep for 20 or 30 minutes.
- Rest for 20 or 30 minutes every two hours or every 200 kilometres, even if you do not feel fatigue or drowsiness.
- If you have a fatigue detector, obey what it tells you. It is a system that works through sensors and recognises if the person is very tired or about to fall asleep. In those cases it warns you by a light, sound, or vibration in the steering wheel so that you stop the vehicle and do not have an accident.

Sensor. Devices that capture information about things that happen outside the vehicle.

Alcohol

It is very dangerous to drink alcohol when you are going to drive, even if you drink a small amount.

Alcohol spreads throughout your body through the blood and affects, above all, the brain and eyesight. Alcohol is the cause of many road traffic accidents.

Blood alcohol level

Blood alcohol level is the total amount of alcohol in the blood after drinking. The blood alcohol limit is the amount of alcohol in each litre of blood. It can be calculated in two ways:

- Grams of alcohol in each litre of blood.
- Milligrams of alcohol in each litre of air that we expel from the lungs when breathing.

The permitted blood alcohol limit depends on the type of vehicle and the driving licence.

1. For people who have obtained the driving licence less than two years ago. And for vehicles that transport:
 - Goods with a weight greater than 3,500 kilos.
 - More than nine people.
 - Minors.
 - People in emergency services.
 - Dangerous loads.

The permitted blood alcohol limit is:

- 0.15 milligrams of alcohol per litre of air.
 - 0.3 grams of alcohol per litre of blood.
2. For any other vehicle and driver, the permitted blood alcohol limit is:
 - 0.25 milligrams of alcohol per litre of air.
 - 0.5 grams of alcohol per litre of blood. In any case, the only blood alcohol level that is safe for driving is 0.0, that is, not drinking alcohol.

What factors influence the blood alcohol level?

Alcohol does not affect all people in the same way. The factors that influence the blood alcohol level are:

Amount of alcohol

Sex of the person

Type of drink

Speed at which you drink

Age

Personal circumstances

Drinking alcohol without eating

Weight of the person

Sleeping after drinking

Time since you drink alcohol

- The amount of alcohol you drink. The more alcohol you drink, the higher the blood alcohol level will be.
- The weight of the person. Alcohol affects thin people more.
- The sex of the person. Alcohol usually affects women more.
- The time that passes since you drink alcohol. The moment when the blood alcohol level is highest is one hour after drinking alcohol. After that, the effects of alcohol go down very slowly.
- The type of drink and the way of drinking it.

Alcohol reaches the blood faster after drinking some drinks such as gin or whisky than after drinking other drinks such as wine or beer.

Also, alcohol reaches the blood faster when it is mixed with tonic or some soft drinks.

- Sleeping after drinking.

Alcohol is eliminated more slowly when we sleep. For that reason, it is not safe to drive after drinking a lot of alcohol and sleeping a few hours.

- The speed at which you drink. The body eliminates alcohol better when you drink slowly than when you drink fast.
- Drinking alcohol without eating.

Food helps alcohol reach the blood more slowly.

- Age.

Alcohol usually affects people under 18 years old and people over 65 years old more.

- Personal circumstances. There are circumstances that can make alcohol affect a person more. For example, being pregnant, stress, or having an illness.

Effects of alcohol while driving

On behaviour

- False confidence in yourself.
- You take more risks.
- You commit more faults that cause accidents.
- You may treat other drivers in a more aggressive or impulsive way.

On the way you see the surroundings

- You see traffic signs and traffic lights worse.
- You judge the distance to other vehicles worse.
- Less ability to see what happens to one side and the other.

- You are dazzled more by vehicle lights.
- Possibility of being distracted by elements in the surroundings. On movements
- Difficulty coordinating your body movements. On decisionmaking
- You need more time to react.
- Higher probability of making bad decisions or not knowing how to carry them out.

Other drugs

Taking drugs before driving is very dangerous.

One in every 10 people who die in a road traffic accident had taken drugs before driving. It is prohibited to drive any type of vehicle when a person has taken drugs and they are still in their body. You may only drive after taking substances prescribed by the doctor and that do not affect driving. How do drugs affect driving?

All drugs are dangerous and it is prohibited to drive when they have been taken. But each one produces effects that are different, and these are the cause of the danger.

Type of drug Possible dangers while the person is driving

Cannabis

- Seeing the surroundings in a different way. For example, colours look different.
- Judging distances worse
- Losing concentration.
- Needing more time to react.
- Falling asleep while driving.

Cocaine

- Becoming more impulsive or aggressive.
- Losing the sense of danger.
- Driving in a more dangerous way.
- Seeing the surroundings in a different way.
- Losing concentration.

Watch video

Ecstasy

- Having hallucinations.
- Being more sensitive to light or seeing blurred.
- Losing concentration.
- Suffering depression or anxiety.
- Feeling fatigue when the effects of the drug wear off.

LSD

- Having hallucinations.

- Reacting aggressively.
- Having anxiety and even panic.

Amphetamines

- Losing patience.
- Having impulsive and violent behaviour.
- Having little sense of danger.
- Delayed fatigue and sleep. This can cause the person to suddenly feel very tired and fall asleep unintentionally when the effect of the drug wears off.

Tests to detect alcohol and drugs

Driving after drinking or taking drugs is prohibited and punished by law. The person who does it will have to pay a fine and may even go to prison. To know whether a person who is driving has taken alcohol or drugs, a test is done. What does the test consist of? To detect whether a person has drunk alcohol the traffic officer asks them to blow into a device that measures the amount of alcohol in the blood. If the test is positive or the person shows signs of having drunk the test is repeated to confirm it. To detect whether a person has taken drugs a saliva sample is taken and the sample is put into a device.

If the person does not agree with the results they may ask for a blood test.

Test results

If the alcohol or drug tests are positive, the person who is driving will have committed a very serious offence. The consequences are:

- Pay a fine.
- Lose between 4 and 6 points from the driving licence.
- Possibility of losing the driving licence for a period of time.
- Possibility of going to prison if they have put at risk the lives of other people. The traffic police officer may prohibit the person from continuing to drive. They will leave the car immobilised until the effects of alcohol or drugs wear off.

Who must take the alcohol and drug detection tests? People who are driving a vehicle and:

- Have a road traffic accident.
- Show symptoms of driving under the effects of alcohol or drugs.
- Are reported for breaking a traffic rule or committing some other offence.
- Go through a checkpoint to prevent the use of alcohol and drugs.

Pedestrians who may cause a road traffic accident because of their behaviour must also take the test. The law imposes fines and penalties on people who refuse to take these tests when security officers ask them to.

Illnesses and medicines

Some illnesses and medicines make the person lose abilities to drive safely. When you have an illness or take medication you must always ask doctors if you can drive. It is also important to go to the Driver Medical Assessment Centre. There they will tell you if it is safe to drive with the medication you are taking.

If you have a chronic illness you can follow this advice to avoid accidents while you drive:

- Know the illness well.
- Know the effects of the medicines you take.
- Know how to act in case of a crisis.
- Avoid driving when you feel unwell.
- Do not stop the treatment until the doctor authorises it.
- Do not drink alcohol while you are taking the medication.

What illnesses can affect driving?

Illness

Advice for driving

Respiratory allergy

- Drive with the windows closed.
- Set the air conditioning low.
- Keep the vehicle and the ventilation ducts clean.
- Do not mix alcohol with medicines.
- Do not do long journeys driving.
- Use sunglasses. The sun, many times, makes you sneeze.
- Do not drive at dawn or in damp areas.
- Do not take medication without a doctor's prescription.

Stress

- Do not drive in the phases of greatest stress.
- Do not drive if you take medication for stress.
- Seek help to reduce stress.

Depression

- Do not take drugs or alcohol to improve the state of depression.
- Take only the medication prescribed by the doctor.
- Put yourself in the hands of specialists and follow the treatment.
- Do not drive in the periods of greatest depression.

What medicines can be dangerous for driving?

Medicine	What is it for?	Possible effects
Painkillers	Helps the pain decrease or disappear.	Sleepiness. Vertigo. Lack of concentration.
Cough suppressants	Calm a cough.	Mood changes. Loss of reflexes.
Antihistamines	Treat allergies.	Sleepiness. Depression. Loss of reflexes.
Psychotropic medicines	Treat depression, anxiety and sleep disorders.	Sleepiness. Loss of reflexes. Dizziness. Blurred vision. Confusion.

Diet

Eating a large amount and foods that are hard to digest before driving can cause sleepiness and fatigue. On the other hand, eating too little can cause dizziness. The most suitable thing is to have several light meals instead of one very heavy one. It is also recommended to wait a while (between 15 and 20 minutes) after finishing eating before driving. It is important to drink water or juices so that the body is hydrated and to have less chance of feeling sleepy or fatigued.

Suitable clothing

You must wear comfortable clothes when you drive. Clothes that are too tight will not let you move well. In winter, you must take off your coat before you start driving.

Footwear has to be comfortable and light to use the pedals better. It is not advisable to drive with high-heeled shoes, flip-flops or shoes with very thick soles.

Watch video

Motorcycle

You must pay special attention to the clothes you wear when you ride a motorcycle. In this vehicle there is no bodywork to protect you in an accident.

Bodywork. Metal part that covers most parts of a vehicle. The most suitable clothing to ride a motorcycle is:

- A leather suit or a similar material. This suit must fit your body well.
- Leather gloves with protection.
- Strong boots that support the foot and ankle.
- Wear bright and eye-catching colours.

Accidents due to distractions

Distractions while driving happen when the person who is driving looks at something that happens inside or outside the vehicle and that has nothing to do with driving. For example, picking up the mobile phone or looking at a shop window. When are accidents due to distractions more frequent?

- In young drivers between 18 and 25 years old and in those over 70 years old.
- When there are several people inside the vehicle.
- In the summer months.
- On weekend trips and during the day.
- On roads more than in the city.
- On motorways and dual carriageways.

Causes of accidents due to distraction

Distractions related to the road	Distractions of the person who is driving
Knowing the road well and being overconfident. Signs are hidden or cannot be seen well for some reason. Complicated traffic situation with many vehicles, pedestrians and signs to pay attention to. Not seeing well because it is night or because the lights of other cars dazzle.	Suffering fatigue or sleepiness. Suffering stress, anxiety or depression. Being very old. Drinking alcohol, taking drugs or taking medicines. Some behaviours such as: using the mobile phone, lighting a cigarette or using the GPS sat nav. Looking at a map, throwing an insect out of the vehicle, eating or drinking while driving.

Smoking in the vehicle

People who smoke while driving have twice as many accidents as people who do not smoke.

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The reasons are:

- Lack of attention to the road while you look for the cigarette and light it.
- One hand is occupied with the cigarette.
- You see the road worse because of the smoke that comes from the cigarette.
- The air inside the car is of worse quality and can affect the abilities to drive.

Using the mobile phone

It can be very useful to carry a mobile phone in the car in case there is a breakdown or emergency. But using it wrongly can cause accidents. The risk of an accident when the mobile phone is used in the vehicle is four times greater than when it is not used.

The reasons are as follows:

- You pay more attention to the conversation than to the road.
- You judge distances worse and you stop seeing some signs because of the lack of attention to driving.
- Leaving the road and invading the opposite lane because you pay attention to the conversation.

- Difficulty handling the steering wheel when you hold the mobile phone in your hand or on your shoulder.
- Disorientation and loss of sense of time. In Spain it is prohibited to use the mobile phone while you drive. But it can be used through a hands-free device that allows you to talk on the mobile phone without touching the screen.

However, hands-free devices are also dangerous because the person who is driving pays part of their attention to the conversation and not to the road.

Recommendations for using the mobile phone and the hands-free device

- Stop the vehicle in a safe place when you need to make a call. Continue driving only when the call ends.
- When you speak through the hands-free device, tell the other person that you are driving.
- Do not have conversations that last more than one minute.
- Never call a person if you know that they are driving and do not have hands-free.
- When you drive, be careful with pedestrians who are talking on the mobile phone. They may be distracted.

GPS sat navs

It is a system that calculates the route from one place to another in real time. It helps you find your way and know where you have to go. If you make a mistake, the GPS sat nav recalculates the route and tells you again where you must go.

Tips to use the GPS sat nav well:

- Set up the device before you start driving. Do not do it while you drive.
- Leave it fixed in one place. So that it does not move or roll around the vehicle.
- Place it in a place where you can see it without taking your eyes off the road.
- Place it in a place that allows the airbags to open if necessary.

Watch video

Airbag. Safety device that is fitted in the front of a car and on the sides to protect passengers in case of an accident.

Topic 4. Obligations of drivers and pedestrians

Contents

Road safety

Obligations of drivers

- General obligations
- Two- or three-wheeled vehicles
- Special vehicles

Obligations of pedestrians

- General obligations
- Personal mobility vehicles

Animals on the road

Road safety

Road safety depends on drivers and also on pedestrians.

Watch video

That is why it is necessary for everyone to work together by respecting others and travelling in an orderly way when we are walking or in any vehicle. To achieve this, it is important that we follow all the instructions and obligations set by the law. For example, it is forbidden to throw objects and materials onto the road or leave them abandoned. Especially objects that can damage the road, make it difficult for vehicles to pass or cause an accident.

Drivers' obligations

General obligations

When you drive a vehicle you are obliged not to get distracted to prevent harm or accidents. In this way, you will avoid dangers for you, for the people travelling with you, those travelling in other vehicles and pedestrians. In particular, you must pay attention to pedestrians, especially children, older people, people who cannot see or who use a wheelchair.

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It is forbidden for all drivers:

- To drive a vehicle that gives off more noise, gases or fumes than allowed.
- To keep the vehicle doors open or open them before it has come to a complete stop.
- To open the doors and get out of the vehicle before making sure there is no danger for you or for other vehicles and pedestrians. For example, it is dangerous to open the car door without looking because at that moment a bicycle may pass in front and you can cause an accident.
- To use headphones connected to the mobile phone or to another device to listen to music or any type of sound.

Only the use of some specific headphones in the helmet of motorcycles and mopeds is allowed when the headphones are used as GPS navigators.

Watch video

- To use a mobile phone. You can use it through hands-free devices.
- To refuel the vehicle when the engine is running, the lights are on or the radio is connected.

Everything must be switched off or disconnected to refuel the vehicle.

Watch video

Fuel. Material that when combined with oxygen gives off heat. Many vehicles need fuel to work. For example, petrol.

Terms related to safe driving

Defensive driving A way of driving in which the person is alert to anticipate what other road users are going to do (drivers and pedestrians) and react appropriately. For example, taking into account that another driver may need to change lane and being prepared if this happens.

Danger zone

Space in which unexpected events may arise from other drivers and pedestrians. For example, a child runs out onto the road behind a ball or a bicycle crosses in front of your vehicle suddenly. These unexpected events can arise in front, behind and to the sides of your vehicle.

Watch video

Two- or three-wheeled vehicles

These vehicles are less visible, less stable and also more fragile than other cars. That is why their passengers are more likely to suffer injuries in the event of accidents. Which vehicles are these?

- Motorcycles and motorcycles with sidecar.
- Three-wheeled vehicles, quads and heavy quadricycles.
- Mopeds.

Whenever you travel in these vehicles as a driver or passenger you must wear the safety helmet properly fastened and fitted to your head. When you ride a bicycle at night or if visibility is poor for another reason, you have to switch on the vehicle lights and wear a garment or object that shines and can be seen from a distance of 150 metres. You must wear this bright garment or object if you are the driver and also if you are travelling as a passenger. Some two- or three-wheeled vehicles are required to have a seat belt. In those cases, you have to fasten it. In vehicles with a seat belt you do not need to wear a helmet.

Special vehicles

Vehicles that enter the roads to carry out works or provide a special service must have a yellow light switched on while they carry out improvement works or road maintenance. When they work on motorways or dual carriageways they must switch on the yellow light from when they enter the motorway or dual carriageway until they reach their destination. This light is used so that other drivers know that the vehicle is there. They do not have right of way to pass or overtake. The maximum speed at which these vehicles may travel is 40 kilometres per hour.

Pedestrians' obligations

General obligations

When walking in towns and cities When walking on the road When crossing the road When walking in towns and cities

- Walk on the pavement and not on the road whenever you can. This rule must also be followed by pedestrians using skates, skateboards or other similar devices that are not electric.
- Always keep to the right, in town and on the road, when you pull or push a bicycle or a two-wheeled moped, handcarts or a similar device.

People who use a wheelchair must also keep to the right.

- People carrying a very large object or a small vehicle without an engine may walk on the carriageway if the hard shoulder or the pavement are narrow and get in pedestrians' way.

Hard shoulder. Sides of the road that may be on the right and on the left. On the one on the right only some vehicles may travel, such as, for example, mopeds.

Watch video

When walking on the road

- On roads that are outside towns or cities you must always walk on the left of the road. Unless it is safer to walk on the right.
- At night you must wear a luminous or bright element that can be seen from a distance of 150 metres. For example, a vest that reflects light, a luminous bracelet...
- Groups of people must walk in single file, one behind the other. The people at the front must carry a white or yellow light that shines and the people at the back must carry a red light that shines as well. This way drivers will be able to know where the line of pedestrians starts and ends.

Watch video

When crossing the road or the street

- Cross in a straight line without stopping until you reach the other pavement.
- At pedestrian crossings, make sure that cars have stopped or are coming slowly before you start to cross.
- Go around squares or roundabouts. Do not cross them on the road.

Personal mobility vehicles

Electric scooters, electric unicycles and other personal mobility vehicles are forbidden to travel on:

Unicycle. Vehicle that has a single wheel attached to the seat by means of a metal bar.

- Pavements and pedestrian zones.
- Tunnels.
- Through roads.
- Motorways and dual carriageways.
- Roads that are outside the town or city.

What obligations do you have if you ride these vehicles?

- You cannot use headphones while riding.
- You cannot use a mobile phone while riding.
- You must take alcohol and drug tests when a traffic officer asks you to, like the rest of drivers.
- You must wear bright and luminous clothing if you ride at night or in places where visibility is poor.
- You must wear a helmet when the law requires it.
- You cannot carry passengers.

Animals on the road

Livestock, animals that carry loads or those that transport people, may only travel on the road when there are no other paths to get from one place to another. They must always be guided by a person over 18 years old.

Livestock and pack animals must always go on the right hand shoulder. When there is no hard shoulder they must go close to the right edge of the road.

Animals travelling in a herd or flock must also go on the right side of the road, taking up as little space as possible.

No animal may travel on motorways or dual carriageways.

Livestock. Group of animals raised by people to produce meat, milk, wool and other products. For example, cows, sheep, goats and pigs.

Topic 5. Safety devices in the vehicle

Contents

The importance of the vehicle in road traffic accidents

Active safety elements

- Lighting systems
- The brakes
- The wheels
- Other active safety elements

Passive safety elements

- The chassis and the bodywork
- The seat belt
- The airbags
- The head restraint
- The helmet

The importance of the vehicle in road traffic accidents For a vehicle to be safe and have less risk of having accidents it must meet two requirements:

1. Follow the programme of checks and compulsory maintenance for each vehicle.
2. Have a good safety system.

Most new vehicles have safety systems that help to make the vehicle safer and have fewer accidents. But the reality is that new vehicles have the same number of accidents as older vehicles. This can happen for two reasons:

1. Few drivers know what their vehicle's safety systems are for and how they work.
2. Drivers take more risks because they trust too much in their vehicle's safety systems. They do not realise that these safety systems have limits.

A vehicle's safety systems are divided into two types:

Active safety

Passive safety

Watch video

Active safety

Vehicle elements designed to prevent accidents. The driver has to use them for them to work. For example, brakes, wheels and the lighting system.

Passive safety

Elements that help passengers and other users suffer less harm in an accident. They work automatically. For example, seat belt, airbag and head restraint.

Active safety elements

Lighting systems

These are the different lights on the vehicle. They allow you to see what is around you and allow pedestrians and drivers of other vehicles to see you when it is night or visibility is poor. Thanks to the lighting system you can light up roads and streets to see nearby hazards and act in time. It is very important to use the lights properly by switching on the right ones and switching off the ones you do not need. Some lights can dazzle other drivers when they are used incorrectly.

Today's vehicles have new lighting systems to provide more protection. These new lighting systems are:

Xenon and bi-xenon lamps

Automatic activation of lighting

Adaptive lights

Xenon and bi-xenon lamps

Headlamps that light up the road with a powerful bluish-white light more similar to natural light than the headlamps that were used before. They manage to light up the road better, so the person driving can see further ahead and reduce eye fatigue.

Also, they dazzle other drivers less than other lights.

Xenon. Gas used in some lighting systems.

Automatic activation of lighting

System that measures the light outside and switches the lights on or off automatically so that the vehicle can be seen better.

Adaptive lights

System that adjusts the intensity of the lights and where they shine depending on the needs at each moment. It does this taking into account: the distance to the vehicle coming in the opposite lane, the type of road you are driving on and the speed your vehicle is travelling at. Adaptive lights are divided into two types of lights:

1. Cornering lights. They switch on when you turn the steering wheel to show the side your vehicle is

going to turn towards. This light helps you see better on bends.

At roundabouts it allows other drivers to know where you are going to turn.

Roundabout. Circular square that is a junction between streets.

2. Automatic main beam headlights.

System to switch on vehicle lights that light up more of the road when there is little light outside. For these lights to switch on your vehicle has to be travelling at a speed higher than 45 kilometres per hour and it must not detect another vehicle nearby. These lights switch off when they detect another vehicle on the road coming towards you in the opposite lane.

Brakes

Brakes are responsible for reducing the speed of the vehicle until it stops completely. A vehicle's brakes do not usually fail. But sometimes it does happen and this can cause a serious accident.

Watch video

What safety systems are there in brakes?

Engine brake

Anti-lock braking system (ABS)

Autonomous emergency braking (AES)

Emergency braking warning (EBD)

Engine brake

System used to reduce the speed at which a vehicle is travelling without using the wheel brake. It works when you stop accelerating and, in this way, the engine is what holds back the vehicle. If you drive in low gears (1st or 2nd) the engine slows the vehicle more. It is recommended to use engine braking when going down long slopes.

Anti-lock braking system (ABS)

Device that prevents the wheels from locking when braking. This system is very important because it helps you keep control of the vehicle and to brake in less distance. The more weight there is in the vehicle the harder it is for the wheels to lock when braking. For example, it will be harder for the wheel of your motorcycle to lock when you are carrying a passenger with you.

Autonomous emergency braking (AEB)

System that calculates, through a radar, the distance there is and should be between your vehicle and the one in front of you. If there is very little distance between the vehicles, this braking system will warn you by means of a voice and a sound. If you do not pay attention to these signals and there is a risk of an accident the system will perform emergency braking.

Watch video

Radar. System that can detect where an object is, for example another car, and at what speed it is travelling.

Emergency braking warning (EBD)

System that warns the driver of a vehicle that the vehicle in front has to make an emergency stop. When the driver in front presses the brake firmly and quickly, the brake lights of their vehicle flash. In this way, the system warns the driver behind so that they have more time to brake and try not to crash into it. How are brakes used? To brake safely and in a controlled way it is necessary to:

1. Brake gently, little by little and with enough time.
2. Take into account the condition of the road. When it is not in good condition, you must brake more gently and with more time.
3. Do not use the brake too much because it will get hot and will brake worse.

To brake in an emergency in a vehicle without ABS, you must press the brake pedal hard and fully until you notice that the wheel starts to lock. Then lift your foot off the brake a little to prevent the wheels from locking. But do not stop braking. To brake in an emergency in a vehicle with ABS, you must press the brake pedal hard and fully until the vehicle stops completely. What to do when the brakes fail? You can act in different ways depending on what the circumstance is in which the brakes fail.

Problem	Possible solution
The brakes get very hot because they have been used too much in a short time.	Release the brake pedal so that they cool down.
The brakes fail when the vehicle goes down a long and very steep slope.	Do not accelerate. Change down gears so that the engine acts as a brake. If you cannot change down gears, pull the handbrake lever gently.
Other occasions when the brakes fail.	Press and release the pedal several times. This will reduce the speed of the vehicle.

Wheels

The wheels of a vehicle are made up of two parts:

1. Rim. Metal part in the shape of a circle that is the inner part of the wheel.
2. Tyre. Elastic rubber part that is fitted around the rim. A piece of rubber is elastic when it can be stretched. When you stop stretching it, it returns to its original size.

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The rim and the tyre of a wheel have to be compatible with each other so that the tyre fits well on the rim.

Tyres

Tyre failures are a common cause of traffic accidents. That is why it is very important that they meet certain quality characteristics.

On new tyres that are in good condition you can read on the side what their characteristics are.

Watch video

They must not have blisters, deformations or tears. Nor should any of their layers be coming loose or have exposed cords or cracks. The part of the tyre that is in contact with the road is called the tread. The tread of cars must have grooves of at least 1.6 millimetres deep.

These grooves are used so that the tyre:

- Does not get hot.
- Is more flexible and rolls better on the road.
- Grips the road better.
- Removes the water it picks up from the road when it rains.

Watch video
Inflation pressure

Tyres are inflated with air so that they are more resistant and better support the load and the weight of the vehicle. The manufacturer of each vehicle is the one who sets how much air the tyres must have. This amount of air will be the inflation pressure.

What happens when the inflation pressure is not correct?

Inflation pressure lower than recommended	Inflation pressure higher than recommended
The tyre gets hot, deforms and wears out sooner. More fuel is used. The vehicle is less stable. The vehicle has a higher risk of skidding when the ground is wet. There is a greater possibility that the tyre will burst.	The tyre has less contact with the ground. Therefore, it grips the road less. The centre of the tyre wears more than the sides. The vehicle vibrates more because the tyres cannot absorb small stones and other elements. For example, the potholes on the road. The shock absorbers get damaged more.

You must check the tyre pressure of your vehicle at least once a month. You must also always carry a spare wheel with the correct inflation pressure.

Tyre wear

The rubber of tyres can wear out due to friction with the road. For that reason, you must change the tyres of your vehicle every five years, even if they are in good condition. The rubber of tyres can wear out earlier for these reasons:

- Driving abruptly or aggressively.
- Driving very fast.
- The climate.
- Tyres wear more in summer.
- Carrying a lot of load in the vehicle.

- Not having the correct inflation pressure.
- Driving on roads in poor condition.
- Problems with the brakes or with other elements of the vehicle that affect the wheels.

What should you do when a tyre gets punctured? The steps you must take if a tyre gets punctured while you are driving are:

- Reduce speed slowly until you stop the vehicle. Do not stop abruptly.
- Immobilise the vehicle in a safe place. Off the road and off the hard shoulder if possible.
- Change the punctured wheel for the spare one. Some vehicles have other systems to be able to keep driving and there are vehicles that carry a wheel that only serves to get you to the nearest garage. It is called a space-saver wheel. You can travel about 200 kilometres with this wheel at a maximum speed of 80 kilometres per hour.

Watch video

Wheel balancing

Wheel balancing consists of adding weight to the rims so that the wheels support the weight and the load of the vehicle. This balance is very important so that the wheels turn properly.

Wheels can lose balance for these reasons:

- Tyre change.
- The rims lose the counterweights.
- The rims get dented.
- The wheel wears out or gets cuts.

Watch video

How to know that a wheel is losing balance? You may notice one of these things:

- The vehicle makes noises and vibrates.
- The tyres wear more than normal.
- The wheel bolts are loose.

Other active safety elements

Traction control

Electronic stability control

Speed limiters

Cruise control

Adaptive cruise control

Traction control

Safety system that identifies if a wheel is turning faster than the others and brakes it so that it does not skid. This system helps to maintain the stability of the vehicle on bends, when going up a slope or when it is raining.

Electronic stability control

Safety system that helps the vehicle to follow the path set by the steering wheel and not to deviate when it does not turn as much as the driver asks or turns too much. This system helps you keep control of the vehicle in dangerous situations. For example, when swerving to avoid an obstacle or when taking a bend.

Speed limiters

System that allows you to choose the maximum speed that your vehicle can reach. When you activate the speed limiter, the vehicle will not go above the speed chosen by the driver.

Cruise control

System that helps you maintain the speed that you choose to drive without needing to press the pedals to accelerate or slow down. For example, travelling all the time at 100 kilometres per hour on the motorway. This system is activated from the steering wheel. It is deactivated when you accelerate or brake. Then you have to choose the speed again and activate it.

Adaptive cruise control

As well as helping to control speed, this system helps you keep the distance from the vehicle in front.

Your vehicle will reduce speed or brake if you do not keep that distance and you get too close to the vehicle in front.

Passive safety elements

The chassis and the bodywork The chassis is the internal structure of the vehicle on which all the parts that make it up are fitted. It is the skeleton of the vehicle and cannot be seen at first glance. The bodywork is the external metal structure that covers the vehicle. In the event of an accident, the chassis and the bodywork deform and protect the vehicle so that the passengers inside suffer less damage.

Rigid passenger compartment

Deformable parts

The seat belt What is it for? To protect all the people travelling in a vehicle in the event of a sudden impact or if the vehicle rolls over. In these cases, the belt helps people not to move from their seats and not to be thrown out of the vehicle.

Wearing the seat belt correctly means that a person has twice the chance of surviving an accident. In fact, there are fewer deaths from traffic accidents since the seat belt exists. For a seat belt to be useful and safe it must meet the quality requirements and be properly secured to the vehicle bodywork. It must be checked from time to time and taken to the garage to be repaired or replaced if it has any damage. Who must wear the seat belt?

All the people travelling in a vehicle that has seat belts installed.

Drivers must do so and also passengers on all roads and streets.

There are some exceptions in which drivers are not obliged to wear a seat belt, although it is recommended that they always wear it. These exceptions can only be applied when the vehicle is travelling within a town or city. Never when travelling on roads, dual carriageways or motorways. Exceptions:

- Drivers who are parking moving the vehicle backwards.
- Taxi drivers who are on duty.
- Children who are less than 135 centimetres tall and travel in a taxi that does not have child safety systems. In those cases, the children will travel in the rear seats and will fasten the seat belt that is in the seat.
- Delivery drivers who have to get out continuously of the vehicle to collect and deliver orders.
- Drivers and passengers of vehicles in emergency services. For example, ambulances.
- Driving school instructors who take a learner and are in charge of the additional controls of the vehicle.

Only people who cannot wear it for medical reasons or disability may travel without a seat belt in any situation or on any road. These people must carry a medical certificate that explains the reasons.

Seat belts for children

Children who are less than 135 centimetres tall must use a different safety system, more suitable for them. They are called child restraint systems and they are adapted to each child's height and weight. It is best for the child to try the child restraint system before starting to use it to check that it fits their measurements and is comfortable.

To fit a child restraint system in a vehicle you must take into account the following recommendations:

- Do not install the child seat in a seat that has an airbag in front of it.
- For small children, place the child seat facing the opposite direction to the vehicle's travel. That is, facing backwards. It is safer.
- Place the child seat in the central rear seat. This way they will be more protected in case there is an accident from either side.

Watch video

It is important that child restraint systems have passed all quality tests so that they are safe. In vehicles with nine seats or fewer children must travel in the rear seat and have the child restraint system properly secured.

Children may only travel in the front seat when:

- The vehicle has no rear seats.
- All the rear seats are occupied by other children.
- The child restraint systems cannot be installed in the rear seats. On buses, children who are less than 135 centimetres tall and are three years old or more must also use child restraint systems.

If there are none, they will have to fasten the seat belt that is in the seat, as long as it is suitable for their height and weight. How do you fasten the seat belt? You must wear the seat belt properly fastened and adjusted to the body. It must not be too loose or too tight so that you can move easily. The seat belt has two straps that you must wear in the correct place so that they protect you.

Strap chest It must pass over the collarbone, between the shoulder and the neck, and go down the centre of the chest.

Placing the chest strap over the neck or over a breast can cause serious injuries in an accident.

Placing it over the shoulder can make the seat belt slip and protect less.

WRONG

RIGHT

Strap lap It must be placed over the hip bones. Always below the stomach. If it is placed above, it can cause serious injuries inside the body in an accident.

WRONG

RIGHT

After fastening your seat belt pull it slightly upwards to check that it is properly adjusted. Check that it is not caught or twisted in any of its parts.

The submarining effect This effect occurs when in an accident the body slides down, under the seat belt. This happens because the seat is tilted backwards and the seat belt is incorrectly positioned. To prevent this from happening you must:

- Fasten your seat belt properly.
- Check that the seat belt is properly adjusted to the body.
- Do not place towels, cushions or covers on the seat that could make you slip.
- Have a correct driving posture, without tilting the seat too far back.

Submarining effect

Airbags

What are they? A safety device that consists of an air bag that inflates inside the vehicle in the event of an accident to protect passengers.

Airbags are located at the front of the vehicle and on the sides. What are they for? The main functions of the airbag in the event of an accident are:

- To slow down the sudden movement of the body.
- To prevent people from hitting violently against any part of the vehicle.
- To protect people's face and eyes from broken glass and other elements that come loose because of the accident. Precautions for using airbags:
- Always use the seat belt so that the impact against the airbag is smaller.
- Place your chest at a distance of at least 25 centimetres from the steering wheel so that the front airbag does not hit you when it inflates.

- Disconnect the passenger seat airbag if you are going to place a child seat in that seat.

The head restraint A device that protects the neck and the cervical vertebrae in the event of a sudden impact. It is very important to position the head restraint at the height of the passengers' heads, both in the front seats and in the rear seats.

Cervical vertebrae. Bones that are in the upper part of the spine. The top part of the head restraint must be at the same height as the top part of your head. You should try to keep the distance between your head and the head restraint to four centimetres or less.

The helmet

Head injuries are the main cause of death in accidents involving two-wheeled vehicles. Three out of ten people who have an accident in a two-wheeled vehicle save their life because they wear a helmet. What does the helmet do in an accident?

- It protects the head from impacts against the ground, other vehicles or elements of the road.
- It prevents stones, metal or other sharp objects from entering the head
- It spreads the force of the impact over the whole helmet so that it does not concentrate on just one point of the head. This prevents serious injuries.
- It helps prevent the face and head from being burned when sliding along the ground after the fall.

Type of helmet	Full-face model. That is, it also protects the lower part of the face and the jaw. Watch video
Material	Some helmets expire after a few years and can lose properties if they are painted or stickers are put on them.
Size	It must fit the head properly.
Strap fastening	Check that the helmet is properly fastened and does not come off even if you pull hard. Watch video
Colour	Light and bright helmets are safer.
Ventilation	It must have enough holes for air to go in and out.

Visor	It is recommended to wear visors over the face that do not mist up
Approval	and allow you to see well with them. These visors are called anti-fog. The helmet must have a label that explains that it has passed the quality tests and is a safe helmet.

Topic 6. Vehicle components and rules for using them

Contents

Vehicle controls

- Controls of vehicles in general
- Controls of motorcycles

Visibility when driving

- Windscreen wipers
- Windscreen washers
- Heated rear window
- Rear-view mirrors

Rules for driving comfortably and safely

- Rules for driving cars, vans and lorries
- Rules for driving motorcycles

Systems that help you drive safely

Vehicle controls

Controls of vehicles in general

Some vehicle controls are used with the feet and others with the hands.

Controls that are used with the feet

Accelerator pedal

Brake pedal

Clutch pedal

Controls that are used with the hands

Steering wheel

Handbrake

Accelerator pedal

It is used to regulate the amount of fuel that enters the vehicle's engine. The harder you press the accelerator, the more fuel will enter the engine and the faster the vehicle will go. This pedal is pressed with the right foot. If you do not press it, the vehicle receives the amount of fuel needed to keep running and not stop.

Watch video

Brake pedal

It is used to reduce speed or stop the vehicle. The brake pedal acts on all the wheels of the car. This pedal is pressed with the right foot and you must press it gently.

Clutch pedal

It is a mechanism that allows the gearbox to choose the gear you want to use and connect it to the engine. It is used to change gear while driving and to allow the vehicle to go forwards or backwards. When you press the clutch pedal fully the movement of the engine is not transmitted to the gearbox or the wheels. Then you can change gear. The gearbox is used to change gear.

Watch video

When you release the clutch pedal the engine power works again by itself so that you can continue driving.

Steering wheel

Part of the vehicle that allows you to control which direction you are going and where the wheels move.

Watch video

How should you hold the steering wheel?

- With both hands. You should only let go with one hand for just long enough to operate other controls of the car.
- By the outer part. You must never hold the steering wheel by the inner part.
- Firmly, but without using much force.
- Without crossing your hands when turning the steering wheel.

Handbrake

A system that brakes the rear wheels of the vehicle and makes the vehicle stay stopped. The handbrake is a lever that is used with the right hand.

Controls of motorcycles

Motorcycles have the same controls as the rest of motor vehicles. But they are located in a different place and are used differently.

Watch video

With the left hand Clutch. It works by squeezing a lever. Control to emit sounds (horn). Switch to turn on the lights.

With the right hand Front brake. Lever to brake the front wheel.

Accelerator. It is on the handlebar and is activated by twisting the grip.

With the feet

Brake pedal. Activates the brake of the rear wheel. It is usually on the right side of the motorcycle. Gear pedal. It is used to change gear. It is usually on the left side of the motorcycle.

Visibility when driving

It is necessary to see well everything that is around the vehicle so that driving is safe. To achieve this, all the windows must be clean. It is prohibited:

- Sticking films or stickers on the windows that make an area harder to see.
- Fitting coloured windows that are not approved or permitted by law.

Which vehicle components help you see better?

Windscreen wipers

Windscreen washers

Heated rear window

Rear-view mirror

Windscreen wipers

They keep the vehicle's front windscreen clean. There are cars that also have a rear window wiper to clean the rear window. They move from side to side to clean the windows well. They have different speeds. You should try to use the slowest one. The windscreen wiper and the rear window wiper have a wiper blade that you must change when you notice it leaves marks on the glass or it does not get properly clean. You must not use the windscreen wiper when the windscreen is dry because it can scratch it. There are automatic windscreen wipers that activate by themselves when they detect water. They work faster or slower depending on the amount of rain that falls on the vehicle. The driver can activate or deactivate the automatic windscreen wiper.

Windscreen washers

A device that sprays a jet of liquid onto the windscreen so that the windscreen wiper can clean better. There are special liquids that work as windscreen washer fluid, although you can use water mixed with a little detergent. It is important to check that the reservoir where the windscreen washer is has liquid and refill it when it is running low.

Watch video

Heated rear window

Lines that you can see on the vehicle's rear window. They remove ice and vapour that form on this window so that you can see well.

Rear-view mirrors

Mirrors that allow the driver to see better what happens at the sides and behind the vehicle.

Rear-view mirrors in cars, vans and lorries

Cars, vans and lorries that can carry up to 3,500 kilos in weight must have the following rear-view mirrors:

- One rear-view mirror outside the vehicle, on the left-hand side.
- One rear-view mirror inside the vehicle, in the central part, above the front windscreen.
- One rear-view mirror outside the vehicle, on the right-hand side. This rear-view mirror is recommended, but it is not compulsory.

Watch video

You can fit sun blinds on the rear side windows of the vehicle only when the two exterior rear-view mirrors are fitted, on the right and on the left.

Rear-view mirrors on motorcycles

Motorcycles that travel at less than 100 kilometres per hour are required to have one rear-view mirror on the left-hand side of the vehicle.

Motorcycles that travel at more than 100 kilometres per hour are required to have two rear-view mirrors. One on the left-hand side of the vehicle and another on the right-hand side.

Rear-view mirrors on two-wheeled mopeds

These vehicles must have one rear-view mirror compulsory on the left-hand side of the moped. The right-hand mirror is optional.

Rules for driving comfortably and safely

Having a good driving posture helps to avoid fatigue and makes driving safer. Sitting in a good posture in front of the steering wheel helps you respond better and faster to unexpected situations. Driving too close to the steering wheel causes fatigue because you have to strain your body more to make movements.

Driving too far from the steering wheel forces you to lean forwards to reach the controls and to lift your back off the seat backrest. This can be dangerous.

Your head must be above the steering wheel. If this is not possible because you are not tall enough, you can use a suitable accessory securely fixed to the seat, bearing in mind that it cannot be a cushion or similar.

Rules for driving cars, vans and lorries Before you start driving you must check that: The seat and the backrest are properly adjusted The rear-view mirrors are properly positioned The seat belt is properly fastened The seat and the backrest are properly adjusted when:

- You can reach the pedals well and you can press them fully without straining your ankles.
- Your legs are not fully stretched out. They are slightly bent and do not rub against any part of the vehicle.
- Your head is above the steering wheel so that you can see over it and not through it.
- The position of the seat backrest allows you to reach all the controls. To check it, rest your back against the backrest, stretch your arms and check that your wrists can rest on the top part of the steering wheel.
- The head restraint is at the height of your head. The rear-view mirrors are properly positioned when:
- Looking at the rear-view mirror inside the vehicle, you can see the four edges of the rear window.
- Looking at the rear-view mirrors at the sides, you can see the road, the vehicles coming from behind and those at your sides. By turning your neck a little you can see a part of the left rear of your vehicle. If you find that the rear-view mirrors are badly positioned while you are driving, you must stop the vehicle and position them correctly. To do this, try to stop in a flat and straight place. The seat belt is properly fastened

Once you have made these checks fasten your seat belt properly before you start driving.

Rules for riding motorcycles

- Keep a natural body position. Do not force your posture.
- Lean your body just enough to reach the handlebars.
- Keep your arms and hands relaxed. This will help you avoid tiredness.
- When the motorcycle starts moving place your feet on the footrests. Do not let them hang down.
- Lean your body a little to take bends. Do not lean too much, like professional riders in races do.

Watch video

Which postures are not suitable for riding a motorcycle?

- Leaning your body too much.
- Keeping your elbows tucked inwards.
- Having your arms completely straight. These postures cause fatigue and can make it harder for you to react.

Systems that help you drive safely

Traffic sign recognition system

(TSR) It detects speed limit signs on the road so that the person driving can reduce the vehicle's speed if necessary. In some vehicles the system shows the number that corresponds to the maximum speed at which the vehicle may travel. This number appears on the instrument panel. In other vehicles, as well as warning, the system reduces the speed by itself when the driver is going faster than they should.

Instrument panel. Set of indicators that are in front of the driver and that inform about the condition and operation of the vehicle.

Lane departure warning (LDW)

This system warns the driver when they make an unexpected lane change because they are distracted or because they have fallen asleep. The system warns by means of a light on the instrument panel, a sound or by making the driver's seat or the steering wheel vibrate.

Watch video

Lane keeping assist (LKA)

A system that detects the white lines on the road to help the driver steer their car and not leave their lane. It warns the driver when they leave the lane.

Watch video

Reversing camera and 360-degree camera

The reversing camera is located at the rear of the vehicle. Its function is to show the driver the obstacles behind the car when reversing. For example, when they have to reverse to park the vehicle. What this camera sees appears on the screen at the front of the vehicle. The 360-degree camera captures everything around the vehicle and projects it on the screen at the front. These types of cameras are very useful for parking the vehicle.

Parking assist

A system to help the driver park or leave a parking space. There are different parking assists with different functions such as:

Finding a place to park, showing what is around the vehicle, or turning the steering wheel without the driver's help while the vehicle parks. The driver must pay close attention to make sure that these movements can be done safely and can stop the manoeuvre at any time.

Hill start assist system

This system helps the vehicle not roll backwards when the driver moves off on a very steep slope. Or when the driver moves their foot from the brake to the accelerator to continue driving on a slope.

Rear cross traffic alert system (RCTA)

This system warns the driver if there are other vehicles behind when reversing. For example, when leaving a parking space. The system will make a sound when it sees that there is a vehicle behind that is too close and there may be a risk of a collision.

Topic 7. Vehicle lighting system

Contents

General rules for using the vehicle lights

Vehicle lights

- Sidelights
- Parking light
- Clearance lights
- Main beam or main beam headlights
- Dipped headlights
- Fog lights
- Reflector
- Direction indicator
- Hazard warning lights
- Brake light
- Third brake light
- Reversing light

Audible warnings

General rules for using the vehicle lights When must you switch on the vehicle lights? At night you must have the lights switched on that apply at each moment whenever it is night. During the day, you must switch them on in these cases:

- It is a dark day with little natural light.
- You go through a tunnel.
- You drive in a reversible lane Reversible lane. A lane that is sometimes open in one direction and other times in the opposite direction.
- You drive in an additional lane or one that goes in the opposite direction to the one it normally has. For example, an additional lane is put in place or the direction of a lane is changed for a time because of roadworks or because of an accident in that area. Additional lane. A new lane that is opened temporarily because of roadworks, queues, an accident...

If some lights break down while you are driving, you must reduce the car's speed and switch on the lights that work, even if they are less visible. When you reach a lit area, you must park the vehicle and not continue driving until the lighting system is repaired. A road is considered poorly lit when:

- You cannot read the number plate of the vehicle in front at a distance of 10 metres.
- You cannot see a vehicle painted a dark colour at a distance of 50 metres.

Vehicle lights

1. Front sidelight
2. Dipped headlights
3. Main beam
4. Direction indicators
5. Front fog light
6. Daytime running lights
7. Rear sidelight
8. Brake light
9. Rear fog light
10. Direction indicators
11. Reversing light
12. Third brake light

Sidelights

They are used so that the vehicle can be seen well and so that its width is known. There are sidelights at the front, at the rear and on the sides of the vehicle.

Watch video

What colour are the sidelights? The front lights are white. The rear lights are red. The side lights are amber. When are they used?

- When you drive at night.
- When going through a tunnel.
- On dark days when there is little natural light for driving. When the vehicle is moving, the sidelights are always switched on together with another type of lights.

Watch video

The sidelights are only switched on on their own, without any other type of lights, when the car is stopped for some reason.

For example, a car that has to stop because of a breakdown on a poorly lit road.

Parking light

It is used when a vehicle has to stop in an area that is poorly lit. It can be used instead of the sidelights. They are fitted at the front and at the rear of the vehicle. They are the same colours as the sidelights.

Width lights

They are used to clearly indicate what the total width of the vehicle is. They are two front lights and two rear lights that are fitted on the outer edges of the vehicle, in the highest area, so that they can be seen well. They are used in the same cases as the sidelights.

Width. Width of a vehicle

Watch video

What colour are they? The front lights are white. The rear lights are red.

Which vehicles must have them? It is compulsory for cars, trailers and semi-trailers that measure more than 2.10 metres wide. It is optional for cars that measure between 1.80 metres and 2.10 metres wide.

Main beam or main beam headlights

It is used to light a long distance in front of the vehicle travelling on the road. This light is compulsory for all cars and optional for mopeds. It is a powerful white light that can dazzle other vehicles.

Watch video

When is it used? On roads that are outside towns or cities. It is compulsory to have it on:

- When you drive on a road at night, there is little natural light and you are going at more than 40 kilometres per hour.
- At any time of day in tunnels outside a municipality when the tunnel is not well lit. It is optional to switch it on when you drive at less than 40 kilometres per hour. When is it forbidden to switch on the main beam?
- Inside towns and cities.
- When the vehicle is stopped. You can use the main beam to warn other drivers of a danger. You will do this by switching the lights on and off intermittently so as not to dazzle.

Dipped headlights or dipped beam

It is used to light the road in front of the vehicle without dazzling or bothering the drivers of other vehicles. It is white.

It is compulsory for:

- Cars: they must have two lights.
- Two- and three-wheeled mopeds: they must have one or two lights.

Motorcycles, three-wheeled vehicles and heavy quadricycles may have one or two. When is it compulsory to use it? At night, on all types of roads, inside and outside towns and cities. During the day, it is used when driving through tunnels or through temporary lanes that are put in place because of roadworks, accidents or other circumstances.

These lights must also be used when:

- The vehicle does not have main beam.
- You drive at less than 40 kilometres per hour.

- There is a possibility that you could dazzle the drivers of other vehicles when using the main beam. Motorcycles must always have dipped headlights on during the day when they travel on any street or road.

Watch video

Fog lights

They are used to see the road better in case of fog, snow, heavy rain and smoke. There are front and rear fog lights. The front lights are white or yellow. The rear lights are red

How are they used?

Front fog lights are used with the sidelights. You can also use them at the same time as the main beam and dipped headlights.

Rear fog lights you must only use in cases of great need, when the weather and visibility conditions are very bad. They can be switched on at the same time as the main beam and dipped headlights and the front fog lights.

Watch video

Reflector

It is a device that reflects external light from other vehicles. It is used so that the vehicle can be seen. These types of lights are on the front, rear and side of the vehicle. Reflectors must be the colour:

- White at the front.
- Red at the rear.
- Yellow at the sides.

Watch video

Trailers and semi-trailers must have them in a triangular shape at the rear.

Direction indicator

It is used to warn that a vehicle is going to turn and move to the right or to the left. You must switch it on whenever you are going to change direction, do a U-turn or change lane. You must switch it off when you finish that movement. Its light is yellow and flashes intermittently. It is compulsory for all motor vehicles, trailers and semi-trailers.

Watch video

Hazard warning lights

They warn that the vehicle has a problem and may be a danger to other vehicles that are on the road. To show the hazard warning lights all the vehicle's direction indicators are switched on. These lights are compulsory for cars, their trailers and semi-trailers.

Watch video

When are they used? They are used in the following cases both by day and by night.

- When a vehicle has a breakdown and cannot reach the minimum speed that must be driven on that road.
- When a vehicle is making an emergency journey. For example, taking to hospital a woman who is

about to give birth.

- To warn that another vehicle is stopped because of a problem or an emergency on any road or in a tunnel.
- When picking up and setting down passengers in school transport vehicles.

Brake light

It indicates that the vehicle is braking. The driver must warn with these lights whenever they are going to brake the vehicle, even if it is for a short time. For example, stopping at a red traffic light. Its light is an intense red and they are fitted at the rear of the vehicle. It is compulsory for all cars, mopeds, trailers and semi-trailers.

Watch video

Third brake light

It is a single light that is fitted at the rear of the vehicle, above the brake lights. It switches on at the same time as the brake lights and is the same colour. This light is optional for cars. Motorcycles cannot have it.

Reversing light

It lights the road from the rear of the vehicle to warn that that vehicle is going to travel backwards. They are one or two white lights that switch on automatically when the vehicle starts to go backwards. This light is compulsory for all cars, except for motorcycles, which are forbidden to have it. It is optional for three-wheeled vehicles and heavy quadricycles.

Watch video

Audible warning devices

They are sounds used to warn other drivers that a vehicle is there. They can only be used in the following cases.

- To avoid a possible accident.
- On roads that are outside a town or city, to warn a driver that you are going to overtake them.
- To warn other vehicles that you need to pass or park there because you are carrying out a priority service. For example, when you are taking a seriously injured person to hospital.

Topic 8. Traffic signs

Contents

Types of signs

- General rules
- Traffic officers
- Temporary signs
- Traffic lights

Vertical signs

- Warning signs
- Regulatory signs

- Information signs

Lines and markings on the roads

- White lines and markings
- Coloured lines and markings

Types of signs

General rules

- It is forbidden to change road signs without a justified reason.
- The instruction of each sign must be followed across the full width of the street or road. In cases where there is a different instruction for each lane there will be road markings that indicate this.
- All drivers and pedestrians must obey the instructions of traffic signs.

Which traffic signs must you obey first?

1

Signals and orders from traffic officers.

2

Signs that indicate that the road has been modified temporarily for some reason and signs that warn of bends and obstacles.

3

Traffic lights. 4 Vertical traffic signs. 5 Road markings and lines.

Watch video

Signs on the right and left

- You must obey the vertical signs and traffic lights that are on your right on the road.
- When you want to turn left or go straight ahead and there are no vertical signs or traffic lights on the right you must obey the signs that are on the left.
- You must obey the signs on your left when you want to turn left or go straight ahead and there are vertical signs or traffic lights with different instructions on each side.

Contradictory signs

Sometimes there are signs placed together, but they give different instructions.

Different signs	Which one must you obey?
STOP sign and traffic light on green	Traffic light on green
STOP sign and give way sign	STOP sign
Traffic light on green and no left turn sign	You must obey both. You can go straight ahead or turn right.

Traffic officers
Signals and orders from traffic officers

The signals given by traffic officers
must always be obeyed. To give instructions they will use objects and clothing that can be seen well from 150 metres away.

Traffic officers will give instructions by the following means:

Arm signals	Sound signals Signals from with a whistle a vehicle
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Arm signals

Arm raised vertically

All drivers approaching the officer must stop. When this signal is made at a junction drivers who were already inside the junction may continue.

Arm or arms extended horizontally

All drivers approaching the officer must stop. The officer’s arm or arms act as a barrier for vehicles approaching. This order must be obeyed until the officer gives another instruction, even if they lower their arms.

Swings a red or yellow light with one arm

Drivers towards whom the officer directs the light must stop.

Arm extended moving up and down

All drivers approaching the officer from the side from which the officer makes the arm signal must reduce the vehicle’s speed.

Sound signals with a whistle

Several short, repeated whistle blasts One long whistle blast	Continue driving
Arm extended downwards and held still Police vehicle with a flashing red or yellow light and emitting sounds	on the right-hand side for the drivers indicated by the arm. You must stop the vehicle on the right-hand side, in front of the police vehicle, and stay inside the vehicle following all the officer’s instructions.

Temporary signs

Panels with messages that change

Beacons

Panels with messages that change

They are panels placed on roads that change the information depending on traffic circumstances. They are used to:

- Give information to drivers.
- Warn of possible dangers.
- Give recommendations and instructions that must be followed.

Beacons

They are lights, signs and devices to highlight roadworks. They indicate the direction you must follow on a street or road and the obstacles you may find on them.

Temporary direction panel It prohibits entry and tells you which way you must go.

Flags and cones

They prohibit passing through them and through the space between each flag or cone.

Steady red light

The road is closed to traffic.

Steady or flashing amber lights They prohibit passing through the space between the lights.

Permanent direction panels

Devices that show which way you must go in a place where there is always a possible danger. The number of panels warns how much danger there is in that area:

- One panel means moderate danger.
- Two panels mean quite a lot of danger.
- Three panels mean a lot of danger.

Traffic lights

There are different types of traffic lights: For pedestrians

Circular for vehicles

Lane

For some vehicles

Traffic lights for pedestrians

Their indications are for pedestrians.

Not for drivers

Watch video

Circular traffic lights for vehicles

Steady red light	No entry.
One or two flashing red lights	No entry for a period of time. They are placed before a level crossing or a bridge.
Steady amber light	No entry, the same as a steady red light. Although it does allow vehicles to pass that cannot stop for safety reasons.
One or two flashing amber lights	You must give way to vehicles coming from the right and from the left. Also to pedestrians.
Steady green light	It allows vehicles to pass and also gives them priority at the junction.

Black arrow

The vehicle can only go to the side shown by the arrow.

Also, you must obey the indication of the colour behind the arrow. For example, do not go through if it is red.

Green arrow

It allows vehicles to go ahead in the direction shown by the arrow. You must do it carefully, watching for pedestrians crossing the road and vehicles entering that lane.

Watch video

Square traffic lights for vehicles or lane traffic lights

Only vehicles travelling in the lane where the traffic light is placed must obey them.

Meaning of their lights

Red light in the shape of a cross You must not use that lane.

All drivers must leave it as quickly as possible.

Green light in	It is permitted to drive
the shape of a downward arrow	in that lane. Drivers must obey the other signs in that lane.
White or amber light in the shape of a downward arrow	It tells drivers that they must move to the lane shown by the arrow because the lane they are in now is going to be closed.

Traffic lights for some vehicles

Traffic lights for cycles and mopeds

A cycle is shown on the traffic light.

Their indications do not apply to other vehicles.

Traffic lights for trams, buses and other vehicles They have a white line on a black circular background. Their indications do not apply to cars.

Vertical signs

They are plates with information that stay fixed on posts or on other structures on the road and in the streets.

Vertical signs depending on what their function is:

1. They warn of dangers
2. They require or prohibit
3. They inform and guide Remember... A road is the place where you travel, it can be a street, a road or a track. The carriageway is the part of the road where vehicles travel.

Warning signs

The name of these signs starts with a P for danger.

P-1. Priority to pass.

Vehicles travelling on that road have priority to pass before vehicles coming from the roads or streets at the sides.

P-1a Priority to pass before vehicles coming from the road on the right.

P-1b Priority to pass before vehicles coming from the road on the left.

P-1c Priority to pass before vehicles that want to enter that road from the right.

P-1d Priority to pass before vehicles that want to enter that road from the left. P-2 Junction.

Vehicles coming from the right have priority to pass. P-3 Traffic lights.

Nearby there is a junction or a section of road with traffic lights to regulate traffic. You must drive carefully because there may be vehicles queued and stopped at the traffic lights.

P-4 Junction with roundabout traffic.

Vehicles can only turn in the direction of the arrows. P-5 Movable bridge. There is a bridge nearby that can be raised or turned. When it is raised or turned, traffic is stopped and vehicles must wait. P-6 Tram crossing.

Danger because nearby there is a junction with a tram line that has priority to pass. P-7 Level crossing with barriers nearby. A level crossing is the place where railway tracks cross a track or road. On the road, before reaching the railway track, you will find a barrier. P-8 Level crossing without barriers nearby. There will be no barrier separating the road you are travelling on and the railway track. P-9a and P-10a There is a level crossing, movable bridge or quay about 300 metres away. P-9b and P-10b There is a level crossing, movable bridge or quay about 200 metres away. P-9c and P-10c There is a level crossing, movable bridge or quay about 100 metres away.

P-11 Level crossing without barriers

At that same place.

P-11a Level crossing without barriers where there is more than one railway track. At that same place.

P-12 Airport

Danger due to unexpected noises that aircraft can cause. P-13a Dangerous bend to the right. P-13b Dangerous bend to the left. P-14a Several dangerous bends nearby. The first goes to the right. P-14b Several dangerous bends nearby. The first goes to the left. P-15 Uneven road surface. There are humps, speed cushions, or the road is in poor condition.

Humps are parts of the road sloping upwards that stick out to force vehicles to reduce speed. Speed cushions are also parts of the road, they are potholes or trenches in the road. P-15a There are humps in the road. P-15b There are speed cushions in the road.

P-16a Very steep downhill slope. P-16b Very steep uphill slope. P-17 The carriageway becomes narrower.

P-17a The carriageway becomes narrower on the right.

P-17b The carriageway becomes narrower on the left. P-18 Roadworks. P-19 Slippery carriageway. P-20 Pedestrians.

Careful, you are approaching a place where there are usually pedestrians. P-21 There are children near that place. For example, the exit of a school.

P-22 Place where cyclists often pass or cross. P-23 Place where domestic animals may often pass. For example, cows, sheep...

P-24 Place where animals at large may cross.

P-25 Vehicles may travel in both directions.

P-26 There may be stones or other obstacles on the road because it is an area where objects fall. For example, a road below a mountain.

P-27 The road ends at the quay of a port or at a watercourse. P-28 Small stones jump up from the road surface, called gravel.

P-29 There is strong side wind. P-30 The carriageway is at a different height on one side because there is a step. P-31 There is heavy traffic on a section of the road.

P-32 There are vehicles that hinder traffic due to a breakdown, accident or other causes. P-33 There is fog, snow or smoke on the road and visibility is worse.

P-34 An area of the carriageway is very slippery due to ice or snow.

P-35 Warning of a danger different from all the previous ones.

Regulatory signs

They are signs that inform drivers and pedestrians of their obligations, restrictions and prohibitions on the road.

Regulatory signs placed beside or above the sign that shows the name of the town or city mean that you must comply with that rule throughout the whole town or city. For example, a circular sign with the number 40 above the name of a town means that it is prohibited to drive at more than 40 kilometres per hour throughout the whole town. If the circular sign is before or after, it means that you must not drive at more than 40 kilometres per hour on that section, until you find another sign.

The names of regulatory signs start with the letter R and there are different types:

Priority signs

R-1 Give way. You must give way to all vehicles already travelling on the road you want to enter or that are in the lane you want to move into. R-2 STOP. You must stop the vehicle and allow vehicles already travelling on the road you want to enter to pass. R-3 Priority road. While you are on that road you have priority to pass at junctions before drivers arriving at the junction from another road. R-4 End of priority road. The road you are travelling on stops having more priority than other roads.

R-5 No entry to a narrow passage if, when going through it, you obstruct vehicles coming towards you.

R-6 You have priority to pass through a narrow passage before vehicles coming in the opposite direction.

Signs that prohibit entry

R-100 No traffic. It prohibits traffic for all vehicles, in any direction. R-101 No entry. It prohibits entry for all types of vehicles.

Vehicles may be travelling in the opposite direction to you because this prohibition may apply only in one direction. R-102 No entry for motor vehicles. Mopeds may enter. R-103 No entry for motor vehicles. Two-wheeled motorcycles without sidecar may enter. R-104 No entry for motorcycles.

R-105 No entry for mopeds and for vehicles for people with reduced mobility.

R-106 No entry for vehicles carrying goods, even if they are not carrying much load. For example, vans and lorries.

R-107 No entry for vehicles carrying goods and that can carry a load of tonnes greater than the number shown on the sign. These vehicles are prohibited from entering, even if they are empty at that time.

R-108 No entry for vehicles carrying dangerous goods.

R-109 No entry for vehicles with goods that can explode or catch fire.

R-110 No entry for vehicles carrying more than 1,000 litres of products that can pollute water.

R-111 No entry for tractors and other motorised agricultural machines.

R-112 No entry for motor vehicles with trailer.

Articulated vehicles and single-axle trailers may enter.

R-113 No entry for vehicles drawn by animals. For example, horse-drawn carriages. R-114 No entry for bicycles.

R-115 No entry for handcarts pushed by people. R-116 No entry for pedestrians.

R-117 No entry for animals ridden by people.

Signs that restrict passage

R-200 You must stop the vehicle at the place where it is placed to comply with the signs that apply in each case. For example, a toll booth on the motorway or a police checkpoint.

R-201 No entry for vehicles carrying a load that weighs more than the number of tonnes shown on the sign.

R-202 No entry for vehicles with an axle load greater than the number of tonnes shown on the sign. In this case, vehicles must not pass with an axle load greater than 2.4 tonnes.

R-203 No entry for vehicles that are longer than the measurement shown on the sign.

R-204 No entry for vehicles that are wider than the measurement shown on the sign.

R-205 No entry for vehicles that are higher than the measurement shown on the sign.

Other prohibition or restriction signs

R-300 Minimum separation from the vehicle in front.

Obligation to keep, from the vehicle in front, the number of metres of distance shown on the sign. R-301 Maximum speed permitted. It is forbidden to drive at a higher speed, in kilometres per hour, than the one shown on the sign. This applies until you see another sign for end of speed limit, end of prohibitions, or another maximum speed sign. R-302 No right turn.

R-303 No left turn and no U-turn.

R-304 No U-turn. You must not turn around and go in the other direction. R-305 No overtaking other vehicles. You may only overtake two-wheeled motorcycles if you do not enter the oncoming lane. You must obey this sign until another one appears that indicates that you can overtake in that area.

R-306 No overtaking for lorries that can carry loads of more than 3,500 kilos. They may only overtake two-wheeled motorcycles if they do not enter the oncoming lane. R-307 No stopping and no parking in that place.

R-308 No parking on that side of the carriageway. You may stop, but not park.

R-308a No parking on that side of the carriageway on odd days of the month. You may stop, but not park.

R-308b No parking on that side of the carriageway on even days of the month. For example, days 2, 4, 6... You may stop, but not park.

R-308c No parking on that side of the carriageway during the first 15 days of each month. You may stop, but not park.

R-308d No parking on that side of the carriageway between day 16 and the last day of each month. You may stop, but not park.

R-308e No parking in front of a dropped kerb. Dropped kerbs are spaces in the street reserved so that some vehicles can enter their garages, homes, or shops. R-309 Limited-duration parking zone. The driver is obliged to indicate the time when they left the vehicle parked.

R-310 Remember that you should only make sounds and horn signals with the vehicle when they are necessary to avoid accidents.

Mandatory signs

R-400a Must go to the right.

R-400b Must go to the left. R-400c Must go straight on. R-400d Must turn right. R-400e Must turn left.

R-401a Must pass on the side shown on the sign.

R-401b Must pass on the side shown on the sign.

R-401c Must pass on the sides shown on the sign.

R-402 Must turn in the direction shown by the arrows.

R-403a Must turn right or go straight on. You must not make a U-turn.

R-403b Must turn left or go straight on. You must not make a U-turn.

R-403c Must turn right or left. You must not make a U-turn.

R-404 Cars must use that carriageway.

Motorcycles without a sidecar are not obliged.

R-405 Motorcycles without a sidecar must use that carriageway.

R-406 Lorries, vans and light vans must use that carriageway. R-407a Road reserved only for bicycles. No one else may use this road. R-407b Road reserved only for mopeds. No one else may use this road.

R-408 Vehicles drawn by animals must use that path. Others may also use that path.

R-409 People riding an animal must use that path. No one else may use that path.

R-410 Pedestrians must use that path. No one else may use that path. R-411 Minimum speed.

Vehicles must travel at least at the speed shown on the sign. You must not drive more slowly. You must follow this rule until you find another sign that allows driving at a lower speed.

R-412 Must fit snow chains on the vehicle to continue driving. R-413 Must switch on dipped headlights. You must keep them on until another sign tells you that you can switch them off.

R-414 Vehicles carrying dangerous goods must use that road.

R-415 Vehicles carrying more than 1,000 litres of products that can pollute water must use that road.

R-416 Vehicles carrying explosive material or material that can catch fire must use that road.

R-418 Lane only for vehicles that have an electronic toll device.

Electronic toll is an electronic device that is placed on the windscreen and that recognises your vehicle's registration plate so that the barrier on the motorways opens automatically.

Signs that end the prohibition or the obligation R-500 End of all prohibitions. It indicates the place from which all prohibitions end. R-501 End of speed limit. You may go faster. R-502 End of no overtaking. You may overtake other vehicles.

R-503 End of no overtaking for lorries.

Lorries may overtake other vehicles.

R-504 End of limited parking zone. Vehicles may park in that place. R-505 End of road reserved for bicycles.

R-506 End of minimum speed. You may drive more slowly than the sign indicates.

Signs that give information of interest

They give information of interest to drivers and pedestrians. They are called information signs. There are different types of information signs.

General information signs

S-1 Place where a motorway begins. S-1a Place where a dual carriageway begins. S-2 Place where a motorway ends. S-2a Place where a dual carriageway ends.

S-3 Start of a road reserved for cars to use.

S-4 End of a road reserved for cars. S-5 A tunnel begins. S-6 End of the tunnel. S-7 Recommended maximum speed. It recommends the maximum speed at which you should always drive on that road. S-8 End of recommended maximum speed. The stretch of road ends where it is recommended to drive at that speed or more slowly. S-9 Recommended speeds. It recommends driving between those two speeds shown on the sign. S-10 End of recommended speeds. It indicates the place from which it is not recommended to continue driving at those speeds.

S-11 One-way carriageway. They indicate the direction in which vehicles must travel on that road. They also forbid making a U-turn (turning around). The arrows indicate the number of lanes S-12 Stretch of one-way carriageway. It indicates that on that stretch of street or road you must travel in the direction pointed to by the arrow. It is forbidden to do it in the other direction. S-13 Pedestrian crossing. S-15a No through road.

Vehicles can only leave that carriageway by the same place they entered. S-16 Emergency escape lane. It indicates an area where vehicles can pull off and stop when the brakes fail. S-17 Place where vehicles may park.

S-18 Taxi rank. S-19 Bus stop. Mountain pass. Road that crosses a mountain. S-21 Driving through a mountain pass. It indicates the conditions of the mountain pass and whether you can drive through it or not. The sign has three panels and each one indicates one thing: S-21.1 a, b, c, d and e. Panel number 1. It can be different colours White, with the word OPEN. All vehicles may pass.

Green. All vehicles may pass. But lorries that can carry more than 3,500 kilos must not overtake other vehicles.

Yellow. Cars and buses must travel at a maximum speed of 60 kilometres per hour.

Lorries that can carry more than 3,500 kilos and articulated lorries must not pass.

Red. Vehicles must fit chains and travel at a maximum speed of 30 kilometres per hour. Lorries and buses must not pass. Black, with the word CLOSED.

No vehicle may pass on that road.

S-21.2 a, b, c and d. Panel number 2. It may have the following signs: R-306 when panel 1 is green.

R-106 and R-301 you must drive at less than 60 kilometres per hour when panel 1 is yellow. R-107 with the inscription 3,5 t and R-412 when panel 1 is red. S-21.3 a and b. Panel number 3. It may show the name of the place from which you must follow the indications of panels 1 and 2. S-22 You may make a U-turn. S-23 There is a hospital nearby. It warns drivers to make little noise with their vehicles when passing in front of it. S-24 You may stop using dipped headlights.

S-25 You may make a U-turn using a road that is higher or lower.

S-26 a There is a motorway, dual carriageway or road for cars exit in 300 metres. S-26 b There is a motorway, dual carriageway or road for cars exit in 200 metres. S-26 c There is a motorway, dual carriageway or road for cars exit in 100 metres.

S-27 There is an emergency assistance post on the road to ask for help in case of an accident or breakdown.

S-28 Zone with priority for pedestrians. You must not drive faster than 20 kilometres per hour. S-30 Zone where pedestrians may walk. You must not drive faster than 30 kilometres per hour.

S-31 End of the zone with priority for pedestrians. You may drive at more than 30 kilometres per hour. S-32 Electronic toll.

Vehicles may pay the toll through electronic toll as long as the vehicle has the necessary technical means installed.

S-33 There is a path reserved for pedestrians and bicycles, separated from motor traffic, in parks, gardens and woods. S-34 There is a place where a vehicle may park inside a tunnel in case of emergency or breakdown. S-34a It indicates that this place has an emergency telephone.

Signs that give information about lanes

S-50a, S-50b, S-50c, S-50d and S-50e

Only vehicles travelling at that speed or faster may use the lanes marked with a number.

S-51 Lane reserved for buses.

S-52, S-52a and S-52b

It warns that a lane can no longer be used and indicates the lane you can continue in.

S-53c Road that changes from having two lanes to having three lanes. It also indicates the maximum speed at which you may drive in each of them.

S-60a. On a two-lane road it indicates that the left lane is going to deviate to the left.

S-60b. On a two-lane road it indicates that the right lane is going to deviate to the right.

S-61a. On a three-lane road it indicates that the left lane is going to deviate to the left.

S-61b. On a three-lane road it indicates that the right lane is going to deviate to the right. S-64 Cycle lane or cycle track on the road. Only cycles may use that lane.

Signs that give information about services

S-100 First aid post where you can receive urgent treatment.

S-101 Place where there are ambulances to assist and transport people injured in road traffic accidents.

S-102 Place where you can have vehicle inspections. S-103 There is a vehicle repair workshop. S-104 There is a telephone.

S-105 Place where you can refuel the vehicle.

S-106 Place where you can repair the vehicle and refuel.

S-107 Place where you can camp. S-108 There is a water fountain. S-109 There is a nice place to see. S-110 There is a hotel or a motel. S-111 There is a restaurant. S-112 There is a bar or café. S-113 Caravans can park.

S-114 You can stop to eat.

S-115 From that place you can start a walking excursion.

S-116 You can camp in that place with a tent and with a caravan. S-117 There is a youth hostel. S-118 There is a tourist information office. S-119 Area of a river where you need a special authorisation to fish. S-120 There is a national park.

S-121 There are monuments to see and visit. S-122 Sign to indicate other services.

S-123 Rest area to stop to rest.

S-124 Area to park vehicles that connects with a train station.

S-125 Area to park vehicles that connects with an underground station.

S-126 Area to park vehicles that connects with a bus station.

S-127 Service area within a motorway or dual carriageway. In service areas there are usually petrol stations, restaurants and places to rest for a while.

Direction signs

S-200 Indicates all the exits there are at the next roundabout. The exits shown in blue mean that they lead to a motorway or dual carriageway.

S-222a Indicates the direction of a road and the exits that lead to a motorway or dual carriageway.

S-232 Indicates where the entrance is to the motorway and at what distance.

S-235a Indicates which is the next exit within a motorway or dual carriageway and how far away it is.

S-300 Indicates the names of towns and cities on a road and how far away that town or city is.

S-301 Indicates the names of towns and cities on a motorway or dual carriageway and how far away that town or city is. S-322 Indicates in which direction there is a cycle route or cycle path and the distance to it.

S-360 Indicates what the number is of the road, the place that road goes to, and the name of the town or city at the next exit towards another road.

S-368 Indicates the number of the motorway or dual carriageway, where they go, and the name of the town or city at the next exit towards another road.

Signs to classify roads

S-400 European road.

Name given to that road in Europe. S-410 Motorway or dual carriageway. S-410a Toll motorway. S-420 National road, of the Spanish State.

S-430 Most important road of an Autonomous Community. They are called Autonomous Community roads of first level.

S-440 Roads that connect towns within an Autonomous Community or are used to reach the first-level roads. They are called Autonomous Community roads of second level.

C-241 Roads that connect small towns. They are called Autonomous Community roads of third level.

Signs that indicate where you are

S-500 Entry to a town or city. S-510 Exit from a town or city.

S-574 Indicates the kilometre where you are on a motorway or dual carriageway, and you will see them every 10 kilometres. That is, from the start of the motorway or dual carriageway it marks kilometre 10, kilometre 20, kilometre 30, until the end.

S-574a Indicates the kilometre where you are on a road, and you will see them every 10 kilometres, the same as on the motorway and dual carriageway.

S-574b Indicates the kilometre where you are on a toll motorway, and you will see them every 10 kilometres, the same as the rest.

S-575 Indicates the kilometre where you are on an Autonomous Community road, and you will also see them every 10 kilometres. The colour corresponds to the level of importance of the road.

Supplementary panels

S-800 Indicates the distance there is from the sign to a danger or warning.

S-810 Indicates how long the dangerous section is. For how many metres or kilometres there is danger.

S-820 and S-821 These signs are placed below a prohibition sign. They indicate how many metres that prohibition lasts following the arrow.

S-850 to S-853 Indicates in which direction and direction of travel you can drive with priority. It is placed next to sign R-3, which means road with priority.

S-840 It is placed below the give way sign. It indicates the distance at which you must stop. S-870 It is placed below another sign. It indicates that the prohibition or warning on the other sign only applies to the lane or slip road it points to. S-880 It is placed below another sign. It indicates that the prohibition or warning on the other sign must only be complied with by the vehicles shown on this sign. S-890 It is placed below another sign. It indicates that the prohibition or warning on the other sign must be complied with when there is snow, rain or fog. S-960 There is an emergency telephone.

S-980 There is an emergency exit.

S-990 Indicates where the emergency exit is in tunnels.

Lines and road markings

Lines and white markings

Continuous line

This line prohibits drivers from:

- Crossing the line.
- Driving on it.
- Driving to the left of the line when the road is two-way. When the line is a little wider than normal it indicates that there is a special lane. For example, a lane that only some vehicles may use.

Broken line

This line prohibits drivers from driving on it. You may only drive on it when the lane is less than three metres wide and it is necessary. When the gap between the broken lines is shorter than normal it means that a continuous line is near or a dangerous situation. For example, a bend with poor visibility. When the gap between the lines is wider than normal it means that a special lane is near.

See video

Broken line. Line made up of many smaller lines in a row, with gaps between them.

Double broken lines

Double broken lines on both sides of a lane mean that in this lane sometimes you can drive in one direction and other times in the other. There will be traffic lights in that lane or other means to indicate in which direction you may drive and prevent accidents.

Continuous and broken lines together

In this case, each driver must only take into account the line that is closest to the side they are driving on.

Edge and parking lines

Lines that mark where the road ends or places where you can park. They may be crossed or driven on.

Continuous transverse line

All vehicles must stop at this type of line when the following signs indicate it: The signs that indicate that it is compulsory to stop.

- Pedestrian crossings.
- Traffic lights.
- The instructions of traffic officers.
- The signals at a level crossing or movable bridge.

STOP

In any case, the marking does not by itself impose an obligation.

Transverse line. Line that crosses the road across its width, instead of along it.

Broken transverse line

Vehicles must not cross it if they must give way to other vehicles following the right of way rules and obeying the indications shown by these signs:

- Give way sign.
- Green turning arrow of a traffic light.

Pedestrian crossing

Wide lines on the carriageway that indicate that drivers must give way to pedestrians.

Cycle crossing

Broken and parallel lines on the carriageway that indicate that cyclists have priority to ride through that place.

STOP sign written on the carriageway

The driver must stop their vehicle before the stop line to give way to drivers travelling on the carriageway they want to enter.

STOP

Speed limit sign written on the carriageway

No vehicle must travel at more kilometres per hour than the number shown by the sign in that lane. You must follow that indication until another sign says that you may drive at another speed.

Arrow to choose lane

All drivers must follow the direction or one of the directions shown by the arrows in the lane they are driving in. If the signs allow it, you may also change lane.

End of lane arrow

Warns that the lane you are in is ending and indicates where you must go.

Return to lane arrow

Warns that a continuous line is coming and indicates to drivers that they must move as soon as possible to the right of the lane.

Level crossing marking

The letters P and N written on the ground with two lines in the shape of a cross indicate that there is a level crossing nearby.

Reserved lane or area

Indicates that that area or lane is reserved for certain vehicles to drive or stop. For example, buses and taxis.

BUS

Start of reserved lane marking

Indicates where the reserved lane begins for some vehicles.

Cycle route or cycle path marking

Indicates that that lane is reserved for bicycles to ride.

Zebra markings

Prohibits vehicles from driving in that area.

Only vehicles that are obliged to drive on the hard shoulder may drive in that space.

STOP

Coloured lines and markings

Yellow line with zigzag markings

Indicates that parking is prohibited in that area.

Continuous yellow line

Indicates that stopping and parking are prohibited in that area.

Broken yellow line

Indicates that parking is prohibited in that area.

Marking made up of yellow squares

Indicates to drivers that it is prohibited to enter that area if they could end up stopped in it obstructing traffic.

White and red squares

Area for emergency braking. It is the only reason why a vehicle may enter that area.

Blue parking markings

At some times of the day you may only park for a limited time and you have to pay.

Topic 9. The road

Contents

The road for vehicles and pedestrians

- Parts of the road for vehicles
- Other areas of the road
- Direction and direction of travel

Types of road

- Depending on where it is
- Depending on its characteristics
- Colours that indicate the condition of the road

Lanes

- Where must vehicles drive?
- Rules for using lanes
- Reserved lanes
- Special lanes
- The hard shoulder

Driving on motorways and dual carriageways

- Emergencies on the motorway and dual carriageway

The road for vehicles and pedestrians What is the road? A road is each street, road and track, public and private, that vehicles and pedestrians may use.

Parts of the road for vehicles

The parts of the road that matter to the driver are:

Lane

Roadway

Carriageway

Hard shoulder

Roadway

It is the whole area of the road that vehicles can use to drive and to stop. Within the roadway are the carriageway and the hard shoulder.

Carriageway

Part of the road that vehicles drive on. The same road may have several carriageways. A narrow carriageway is one that measures less than 6.5 metres.

Hard shoulder

Paved area that is on both sides of the carriageway that most vehicles do not drive on. Some roads do not have a hard shoulder. A road may have several roadways.

Lane

It is one of the parts into which the carriageway may be divided.

Its width must allow a line of cars to travel. On the same carriageway there may be several lanes, depending on how wide it is. They are usually separated by a white line. When the lane measures less than 3 metres wide, it is called a narrow lane.

Other areas of the road

Straight

Section of road that does not change direction.

Bend

Section of the road that changes direction to the right or to the left. On some bends you cannot see the full width of the road. In that case it is said to be a bend with reduced visibility.

Crest

It is the slope of the road. A change of crest means that a section of the road changes its slope. There are changes of crest with reduced visibility where the road is very steep and you cannot see the vehicles in front of you or coming from the opposite direction.

Central reservation

Space between two separate carriageways where vehicles do not drive.

Lay-by

Part of the carriageway that widens so that vehicles can stop if they need to without affecting the other vehicles driving on that carriageway.

Emergency braking area

Part of the road prepared so that vehicles can stop if their brakes fail. When entering that area, the vehicle stops even if the brakes do not respond.

Junction

Place where several roads, streets or tracks cross.

Roundabout

Junction where there is a circular structure in the centre and vehicles must drive around that structure. You must not drive through the centre.

Level crossing

Place where a road or track crosses the railway tracks.

Traffic island

Area of the carriageway slightly raised or painted that is used to guide traffic. They are usually placed at junctions. The parallel white lines that form the traffic island are called hatched markings.

Cycle route

Route prepared for cycles to use.

There are several types of cycle route:

Cycle lane

It is next to the carriageway. It can operate in two directions or in one direction only.

Protected cycle lane It is separated from the carriageway or from the pavement by physical elements to give more protection to the people who use it.

Cycle pavement It is signed within the pavement.

Cycle track

There is a space separating the cycle route and the road.

Cycle path

Route located in parks, gardens and woods where only pedestrians and cycles may travel.

Pedestrian zone

Part of the road reserved for pedestrian traffic. The pedestrian zone includes the pavement, the promenade and the refuge. The refuge is a pedestrian zone located on the carriageway that vehicles must not enter. For example, an area to stop between two pedestrian crossings.

Built-up area

Areas where there are buildings. That is, towns and cities. On the roads entering and leaving built-up areas there are signs with their name written on them.

Watch video

Direction and traffic flow

Direction

Straight or curved line that connects two places.

Traffic flow

The two possibilities, outward and return, that can be used to go in the same direction. For example, two cars driving on the same road, one in an outward lane and the other in the return lane, are going in the same direction, but in a different traffic flow. There are one-way roads and others where you can drive in both directions, in different lanes.

Exit

Entry

A

B

TWO-WAY ROAD

ONE-WAY ROAD

A

B

Types of road

Depending on where it is

Urban road

Streets and tracks within the built-up area.

Inter-urban road

Roads and tracks outside the built-up area.

Through road

Road that passes through a built-up area.

Depending on its characteristics

Motorway

It is a road where only motor vehicles may drive and it must meet the following characteristics:

- You must not enter it from any land or property next to the motorway.
- You may only enter via prepared and authorised access points.
- No railway line, tramway or path crosses it, and no other road crosses it.
- It has separate carriageways so that vehicles drive in one direction or the other. Motorways can be toll motorways. That is, you have to pay to drive on them.

Dual carriageway

Road that has the same characteristics as the motorway with two main differences:

- You can enter them from some properties next to the dual carriageway.
- It has more access roads.
- You do not have to pay to drive on it.

Road for motor vehicles

Road where only motor vehicles may drive. It has a single carriageway. As on motorways, you must not enter it from any land or property next to it. It is signed with signs S-3 and S-4.

Conventional road

These are all roads that do not meet the characteristics of motorways, dual carriageways and roads for motor vehicles.

Service road or service lane

It is a road that runs parallel to the main road and connects this road with houses, petrol stations and other properties.

Colours that indicate the state of the road

White

You can drive normally.

Green

You cannot reach the maximum speed allowed on that road.

Yellow

There are stops and queues of vehicles or traffic is slow on some sections.

Red

There are too many vehicles and there are many stops and queues.

Black

The road is closed and you cannot drive.

Lanes

Where should vehicles drive? They must drive on the right and as close as possible to the edge of the carriageway, especially on bends and changes of crest with reduced visibility.

Watch video

You must always keep clear the part of the carriageway that corresponds to the opposite direction. On roads with two carriageways, vehicles must use the carriageway on their right. On roads with three carriageways the middle one may have one or two directions and the side ones one direction only.

When you find on the road refuges or traffic islands, you must drive on the right-hand side of the refuge or traffic island. If the road is one-way you may drive on either side of the refuge or traffic island.

Motor vehicles normally must not drive on the hard shoulder, except in case of breakdown or some other unforeseen event.

Watch video

Rules for using lanes

General rules

Type of road

Where should you drive?

Two-way with two lanes On the right.

Two-way with three lanes On the right. You must never use the left lane. The centre lane is only used for overtaking and to change direction to the left.

Within towns or cities

Motor vehicles and special vehicles may drive in the lane they prefer on roads and streets with two lanes going in the same direction and separated by a white line.

Watch video

In these cases you must take into account that:

- You should only leave that lane to change direction, park or overtake another vehicle.
- You must not prevent passage or be an obstacle to other vehicles. When the lanes are not separated by a white line all drivers must drive on the right.

Other vehicles, such as cycles, mopeds or vehicles for people with reduced mobility, must always drive on the right, even if there is a separation line between lanes.

Outside towns or cities

On roads with two lanes going in the same direction, motor vehicles and special vehicles that with load can weigh more than 3,500 kilos will drive in the lane on their right. They may use the lane on their left when the road or traffic circumstances make it advisable and provided that they do not hinder other drivers.

Watch video

On roads with three or more lanes going in the same direction, some vehicles are obliged to always use the right-hand lane and are forbidden to use other lanes. These vehicles that must drive in the right-hand lane may use the one next to it, and must not use any more, and under the same conditions

as the previous point. These vehicles are:

- Lorries, vans and special vehicles that with load can weigh more than 3,500 kilos.
- Vehicle combinations that measure more than seven metres in length.

Reserved lanes

High occupancy vehicles (VAO)

Bus and taxi To carry out manoeuvres

Lanes reserved for certain vehicles

Some lanes are reserved for a particular type of vehicle and the other vehicles must not use them. These lanes are:

Lane reserved for buses

Driving in that lane is prohibited for the other vehicles.

Taxis may drive when the word TAXI is written in the lane. This lane will be separated from the other lanes by a wide white line.

Lane for high occupancy vehicles (VAO)

For a motor vehicle to be a high occupancy vehicle it must meet three requirements:

1. It may only carry people.
2. It must not carry more than 3,500 kilos.
3. The number of people travelling in the vehicle is the one stated in the VAO lane indications. The following vehicles may also use this lane:
 - Motorcycles, cars or adaptable mixed-use vehicles when they meet the three requirements needed for VAO motor vehicles (see previous paragraph).
 - Vehicles for people with disabilities even if only the driver is travelling.
 - Buses (that weigh more than 3500 kilos).
 - Vehicles providing emergency services.
 - Vehicles carrying out road works.

Lanes reserved for some manoeuvres

Manoeuvre. Movement or action carried out when driving a vehicle.

Watch video

Entry lane (acceleration) For vehicles that want to enter a road.

Exit lane (deceleration) For vehicles that want to leave a road. You must take a series of precautions when entering this lane:

- Move into the lane closest to the exit.
- Enter the exit lane as soon as possible.
- Drive more slowly within this lane.
- Pay attention to the conditions of the new road you are joining. For example, when changing from a dual carriageway to another road where you must drive more slowly.

Special cases for using lanes

Reversible

In the opposite direction to the usual one

Additional

Reversible lanes

These are lanes where you can drive in one direction or the other, depending on the circumstances. They are marked with broken white lines on both sides of the lane.

All drivers must use dipped headlights when driving in this lane, by day and by night.

Lanes that go in the opposite direction to the usual one They can be put in place when there are many vehicles that want to go in the same direction on the road. This means there are fewer traffic jams and vehicles can drive better.

Only motorcycles and cars may use these lanes. How should you drive in these lanes?

- With dipped headlights on, both by day and by night.
- At a speed between 60 and 80 kilometres per hour.
- Without changing into other lanes. Not even to overtake. The vehicles that are in the lane next to it, in the normal direction of the road, must take the same precautions.

Lanes can also be put in place in the opposite direction to the normal one when road works are being carried out. In that case, all vehicles authorised to drive on a road with road works may drive in that lane.

Additional lanes

On carriageways with two directions of traffic, using the hard shoulders, they are sometimes put in place when there are many vehicles on the same road so that there is one more lane to drive in. They are marked with beaconing signs. How should you drive in these lanes?

- Carefully, so as not to move or drive over the beaconing signs.
- At a speed between 60 and 80 kilometres per hour. Or more slowly if the signs indicate it.
- With dipped headlights on, both by day and by night.
- Without changing into other lanes. Not even to overtake.

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The hard shoulder If there is no road or part of it intended for them to travel on, the following vehicles must travel on the hard shoulder:

- Cycles.

- Vehicles following cyclists.
- Mopeds.
- Animal-drawn vehicles and ridden animals.
- Special vehicles that, when loaded, do not weigh more than 3,500 kilos.
- Vehicles for people with difficulty moving. When the road has no hard shoulder or it is not possible to travel on it, these vehicles may use only the essential part of the lane.

Cyclists may leave the hard shoulder and travel on the right-hand side of the lane on downhill slopes with bends.

See video

All vehicles travelling on the hard shoulder must do so in single file, one behind the other. Except cyclists, who may ride two abreast in the same line. But on stretches with no visibility, and when there is heavy traffic, they must also ride in single file, one behind the other.

Vehicles required to travel on the hard shoulder may not overtake other vehicles when the overtaking:

- Lasts more than 15 seconds.
- Needs a distance greater than 200 metres to overtake.

Cyclists may overtake other vehicles without complying with these two rules.

Travelling on motorways and dual carriageways

The following may not travel on motorways or dual carriageways:

- Pedestrians.
- Animals.
- Bicycles.
- Mopeds.
- Vehicles for people with reduced mobility.
- Personal mobility vehicles.

Cyclists who are over 14 years old may travel on the hard shoulder of the dual carriageway, unless there is a sign that forbids it.

On toll motorways you must go through the booths prepared to collect the entry ticket to the motorway or to pay on exit. These booths have a traffic light or green arrow when they are open and a traffic light or red cross when they are closed.

See video

Emergencies on the motorway and dual carriageway

Vehicles that have to travel more slowly on the motorway or dual carriageway due to a breakdown must leave this road at the first exit they come to. A vehicle that has to stop because of an emergency must do so on the hard shoulder or on the central reservation. To request assistance you can use the nearest emergency telephone.

People travelling in the broken-down or crashed vehicle must not walk on the road.

When a vehicle has an accident or a breakdown, it must be removed from the road by another vehicle that is authorised to do so. For example, a tow truck.

Topic 10. Speed and distances

Contents

Types of speed

- Maximum speed
- Minimum speed
- Inappropriate speed
- Appropriate speed
- Situations in which you must drive at low speed

Generic speed

- Speed limits outside towns and cities
- Maximum speed limits inside towns and cities
- Minimum speed limits inside and outside towns and cities
- Why are speed limits necessary?

Types of distance

- Reaction distance
- Braking distance
- Stopping distance
- Lateral separation
- Safe following distance

Types of speed

It is very important to drive at the appropriate speed to avoid dangers and traffic accidents. 2.2 seconds 1.2 seconds 30 metres

50

Km/h

90

Km/h

There are different types of speed when driving:

Maximum speed

Minimum speed

Inappropriate speed

Appropriate speed

Maximum speed

Maximum permitted speed that a vehicle may reach on the road it is travelling on. When the vehicle goes faster than the maximum permitted speed, it is travelling with excess speed. For example, it is excess speed to travel at a speed of 100 kilometres per hour on a road where the maximum permitted speed is 90 kilometres per hour.

Minimum speed

Minimum speed at which a vehicle must travel on the road it is travelling on. When a vehicle travels at a speed lower than the permitted one, it is travelling at an abnormally reduced speed. For example, it is an abnormally reduced speed to drive at 40 kilometres per hour on a motorway where the minimum permitted speed is 60 kilometres per hour.

Inappropriate speed

Speed that is within the permitted limits but is not appropriate because of the driver's conditions, the weather, the condition of the road, or the circumstances of the vehicle. For example, driving at a speed of 70 kilometres per hour on a road where there are ice patches and the vehicle may skid. That speed would be appropriate in normal circumstances. But in that case it is an excessive speed because there is ice on the road.

Driving at an inappropriate or excessive speed is one of the main causes of traffic accidents.

Appropriate speed

Speed that is within the permitted limits and that is appropriate for the weather conditions, the road, the circumstances of the vehicle and the driver's condition.

Appropriate speed allows you to control the vehicle better in unexpected situations.

People are more likely to die in a traffic accident when the vehicle is travelling at excessive speed than when it is travelling at an appropriate speed. What must you take into account to travel at an appropriate speed? You must always respect the speed limits. But, in addition, you must take into account the circumstances that occur at each moment. These circumstances are:

- Your physical and psychological condition. For example, if you are very tired.
- The characteristics of the vehicle and the load you are carrying. It is better to drive more slowly when you are carrying a lot of weight in the vehicle.
- The condition of the road. You must go more slowly if the road has speed humps or is badly surfaced.
- The traffic situation. You must drive more slowly when there is heavy traffic.
- Weather conditions. You must be more careful when it rains, snows, there is fog or ice on the road.

Situations in which you must drive at low speed You must drive slowly and even stop the vehicle, in the following cases:

- When there are pedestrians on the road or they may cross it. Especially when there are children, older people, people who cannot see or with another type of physical disability.
- Near roads reserved for cyclists and in all places where cyclists travel.
- At pedestrian crossings where there is no traffic light or traffic officer.
- Near markets, schools and other places where there may be children.

See video

- On roads or tracks where there are animals or they may appear.
- On stretches where there are buildings right next to the road. People or animals may come out of those buildings.
- When approaching a bus that is stopped. Especially if it is a bus that carries children.
- When approaching any vehicle stopped on the road, emergency service vehicles, and bicycles travelling in your lane or on the hard shoulder.
- When the road is slippery or there is water on it, gravel or other materials.
- When approaching level crossings, roundabouts or junctions where you do not have right of way. In these cases, the maximum speed must be 50 kilometres per hour.

See video

- When approaching narrow places or places where visibility is poor.
- When passing an oncoming vehicle when the circumstances do not allow it to be done very safely. For example, when you pass an oncoming lorry on a bend on a narrow road.
- When you are dazzled by the lights of another vehicle.
- When there is thick fog, rain, snow or smoke.

Generic speed

What is it?

Speed limit that each type of vehicle must comply with on each road. There are maximum and minimum generic speeds.

Speed limits outside towns and cities

Type of vehicle

- Cars.
- Motorcycles.
- Three-wheeled vehicles.
- Motor caravans that can carry loads less than 3,500 kilos.
- Pick up.

Motorway and dual carriageway	Roads	Tracks
Speed Maximum Minimum	Maximum speed	Speed Speed Minimum Maximum Minimum
120	60 90 On some roads the limit may be 100.	45 30 15

Type of vehicle

- Buses.
- Car-derived vehicles.
- Adaptable mixed vehicles.

Motorway and dual carriageway	Roads	Tracks
Speed Maximum Minimum	Maximum speed	Speed Speed Minimum Maximum Minimum
100	60 90	45 30 15

Type of vehicle

- Lorries.
- Tractor units.
- Vans.
- Motor caravans that can carry loads greater than 3,500 kilos.
- Articulated vehicles.
- Vehicles with a trailer.

Road type	Max speed (km/h)	Min speed (km/h)
Motorway and dual carriageway	90	60
Roads	80	40
Tracks	30	15

Type of vehicle

- Three-wheeled vehicles that are not motorcycles.
- Heavy quadricycles.

- Special vehicles that can reach 60 kilometres per hour on level ground.

Road type	Max speed (km/h)	Min speed (km/h)
Motorway and dual carriageway	70	60
Roads	70	35
Tracks	30	15

Type of vehicle

- Special vehicles without brake lights.
- Special vehicles towing a trailer.
- Motor cultivators.

Motorway and dual carriageway

Roads

Tracks

Maximum and minimum speed

May not travel

25

Types of vehicle

- Other special vehicles

Road type	Speed limit
Motorway and dual carriageway	May not travel
Roads	40
Tracks	30

Types of vehicle

- Cycles

Road type	Speed limit
Motorway and dual carriageway	45 on a dual carriageway. On a motorway they may not travel.
Roads	45
Tracks	30

Bicycles may exceed these speeds if circumstances allow.

Type of vehicle

- Two- and three-wheeled mopeds.
- Light quadricycles.
- Vehicles for people with reduced mobility.

Road type	Speed limit
Motorway and dual carriageway	May not travel
Roads	45
Tracks	30

For vehicles that provide school transport with children or that carry dangerous goods, the maximum speed limit is 10 kilometres less than for the rest of the vehicles. For example, the speed limit on roads for a bus is 90 kilometres per hour. But for a bus that takes children to school the speed limit is 80 kilometres per hour.

Maximum speed limit inside towns and cities

Type of vehicle

- All vehicles in general

Streets without kerbs, where the carriageway and the pavement are at the same height	Streets with a single lane in each direction	Streets with several lanes in each direction
20	30	50

Type of vehicle

- Vehicles carrying dangerous goods.
- Special vehicles that cannot reach

a speed of more than 60 kilometres per hour.

Streets without kerbs, where the carriageway and the pavement are at the same height	Streets with a single lane in each direction	Streets with several lanes in each direction
20	30	40

Type of vehicle

- Two- or three-wheeled mopeds.
- Light quadricycles.
- Vehicles for people with reduced mobility.
- Cycles.

Streets without kerbs, where the carriageway and the pavement are at the same height	Streets with one lane in each direction	Streets with several lanes in each direction
20	30	45

Type of vehicle

- Personal mobility vehicles.
- Special vehicles with trailer

or semi-trailer.

- Vehicles without brake lights.

Streets without kerbs, where the carriageway and the pavement are at the same height	Streets with one lane in each direction	Streets with several lanes in each direction
20	25	25

The maximum speed for all vehicles on motorways and dual carriageways that are within a town or city is 80 kilometres per hour,

if it is not signed in another way.

Minimum speed limits inside and outside towns and cities

- On motorways and dual carriageways the minimum speed allowed is 60 kilometres per hour.
- On the rest of the roads the minimum speed allowed depends on each vehicle. For example, on a road where the maximum speed allowed for a car is 90 kilometres per hour, the minimum speed allowed will be 45 kilometres per hour. It is forbidden to drive below that speed, even if there are no other vehicles on the road. The only cases in which you may drive slower than the minimum speed allowed are:
 - When there is heavy traffic, the vehicle has a breakdown, or the road is in poor condition.
 - When the vehicle is a cycle, is being pulled by animals, or is a special vehicle.
 - In the case of vehicles that accompany other vehicles. For example, vehicles that accompany cyclists in a competition.

SPECIAL
CYCLIST

If your vehicle cannot reach the minimum speed and there is a risk of an accident, you must switch on the hazard warning lights. For example, if your vehicle has a breakdown and you have to drive slower than allowed. Why are speed limits necessary?

Speed limits are set so that all vehicles can travel safely and in an easy and comfortable way. One of the conditions that is taken into account to set the speed is the type of road and its conditions. That is, the maximum speed at which you can drive safely on that road.

Inside towns and cities you must drive at less than 50 kilometres per hour. If you run over a pedestrian, they will have a better chance of surviving if the vehicle is travelling at a low speed.

On roads that are not a motorway or a dual carriageway you must not drive at more than 90 kilometres per hour so that accidents are less serious. On a motorway and a dual carriageway there is a better chance of surviving an accident when the vehicle travels at less than 120 kilometres per hour. Things that can happen if you drive very fast

- It is harder to react to unexpected events. This means that many mistakes are made that can cost people their lives.
- You have less ability to see what happens at the sides. This effect is called tunnel vision because you only see what is in front of you, as if you were going through a tunnel. 0 km/h 65 km/h 130 km/h 180 km/h 70s 30s
- Driving fast for a long time can make you feel fatigue and aggressiveness because you drive with more tension.
- Driving very fast on a road in good condition can make you think that you are going at a lower speed than you really are.

It is forbidden to hold speed competitions on any road, unless the authorities authorise them and take the necessary measures. A person who takes part in races that are not authorised will lose points on their driving licence.

Types of distance

Reaction distance

Braking distance

Stopping distance

Lateral separation

Safe following distance

Reaction distance

Space you travel with the vehicle from when you detect an unexpected event, such as an obstacle, a sign, or a noise, until you can react to the unexpected event. For example, the time that passes from when you see a red traffic light until you press the brake. A person's normal reaction time to an unexpected event is usually 0.75 seconds, almost one second. Although this depends on their reflexes, their physical and psychological state, and the environment they are in.

The faster a vehicle is travelling the less reaction distance it will have because it will cover more distance in less time. Therefore, it will reach the danger sooner. The driver sees the traffic light The driver presses the brake

Reaction distance

Braking distance

Space the vehicle travels from when you press the brake until it stops completely. This distance can change for the following reasons:

- Speed. The faster the vehicle is travelling the more space it will need to brake.
- The load. A vehicle with a heavy load takes longer to stop.
- Technical condition of the vehicle. Condition of its brakes and other parts of the vehicle, such as the tyres.
- The weather. On a wet surface the vehicle may need more braking distance.
- The road conditions.
- Characteristics of the driver. For example, age, physical or psychological state.

Stopping distance

Space you travel with the vehicle from when you detect an obstacle until you stop the vehicle completely. For example, the time that passes from when you see a red traffic light until you stop the vehicle completely.

Stopping distance is the sum of the reaction distance and the braking distance.

Reaction distance

Braking distance

Stopping distance

Lateral separation

Necessary separation distance between two vehicles that pass each other or that drive side by side. There must always be enough separation between them to avoid any danger. You must keep a greater lateral separation distance from other vehicles when:

- You are driving fast.
- The road is in poor condition.
- Many vehicles pass yours.
- There is rain, snow, fog, wind, or smoke.

Watch video

Safe following distance

Minimum space you must leave from the vehicle in front of you so you do not crash in case of sudden and unexpected braking.

You must leave a greater safe following distance from the driver in front when:

- You increase speed.
- The road is hard to see.
- The vehicle does not grip the road well.
- You are not in good physical or psychological condition. You must keep a distance from other drivers that allows you to stop in case of sudden braking in the following cases:
- Inside a town or city.

- In areas where overtaking is forbidden.
- Where there is more than one lane in the same direction.
- When you cannot overtake because there are many vehicles.
- When another vehicle signals to you that it is going to overtake. In places where you are allowed to overtake other vehicles you must leave enough distance to overtake or to allow another vehicle to overtake you comfortably.

Vehicles that can carry more than 3,500 kilos and combinations of vehicles that measure more than 10 metres must follow these rules and also leave a minimum separation of 50 metres from the rest of the vehicles.

Bicycles are the only vehicles that can travel without keeping the safe following distance between cyclist and cyclist.

Topic 11. Manoeuvres

Contents

Safety rule R.S.M.

- Steps of the safety rule R.S.M. (Rear-view mirror, Signal and Manoeuvre)
- Safety rule when joining a road
- Precautions when changing lanes

Overtaking other vehicles

- Which side must you overtake on?
- Steps to follow to overtake another vehicle
- Obligations of the driver who is being overtaken by another vehicle
- When is overtaking forbidden?

Change of direction and U-turn

- Change of direction
- U-turn

Reversing

Stopping the vehicle

Stopping and parking

- Difference between stopping and parking
- Where can you stop and park?
- Dangerous stops and parking
- Where is it forbidden to stop and park?

Safety rule R.S.M. What is it? The set of rules and steps you must follow when you do any manoeuvre with the vehicle:

Rear-view mirror

1

Signal

2

Manoeuvre

3

This rule is divided into three steps. Steps of the safety rule R.S.M.

Observe the road

Rear-view mirror

Warn about the manoeuvre

Signal

Do the manoeuvre

Observe the road

It means looking carefully at the traffic and the road directly, and by using the rear-view mirrors, before doing the manoeuvre. This way you will make sure that you can make the movement at that moment.

Warn about the manoeuvre

After checking that there is no danger to do that manoeuvre, you must warn other drivers about what you are going to do.

You can warn them with the vehicle lights or with your arm. The signals you make with your arm have more validity than the ones you make with the lights, as long as you do them correctly and other drivers can see them. When you make signals with the vehicle lights, these lights must be on for the whole time the manoeuvre lasts and you must switch them off the moment you finish it. When you make signals with your arm, you must make the signal just before starting the manoeuvre, with enough time for other drivers to see it.

How must you indicate the manoeuvre with your arm?

Manoeuvre

Arm movement

Lane change, change of direction or U-turn To move the vehicle to the left: arm horizontal with the hand extended downwards. To move the vehicle to the right:

Arm bent upwards with the hand extended also upwards.

Reversing

Arm extended horizontally with the palm of the hand facing backwards.

Braking

Arm movements up and down with short and quick movements. How must you indicate the manoeuvre with the lights?

Manoeuvre

Lights

Lane change, change of direction or U-turn

Switch on the direction indicator on the side you are going to move towards.

Reversing

Switch on the reversing light.

Braking

Switch on the brake lights several times.

Do the manoeuvre

After observing and signalling you can now do the manoeuvre. You must do it correctly, at the appropriate speed, and without putting other vehicles or pedestrians in danger.

Safety rule when joining a road

Watch video

It is very important to follow the steps set out by the safety rule R.S.M. when joining a road with the vehicle. You must observe the road, always make sure that joining that road does not create a danger for other drivers, and warn about your manoeuvre with your arm or the lights.

When doing the manoeuvre you have to:

- Give way to other vehicles without causing them danger.
- Not be an obstacle for vehicles that are approaching you.
- Travel at the appropriate speed. When you have to give way to another vehicle while you are in the acceleration lane to join a road, stop at the beginning of the lane until the vehicle that is already on the road passes by.

Watch video

Then join the lane that applies to you adapting to the speed of the other vehicles.

All drivers travelling on a road have the obligation to allow other vehicles to join that road and, if they can, they must make the joining manoeuvre easier for them.

Above all, for large vehicles that carry many people and are at a marked stop. For example, a bus that is at its stop picking up passengers and must re-enter the road. To make it easier for these vehicles to enter, the vehicles that are already driving must:

- Move to one side to leave them more space, when possible.
- Reduce speed and even stop to let the vehicle leave the marked stop, when this happens within a town or city.

Precautions when changing lane

Watch video

To change lane safely you must take the following precautions:

- Start the lane change at a sufficient distance from other vehicles or any other obstacle.
- Do the manoeuvre little by little, without sudden movements.
- Do not obstruct the vehicles travelling in that lane and do not cut in front of them. They have priority because they were already in that lane.

Watch video

Overtaking other vehicles

Overtaking is the manoeuvre that consists of passing other vehicles that are travelling more slowly. Which side do vehicles overtake on? You overtake on the left side of the vehicle you want to overtake in prohibited areas. There are some exceptions in which you may overtake on the right side.

- To overtake a tram that is travelling in the central area of a two-way road.
- When the driver you want to overtake

is clearly indicating that they want to change direction to the left or stop on that side.

Watch video

You may also overtake on the right within a town or city when the carriageway has at least two lanes going in the same direction and they are separated by a white line painted on the road. There are some situations that are not considered overtaking, even if you pass the other vehicle. These situations are allowed and are:

- Roads with traffic jams, full of vehicles.

Drivers in one lane move faster than those in another lane and their speed depends on the movement of the vehicles in front. In these situations it is forbidden to change lane to overtake.

- Driving faster in the lanes for entering and leaving the road than the vehicles that are already on it.
- Vehicles that travel faster in lanes that are reserved only for them. For example, a bus that goes in the lane reserved for that vehicle may travel faster than the vehicles in the lane to its left.
- Cyclists travelling in a group and overtaking each other.

Watch video

- Overtaking a vehicle that is stopped on the road due to a breakdown or accident.

Watch video

- Overtaking an obstacle on the road, even if you have to move into the opposite lane to overtake it.

Special cases of overtaking	
Situation	How to act?
Two-way roads with three lanes separated by broken white lines	Overtaking is done in the centre lane as long as it is not occupied by another driver travelling in the opposite direction.
Roads with two, or more, lanes in the same direction	You may use the left lane to overtake. The driver may continue their journey in the left lane if they keep overtaking. But they must return to the right lane if another vehicle comes that is travelling faster than them.
Steps to follow to overtake another vehicle	Requesting right of way End of
Preparation	Checks Overtake (if necessary) overtaking
Preparation	
- Keep an appropriate distance from the vehicle you want to overtake.	

- Assess the distance of the vehicles coming in the opposite lane and the speed at which they are approaching you.
- Take into account the speed limits allowed on that road.

Checks before overtaking

You must check that:

- The driver you are going to overtake has or has not signalled with the lights or the arm that they are going to move to the side on which you are going to overtake them. If they have signalled it, you must wait for them to do their manoeuvre before overtaking.
- You can return to your lane after overtaking without risk to other vehicles.
- The lane you are going to use to overtake is not occupied by another vehicle that also wants to overtake. Requesting right of way to overtake (if considered necessary)
- Give light signals with the vehicle.
- Give audible signals with the vehicle. This may only be done outside towns and cities.

Overtaking

To overtake you must increase the vehicle's speed. But you must not exceed the permitted maximum speed limit. To overtake a moped or a cycle, within or outside built-up areas, you must:

- Keep a separation of at least one and a half metres from the moped or the cycle.
- Move into the lane to your left to overtake. Do not overtake in the same lane in which the moped or cycle is travelling. On roads outside built-up areas, you must also leave a separation of at least one

and a half metres in these cases:

- When you are driving a two-wheeled vehicle and you want to overtake any other vehicle.
- When you are driving any vehicle and you want to overtake people, animals, motorcycles, vehicles stopped on the road and vehicles that are carrying out assistance work. For example, ambulances or breakdown trucks at least, one and a half metres, on inter-urban roads.

In the rest of the cases, you must leave enough space between your vehicle and the one you want to overtake. The space will depend on the speed and the road conditions. You must stop overtaking if you see that another vehicle is coming in the opposite direction or that the vehicle you want to overtake suddenly accelerates. In those cases you have to slow down and return to the lane you were driving in.

End of overtaking

When you have finished overtaking, you must switch on the appropriate lights to indicate that you are going to return to the lane you were driving in. Then you have to return to your lane as soon as possible and without forcing other drivers to change their speed.

Obligations of the driver being overtaken by another vehicle

- Allow the other driver to overtake you and make the manoeuvre easier for them. It is forbidden to increase speed at that moment.

Watch video

- Position your vehicle at the right edge of the lane without moving onto the hard shoulder.
- Reduce speed if any dangerous situation arises while the other vehicle is overtaking you.
- Allow the vehicle that has overtaken to return to its lane.
- Drivers of very large or heavy vehicles that cannot move to the right of the lane must reduce speed and indicate to the driver behind that they may overtake. This indication can be made with the arm extended and moving the palm of the hand backwards and forwards. It can also be done by switching on the vehicle's right direction indicator.

When is overtaking forbidden?

- When you cannot see the whole road well

ahead. Except if the lanes are marked and you do not have to enter the opposite lane to overtake, for example:

- On bends where you cannot see well.
- When the road has crests and there is not good visibility.
- When there is fog, heavy rain or the sun dazzles and does not let you see well.
- Behind a vehicle that is also overtaking and does not let you see the road ahead.
- Several vehicles at the same time. This may only be done when you can return to the right at any time without obstructing the vehicles you have overtaken.
- A vehicle that is already overtaking another, if you have to enter the opposite lane to overtake.
- When you could endanger cyclists travelling in the opposite lane.

- At pedestrian crossings, at crossings with cycle tracks and at level crossings and near them.

At level crossings you may overtake two-wheeled vehicles that allow good visibility because they are small, as long as we warn with audible or light signals. At pedestrian crossings you may overtake:

- When it is done very slowly, so that if a pedestrian appears we can stop the vehicle.
- Two-wheeled vehicles.
- In tunnels with traffic in both directions when the vehicle you are going to overtake has only one lane going in its direction.
- At junctions and near them. You may overtake at a junction when:
 - it is a roundabout,
 - you overtake on the right,
 - you overtake a two-wheeled vehicle,
 - there is a sign at the junction that gives you priority. You may overtake on any path or road:
- Pedestrians.
- Bicycles.
- Cycles.
- Mopeds.
- Animals.
- Vehicles drawn by animals. Before overtaking them, you must make sure there is no danger to them or to other drivers.

Change of direction and U-turn

Change of direction

To change direction with the vehicle, you must make sure that the vehicles approaching in the opposite direction are far away and are travelling at a speed that allows you to change direction without danger.

When they are very close or coming very fast, you cannot change direction. If the change of direction is to the left, but you cannot see well, you cannot do it either. If there is an adapted place on the road such as a turning bay or a similar element, the change must be made there, but if there is a sign, you must follow what the sign indicates. How must you change direction?

Watch video

When the change of direction is to the right you must move as close as possible to the edge of the carriageway. When the change of direction is to the left you must position yourself:

- As close as possible to the left edge of the carriageway when this carriageway is one-way.
- Next to the white line separating lanes when the carriageway is two-way. When there is no white separation line you must position yourself as close as possible to the opposite lane, but without driving on it.
- In the centre lane when the road is two-way and has three lanes separated by white lines. When turning to change direction, you must pass through the centre of the junction and leave it to your left.

You must not cross it diagonally.

Correct Incorrect

Watch video

On inter-urban roads, two-wheeled cycles and mopeds that want to turn to change direction to the left must position themselves to the right of the lane and off the carriageway if possible, and wait until it is safe to make the turn. They must also pass through the centre of the junction.

U-turn

Watch video

A U-turn consists of turning around to continue travelling on the same road or path, but in the opposite direction. A U-turn may be made at a roundabout or a turning bay on the road.

Turning bay. A half-circle-shaped diversion in the road to change direction or make a U-turn.

The U-turn manoeuvre must be done with a single turn of the vehicle and without using reversing.

Reversing may only be used to make a U-turn in circumstances in which nothing else can be done. For example, to get out of a street closed by roadworks. You must start the U-turn next to the white line on the road that separates the lanes going in different directions. When it is not possible to do it from there, you must position yourself close to the right edge to start the turn from there.

You must not make a U-turn if you force drivers who are coming behind you or coming towards you to reduce their speed, or you may be a danger to them. In those cases, you must stop on the hard shoulder on your right and wait until it is safe to make the turn to make a U-turn. If there is not enough hard shoulder or it is not a safe place, you must continue driving until you find a place where you can make a U-turn.

Situations in which it is prohibited to make a U-turn

- Bends and changes of gradient where you cannot see well.
- When there is fog or heavy rain.
- At level crossings.
- In tunnels.
- On motorways and dual carriageways. Yes, you can do it in places prepared for it.
- On one-way carriageways and roads.
- In all places where overtaking is prohibited, except where there is a sign that indicates that you can make a U-turn.

Reversing

Watch video

It is prohibited to drive in reverse on any road or track. You may only drive in reverse in the following cases:

- When carrying out manoeuvres to park the vehicle.
- When it is not possible to drive forwards or make a change of direction or a U-turn. For example, to get out of a street closed due to roadworks. In these two cases you must not reverse more than 15 metres and you must not enter a junction of roads or tracks in reverse. You must drive at a very low speed while you are reversing.

Before you start reversing you must make sure that there is no danger to other drivers and pedestrians. You have to stop the vehicle and stop reversing in any dangerous situation. For example:

- Someone makes audible signals to you with their vehicle horn.
- You notice that another vehicle is approaching, a person, or an animal.
- On motorways and dual carriageways reversing is always prohibited.

Stopping the vehicle
Watch video

It may be necessary to stop the vehicle while you are driving for the following reasons.

- An emergency. For example, an accident, a breakdown, or one of the passengers feels unwell.
- There are many vehicles on the road and traffic jams occur.
- Indication from road signs or traffic officers who ask you to stop.

On motorways, dual carriageways, and in enclosed places or places where you cannot see well, you must switch on the hazard warning lights and sidelights whenever you have to stop the vehicle.

Stopping the vehicle due to an emergency

Vehicles that have an accident or breakdown can be seen in the following situations.

Situation	What to do?
The vehicle is still working, but it cannot reach the necessary speed on that road and it obstructs traffic.	If the vehicle weighs less than 3,500 kilos, drive on the hard shoulder on the right. If there is no hard shoulder, place the vehicle as far to the right of the road as possible. If you are on a motorway or dual carriageway, take the first exit.
The vehicle works, but you need a breakdown assistance service	Leave the road at the first exit using the right hard shoulder.
The vehicle cannot continue driving	Park the vehicle on the right hard shoulder of the road or in the place where it obstructs the other drivers the least. (For example, the left hard shoulder or the central reservation).
Watch video	

People travelling in a vehicle that cannot continue driving must get out of the vehicle and go to a safe place. Before getting out of the vehicle, they must put on the yellow high-visibility vests that all drivers must carry in their vehicles. If there is no safe place, they must stay inside the vehicle with the seat belt fastened. You must switch on the hazard warning lights to indicate that there is a vehicle stopped on the road. It is also compulsory to place on top of the car the emergency warning light (orange light). If you want, you can place the warning triangles. They are placed 50 metres from the vehicle. One at the front and another at the back, on the right side of the road, and they must be visible from 100 metres away.

Vehicles that pick up and tow broken-down vehicles must be prepared for roadside assistance (for example, recovery vehicles). Not just any vehicle can do it.

Stopping and parking

Difference between stopping and parking

Stopping

Leaving the vehicle stopped for less than two minutes. When stopping, the driver does not leave the vehicle. And if they do, they stay close to it.

Parking

The vehicle stays without moving for more than two minutes. The driver may leave the vehicle and go to another place. Where can you stop and park?

Watch video

On roads outside towns and cities, you must stop and park on the right-hand side of the road, leaving the hard shoulder free, as far as possible.

On roads and streets within towns and cities you may park on the carriageway and on the hard shoulder. When the road is two-way, the vehicle must stop or park on the right. When the road is one-way you may do it on the right or on the left. How should the vehicle be positioned? As a general rule, you must stop and park the vehicle in a line parallel to the edge of the carriageway. You must always check that your vehicle allows other vehicles to use the rest of the space. What must you do before leaving the vehicle?

Drivers of a motor vehicle or moped must:

- Stop the engine and disconnect the system that allows the vehicle to move.
- Apply the parking brake.
- Leave first gear engaged if you park on an uphill slope.
- Leave reverse gear engaged if you park on a downhill slope.
- Select the parking position if the vehicle is automatic.

Drivers of vehicle combinations, in addition, have to leave their vehicle properly secured when they park on a slope.

Watch video

They can do it in 2 ways:

1. Placing suitable wheel chocks.

Stones or other objects must not be used.

Chock. Wedge placed between the vehicle and the road so that the vehicle does not move.

2. Resting one of the front wheels against the kerb of the pavement.

On uphill slopes, the wheel turns towards the centre of the carriageway. On downhill slopes, the wheel turns outwards.

Dangerous stops and parking

There are different situations in which it is dangerous to stop the vehicle or park it. What are these situations?

- When a parked vehicle does not allow other vehicles to pass. When a vehicle is less than three metres from the opposite side of the road or from the continuous line that separates directions. Some vehicles will not be able to pass through that space.
- When a parked vehicle does not allow another vehicle that is stopped or parked to rejoin the road.
- When people or animals cannot enter or leave their homes or any other place because a vehicle prevents it.
- When vehicles cannot use a dropped kerb that is signposted.
- When a vehicle is an obstacle so that people with physical disabilities can enter or leave the road via the areas prepared for this.

When a vehicle stops or parks in places that are used to regulate traffic. For example, traffic islands. When a vehicle prevents another vehicle from being able to turn where the sign indicates. Parking is considered dangerous when:

- A vehicle is parked in a loading and unloading area at times when it is not permitted.
- A vehicle is parked double-parked and the driver has left.
- A vehicle parks at a public transport stop. For example, a bus stop.
- A vehicle parks in places where there is a sign that prohibits parking.
- A vehicle parks in a place reserved for emergency and security services. For example, ambulances.
- A vehicle parks in the middle of the carriageway.

Where is it prohibited to stop and park? It is prohibited to stop and park:

- On bends and changes of gradient where visibility is poor.
- In tunnels.
- At level crossings, pedestrian crossings, and cyclist crossings.
- In lanes or parts of roads reserved for other vehicles.
- At junctions or near them when you do not allow other vehicles to turn.
- On tram rails or very close to them.
- In places where the parked vehicle does not allow road signs to be seen.
- On motorways or dual carriageways. You may only stop or park in rest areas or service areas.
- Lanes and areas reserved for buses, taxis, or bicycles.
- Areas reserved for people with disabilities. You may stop, but not park:
- In loading and unloading areas.
- On pavements, promenades, and other areas where pedestrians pass. For two-wheeled vehicles, as long as it is regulated by the municipality and they do not obstruct pedestrians.
- In front of dropped kerbs where there is a dropped kerb sign.
- Double-parked.

- In places where a permit is needed to park at certain times or for a certain period of time. You may park if you have the authorisation that allows it.

Topic 12. Right-of-way rules for driving

Contents

Giving way to other vehicles

- General rules
- Rules for giving way in specific places
- Right of way for cyclists, pedestrians, and animals
- Right of way for vehicles in emergency service

Right of way in narrow sections

- Narrow sections with signs
- Narrow sections without signs
- Narrow sections on a slope

Crossing level crossings and movable bridges

Crossing tunnels and underpasses

- Rules for stopping the vehicle inside a tunnel
- Emergencies inside a tunnel

Giving way to other vehicles

General rules

To give way to another vehicle that has right of way you must:

- Reduce speed gradually so that the driver of the other vehicle realises that you are giving way.
- Stay stopped or drive slowly until the other driver has passed and you can drive without danger.

Rules for giving way in specific places

Junctions without signs

Junctions with signs

Roundabout

Junctions without signs

At junctions without signs you must give way to vehicles coming from your right, even if they come from a narrow road or one in poor condition.

At junctions without signs, there are some exceptions that you must take into account. They have right of way:

- Vehicles travelling on rails over those travelling on the road. For example, a tram has right of way over a car.
- Vehicles travelling on a road that is paved over vehicles travelling on an unpaved road.

Pave. Cover the surface of roads and streets with asphalt and other materials that allow vehicles and people to pass more safely.

- Vehicles that are already inside a roundabout over those that want to enter the roundabout.
- Vehicles that are already on a motorway over those that want to enter that motorway.

Junctions with signs

At junctions with signs you must always follow the instructions shown by that sign. When there are no lines that show which is the entry to that junction, you must wait and give way from the place where you can best see the junction and the vehicles approaching it. You must never enter a junction, a pedestrian crossing or a cyclist crossing if you think your vehicle is going to get trapped in the middle of the junction or the crossing, interrupting traffic. For example, you must not enter a pedestrian crossing when there are vehicles just in front that are not going to let you pass. In that case, you must wait before the pedestrian crossing. If you are stopped at a junction with traffic lights and your vehicle is an obstacle for other drivers, you must leave the junction in any way possible, to continue on the route you wanted to take, as long as no vehicles are coming towards you.

Roundabouts

You must drive in the same way as on the rest of roads, depending on whether they are urban with boundaries or not, inter-urban, etc.

Watch video

Right of way for cyclists, pedestrians and animals

Cyclists

Cyclists have right of way in the following situations:

- When they are travelling in a cycle lane, a cyclist crossing or a well-signed hard shoulder.
- When another vehicle wants to turn right or left and there are cyclists nearby.
- When cyclists are travelling in a group and one of them has already entered a junction. The other vehicles must wait until the whole group of cyclists has passed.

Pedestrians

Watch video

Pedestrians have right of way in the following situations.

- At pedestrian crossings, on pavements and in other pedestrian zones.
- When a vehicle is going to turn to enter another road where pedestrians are crossing.
- When a vehicle crosses a hard shoulder where pedestrians are walking who have no other place to go because there are no pavements.
- When a vehicle needs to cross

a pedestrian zone. For example, the pavement that a car must cross when leaving a garage.

Drivers must also give way to:

- An organised group of people who all go together with a purpose. For example, a group of children

on a trip with their school.

- People getting on or off public transport at the stop. For example, people getting off the bus, up to the nearest pedestrian zone.

Animals

Animals have right of way in the following situations.

- On drovers' roads where there is a sign for "Domestic animals crossing". And below it a panel that says drovers' road.

Drovers' road. Path created so that animals can pass along it.

- When a vehicle is going to turn to enter another road where animals are crossing.
- When a vehicle crosses a hard shoulder where animals are travelling who have no other place to go because there is no drovers' road.

Right of way for vehicles on emergency service

Which vehicles are prepared to provide emergency services?

- Police cars.
- Fire engines.
- Civil protection and rescue vehicles.
- Ambulances and other vehicles providing medical assistance. When these vehicles are on an emergency service, they must warn of their arrival by switching on the siren and the lights provided for this. In these cases, these vehicles have right of way on all roads (streets and roads). The rest of vehicles and pedestrians have to give way to them.

The following vehicles also have right of way:

- Maintenance teams that repair the road.
- Roadside assistance vehicles when they are going to help vehicles that have had a breakdown or an accident.

Drivers of vehicles prepared to provide emergency services may drive faster than the permitted speed and are not obliged to comply with signs. The only signs they must obey are those given by traffic officers. They must ensure that they do not endanger pedestrians or vehicles when passing through a junction or going through a red traffic light. On the motorway and the dual carriageway they may change direction, change direction of travel, reverse and enter the central reservation when they can ensure that there is no danger to other vehicles.

How should you act when faced with vehicles providing emergency services? All vehicles must make it easier for them to pass as soon as they hear the siren or see the lights. They will move to the right and stop if necessary.

Pedestrians will clear the road and wait on the pavement.

Any vehicle providing an emergency service

Watch video

In some circumstances drivers of vehicles that are not prepared to provide emergency services must ask for priority over other vehicles because of some circumstance. For example, they are taking to hospital a person who is very ill or a woman about to give birth. In these cases, the driver of the vehicle may warn of the emergency situation in the following ways:

- Sounding the vehicle's horn intermittently.
- Switching on the hazard warning lights of the vehicle.
- Waving a handkerchief through the window. The driver must comply with the traffic rules. Especially at junctions. Other drivers have to let them pass.

Right of way in narrow sections

Narrow sections with signs

You must always follow the rules shown by vertical signs, traffic lights or traffic officers when passing along a road or very narrow track.

Narrow sections without signs

If there are no signs for passing through a narrow section, the vehicle that entered first has right of way in the narrow section.

If there is doubt about which vehicle entered first, the vehicles with more difficulty in manoeuvring have priority.

Which vehicles have right of way to pass through a narrow section? The order of priority of vehicles is as follows:

1. Special vehicles that exceed the weight and dimensions set by the rules that regulate vehicles.
2. Vehicle combinations.
3. Vehicles drawn by animals.
4. Motor caravans and cars towing a trailer that weighs less than 750 kilos.
5. Buses.
6. Lorries, articulated lorries and vans.
7. Cars and car-derived vehicles.
8. Heavy quadricycles, light quadricycles and special vehicles that have the weight and dimensions set by the rules that regulate vehicles.
9. Three-wheeled mopeds, motorcycles with sidecar and other three-wheeled vehicles.
10. Motorcycles, two-wheeled mopeds and bicycles.

When they are vehicles of the same type or there is doubt about which one should go, the following has priority:

- The vehicle that has to reverse for a longer distance.
- When the reversing distance is the same, the vehicle that is wider, longer or can carry more load has priority to go. The lorry can go first. The car has to reverse. The bus can go first. The lorry has to reverse.

Narrow sections on a slope

On steep slopes, the vehicle travelling uphill has priority to go, unless it has a safe place to stop closer than the other vehicle. If there is doubt, the same rule will be followed as in narrow sections that are not on a slope. The car has priority. The lorry must reverse.

Crossing level crossings and movable bridges

You must reduce speed and drive more carefully when you approach a level crossing or a movable bridge.

You must not cross the level crossing or movable bridge when:

- It is closed.
- Its barriers are moving to open or close the crossing.
- There is a traffic light indicating that you must stop. In those cases all vehicles have to wait in their lane, one behind another, until the crossing is clear. Once the crossing is clear, you must cross quickly. Make sure beforehand that there is no risk of being trapped within the level crossing. If there are no barriers or traffic lights, you must check that no train or tram is coming before crossing. When a vehicle becomes trapped within a level crossing or a movable bridge all the people travelling in it, except the driver, must get out of the vehicle. The driver will try to start the vehicle once the rest of the passengers have got out. If they do not manage it, they will also get out of the vehicle and try to warn of the situation to any trains or trams that may pass and to the drivers of other vehicles approaching the area .

Crossing tunnels and underpasses

No vehicle may enter a tunnel when at the tunnel entrance there is a traffic light that prohibits entry.

Watch video

Inside the tunnel you must keep a distance of at least 100 metres from the vehicle in front (or 4 seconds).

Vehicles that can carry a weight greater than 3,500 kilos must keep a distance of 150 metres from the vehicle in front (or 6 seconds).

Rules for stopping the vehicle inside a tunnel When there is heavy traffic and vehicles cannot move forward inside a tunnel, all passengers must stay inside the vehicle. The driver must switch off the engine and leave on the sidelights. When braking, they will switch on for a moment the hazard warning lights so that other drivers can see them.

Emergencies inside a tunnel

When a driver has to stop the vehicle inside a tunnel because of an emergency, the steps they must follow are:

- Switch off the engine, leave on the sidelights and the hazard warning lights so that other drivers can see the vehicle.
- Take the vehicle to the nearest area reserved for emergencies. If there is no emergency area, they must move it as close as possible to the right-hand edge of the road.
- Switch on and place the warning light, if you have one, or otherwise place the warning triangles on the carriageway.
- Ask for help through the nearest emergency call post inside the tunnel and follow the instructions

you are given.

- All the people travelling in the vehicle must get out of it and go to the nearest refuge or exit.
- When the vehicle can be moved, despite the breakdown, drive until you leave the tunnel or reach the nearest emergency area.

If there is a fire inside a tunnel, drivers of vehicles must follow the following steps:

- Move the vehicle to the right to let emergency vehicles pass.
- Switch off the engine.
- Leave the key in and the vehicle doors

open. That is, not locked.

- All the people travelling in the vehicle must get out of it and go to the nearest refuge or exit. Always in the opposite direction to the fire.

Topic 13. Carrying people and loads

Contents

Rules for carrying people, animals and loads

- General rules
- Carrying people
- Carrying loads

Permitted dimensions for vehicles and loads

- Permitted dimensions for vehicles
- Permitted dimensions for loads
- Marking a projecting load
- Loading and unloading operations

Plates and signs on vehicles

Rules for carrying people, animals and loads

General rules

Weight limit

All vehicles have marked a maximum number of kilos they can carry depending on their characteristics. It is prohibited for vehicles to carry more load than they can, counting the weight of the passengers, the luggage and other materials or loads. When a vehicle carries more weight than it should, some of its parts get damaged, such as: the tyres, the acceleration systems and the braking systems.

Carrying animals

Animals must travel in the rear part of the vehicle and be secured to the seat. When possible, a separating mesh or another device will be fitted between the part where the animals travel and the part where the driver travels.

Animals must never travel loose inside the vehicle.

Carrying people

General rules

Carrying people on a bicycle

Carrying people on a moped and motorcycle

General rules

The number of people who can travel in a vehicle depends on the number of seats that vehicle has. For example, in a car with five seats six people cannot travel.

Watch video

Each person must travel in their seat during the whole journey without moving to other spaces in the vehicle.

However, people may travel in the parts of vehicles intended for carrying loads when the vehicle has the necessary authorisation. The driver must distribute the passengers in the vehicle and place the load so that they have enough space to drive and see the road well from all sides of the vehicle. The driver must pay special attention to:

- Being comfortable in their seat.
- Making sure the other passengers use their seats and do not move from them.
- Placing the objects they are carrying properly so that they do not get in the way. All travellers will get in and out of the vehicle when it is stopped and on the side closest to the pavement or the hard shoulder.

Carrying people on a bicycle

On bicycles designed for one person only the rider may travel. However, when the rider is an adult, they may carry as a passenger a child under seven years old. The child will travel in a permitted additional seat. This seat will be fitted behind the seat of the adult who is riding.

Carrying people on mopeds and motorcycles

Watch video

On mopeds and motorcycles one person may travel with the rider when the person meets the following requirements:

- They are over 12 years old.
- They sit behind the rider.
- They sit correctly and place their feet on the footrests that are on the sides of the vehicle.
- They wear the helmet correctly fastened.

Watch video

As an exception, children who are over seven years old may travel on a motorcycle or moped when they meet two requirements.

- They travel with one of their parents or with an adult who takes responsibility for them.
- They follow all the safety rules that the other passengers must follow.

Carrying loads

A load is any object that is carried in the vehicle. It can be luggage, goods, or anything else.

In vehicles designed to carry people, luggage and other types of loads may be carried as long as they meet these requirements:

- The load is suitable for the characteristics of the vehicle. For example, it does not exceed the weight that vehicle can carry.
- The load is well distributed and secured so that the vehicle does not lose its stability.
- The driver of the vehicle indicates that they are carrying that load, when necessary.
- The load does not cover the lights or the warning signals of the vehicle. It also allows the signals that the driver makes with their hand to other vehicles to be seen clearly.

Objects carried in a vehicle must be placed in the boot.

When it is necessary to place an object in another part of the vehicle, it must be secured so that the object will not move in the event of a crash or when braking. The driver must also ensure that the object allows them to see the whole road clearly. It is prohibited for the loads carried by a vehicle to:

- Move around inside the vehicle.
- Fall off.
- Drag along the road.
- Make noise, give off dust or smoke. The transport of loads that may fall or that give off dust will always be done in special vehicles designed to carry these materials. Motorcycles, three-wheeled vehicles, mopeds, cycles and bicycles may carry a trailer or semi-trailer when it meets the following characteristics:
- The trailer weighs at most half of what the vehicle weighs.
- It travels during the day and is clearly visible.
- The vehicle travels more slowly than the maximum speed permitted for that vehicle.
- No people travel in the trailer.

Permitted dimensions for vehicles and loads

Permitted dimensions for vehicles

Width

The maximum width permitted for a vehicle is 2.55 metres, including its load if it is carrying one. 2.55 m

Height

Vehicle

Maximum permitted height

(Including the load)

Vehicles in general

4 metres

4 m

Buses

4.20 metres

Recovery vehicles that remove vehicles 4.50 metres

Length

Vehicle

Maximum permitted length

(Including the load) Motor vehicles, except buses 12 metres

12 m

Trailer only

12 metres 12 m

Vehicles with trailer 18.75 metres 18.75 m

Articulated vehicles, carrying a semi-trailer

16.50 metres 16.50 m

Combination of vehicles in euromodular configuration 25.25 metres 25.25 m

Permitted dimensions for loads

As a general rule, the load cannot be wider, higher, or longer than the vehicle carrying it. That is, it cannot project beyond the vehicle.

However, the load placed on the vehicle's roof rack may project.

Roof rack. The top part of the vehicle where luggage and other objects can be placed. In cases where the load has to project because there is no other way to carry the object, vehicles must meet different conditions depending on the type of vehicle.

Vehicles that carry loads

Vehicles that carry passengers

Vehicles that only carry loads

In vehicles used to carry goods, loads that cannot be folded or divided may project when they meet certain requirements. Some examples of these loads can be: pipes, beams, or posts.

Type of vehicle	How far may the load project?
Vehicles that are 5 metres long or less	One third of the metres of the vehicle's length. For example, if the vehicle is 3 metres long, the load may project 1 metre. The load may project at the front and at the rear 1/3 Maximum 1/3 Maximum
Vehicles that are more than 5 metres long	The load may project 2 metres at the front and 3 metres at the rear. Maximum 2m Maximum 3m
Vehicles that are at most 2.55 metres wide when carrying the load	The load may project 0.40 metres on each side of the vehicle. NOT PERMITTED It is not permitted to place the load like this. The panels cannot be placed transversely because they take up more. Larger dimension PERMITTED It is permitted to place Larger dimension the load like this. Smaller dimension

Combinations of vehicles in euromodular configuration The load cannot be longer than the vehicle.

Vehicles that also carry passengers

The load cannot project at the front or on the sides of the vehicle. The load may project at the rear of the vehicle when it meets these characteristics.

- Loads that can be divided into smaller loads may project by a maximum of 10 per cent of the vehicle's length at the rear of the vehicle.

For example, in a vehicle that is three metres long, a bicycle placed at the rear may project 30 centimetres at most.

- Loads that cannot be divided into smaller loads may project, by a maximum of 15 per cent at the rear of the vehicle.

For example, in a vehicle that is 3 metres long, a board placed on top may project at the rear by 45 centimetres at most.

- In vehicles that are less than one metre wide, such as, for example, motorcycles, the load:
- Cannot project at the front.
- May project at the rear 0.25 metres (25 centimetres).
- May project on each side 0.50 metres (50 centimetres) from the centre of the vehicle.

Whenever the load projects beyond the vehicle all necessary measures must be taken to prevent accidents or damage to other people and vehicles.

Marking projecting loads

Loads projecting at the front

- During the day they are not marked.
- At night or on days with poor light they are marked with a white light.

Loads projecting at the rear

- They are marked during the day and at night.
- To mark them, a board is fitted with red and white diagonal stripes at the rear of the load.
- Two boards with red and white diagonal stripes are fitted at the rear of the vehicle when the load is as wide as the vehicle.

Each board is fitted at one end of the projecting load.

Loads projecting at the rear must be lit with a red light when the vehicle is travelling at night or on a day with poor light.

Loads projecting on the sides

Loads projecting more than 40 centimetres on the sides of the vehicle must have a light switched on at the ends of the load when it is night-time or a day with poor light. The lights that mark the load at the front of the vehicle will be white and those that mark the load at the rear will be red.

Loading and unloading operations

Loading and unloading objects is always done off the road. When circumstances make it necessary to do it on the road, the following precautions will be taken:

- Not causing damage to other vehicles or pedestrians travelling in that area.
- Respecting the rules about the places where stopping and parking are allowed and the times when it is permitted.
- Stopping the vehicle on the side closest to the pavement or the hard shoulder.
- Taking as little time as possible to load or unload.
- Avoiding noise and disturbance that are not necessary. It is prohibited to leave loads on the road, the hard shoulder, or in pedestrian zones.

Plates and signs on vehicles

Number plate

All motor vehicles, except motorcycles, must have two number plates. One at the front and one at the rear.

Motorcycles and mopeds must have only one plate, at the rear.

Trailers and semi-trailers that can carry more than 750 kilos must have their number plate at the rear and, in addition, the number plate of the vehicle towing it.

Trailers and semi-trailers that can carry less than 750 kilos will only have the number plate of the vehicle towing them at the rear.

Priority vehicle sign

A light signal made up of one or more blue lights. It is carried by vehicles that provide emergency services. These vehicles are: police cars, fire engines, civil protection vehicles, rescue vehicles, and ambulances. These vehicles may use the light signal at the same time as the siren to emit sounds.

Speed limit plate

It indicates that the vehicle carrying it cannot travel at more kilometres per hour than the number shown on the plate. It is placed on the rear of the vehicle.

Slow vehicle plate

It indicates that the vehicle carrying it cannot travel faster than 40 kilometres per hour. It is placed on the rear of the vehicle.

Long vehicle plate

It indicates that the vehicle carrying it is more than 12 metres long. This plate is placed on the rear of the vehicle. It is rectangular, yellow, and with red edges.

Plate for a vehicle carrying dangerous goods These plates may have numbers. The numbers at the top inform about the type of danger that this material can cause. For example, if it is toxic material or if it can burn. The numbers at the bottom indicate the material that the vehicle is carrying.

New driver plate

It indicates that the driving licence of the person driving the vehicle is less than one year old. This sign is placed on the rear window of the vehicle, in a place where it can be seen clearly.

Sign to indicate danger

It indicates that the vehicle is stopped on the road due to any emergency, due to a breakdown, or that the load it is carrying has fallen onto the road surface. It is a yellow sign that is placed on the highest part of the vehicle so that it can be seen clearly.

Vehicle roadworthiness inspection sign

Sticker that indicates that the vehicle has passed the roadworthiness inspection and is in good condition to drive. It also indicates the date when it must have the next inspection. This sticker is placed on the right-hand side of the windscreen at the front of motor vehicles. On trailers and semi-trailers it is placed in a place where it can be seen clearly.

Escort warning sign for special transport Sign that a vehicle carries on the top to indicate that a special transport vehicle is travelling nearby. For example, a combine harvester or an excavator.

SPECIAL

Escort warning sign for cyclists

Sign that a vehicle carries on the top to warn that there are cyclists travelling nearby.

CYCLIST

Topic 14. Driving safely

Contents

The road

- Bends
- Roadworks area

Driving at night

The weather

- Driving in rain
- Driving in snow
- Driving on ice
- Driving in fog
- Driving in clouds of dust or smoke
- Driving in strong wind
- Driving in heat

The road On some stretches of road you must be more careful than on others when driving because more difficulties may appear. These places are:

Bends

Road works area

Bends

Bends are dangerous stretches of road where you must drive more carefully. The reason is that, when taking a bend at speed, a force is created that can push the vehicle out of the bend and send it towards the side it must not go to.

Bends are more dangerous when:

- The bend is very tight.
- The vehicle is carrying a lot of weight.
- The vehicle is travelling very fast. In those cases, the vehicle can continue straight on, as if there were no bend. Because of this, it can leave the road or enter the oncoming lane.

Steps to take a bend properly

Before entering the bend you must:

- Reduce speed or brake if necessary.
- Move as close as possible to the right-hand edge of the carriageway. In the bend you must:
- Turn the steering wheel smoothly.
- Do not accelerate or brake sharply.
- Accelerate little by little. When leaving the bend you must:
- Turn the steering wheel smoothly to drive straight again.
- Increase speed little by little.

Skidding on bends

Skidding happens mainly on bends. The reason is that the force that pushes vehicles on bends can make their tyres not grip the road, slide, and lose control.

Skid. To slide or slip on the road. When a vehicle skids it moves away from the direction it was going and goes to another side. A skid can be caused by:

- Driving too fast.
- Making a sudden turn with the steering wheel.
- Pressing the brake pedal very hard.
- Having the tyres or the shock absorbers in poor condition.
- Distributing the load badly in the vehicle.

How can you make the vehicle stop skidding?

Type of vehicle	Which wheels skid?	What must you do?
Front-wheel drive.	Rear	Do not brake.
The engine power goes to the front wheels		Turn the steering wheel towards the side the wheels are moving to. Accelerate smoothly.
Rear-wheel drive.	Rear	Do not brake.
The engine power goes to the rear wheels		Stop accelerating smoothly. Turn the steering wheel towards the side the rear wheels are moving to.
Rear-wheel drive	Front	Stop accelerating. Straighten the steering wheel until the wheels stop skidding.

Roadworks area

Roadworks can be a danger. Whenever you drive on a road with roadworks, you must follow the instructions of the roadworks staff. To mark that a road has roadworks the following signs are used.

- Vertical danger signs and signs that inform you of the rules you must follow. These signs will have a yellow background.
- Signs painted on the road, yellow in colour.
- Other signs that are only placed when there are roadworks.

Driving at night

Driving at night is more dangerous than driving during the day.

Watch video

The reasons are:

- The road is less well lit.
- Distances, people, objects, and vehicles are seen worse.
- Other vehicles can dazzle you with their lights. You must pay special attention when going from an area that is well lit to another that is poorly lit. It takes the eyes a few seconds to get used to the change in light.

When driving at night you must take special care with:

Speed

Bends

Dazzle

Overtaking

Speed

You must respect the speed limits and drive more slowly if you cannot see the road well. In this way it will be easier to stop the vehicle if there is a danger or something unexpected.

Dazzle

When a vehicle dazzles you you cannot see the road well because too much light enters your eyes.

Watch video

What can you do to avoid dazzling other drivers?

- Keep the lights properly adjusted.

Dipped headlights may bother because they are not properly adjusted.

- Distribute the weight you are carrying in the vehicle properly.

Dipped headlights may be too high and dazzle because there is too much weight at the rear of the vehicle.

- Switch off the main beam and switch on dipped headlights when another vehicle is coming towards you from any direction. When a vehicle dazzles you, reduce speed or even stop the car taking the necessary safety precautions.

Do not move into other lanes when reducing speed or stopping. You can guide yourself by the line on the right-hand edge. Do not use dark or sunglasses at night. Other vehicles may dazzle you less, but you will see the road much worse.

Overtaking

When you are going to overtake another vehicle at night, you must switch off the main beam and switch on dipped headlights.

Main beam can dazzle the other driver through the rear-view mirrors. When you are already overtaking, switch the main beam back on as soon as possible. That is, when you can no longer dazzle the other driver.

If another vehicle is overtaking you, keep the main beam on while they overtake you. Those lights will light up the road for both of you during the overtaking manoeuvre.

Bends

When two vehicles travelling in opposite directions pass each other at night on a bend, the vehicle travelling on the inside of the bend is the one that must switch off the main beam and switch on dipped headlights. The reason is that its lights are the ones that light up the road directly and can dazzle the other driver. The driver travelling on the outside of the bend can keep the main beam on because their lights shine off the road. They cannot dazzle other drivers. They must only change to dipped headlights if they see that at any moment they are dazzling another driver.

The weather

Rain

Snow

Ice

Fog

Dust

Wind

Heat

Driving in rain

When it rains heavily it can be more difficult to drive for the following reasons:

- The road is seen worse.
- The tyres grip the road less. When this happens the vehicle needs more space to brake.

The precautions you must take to drive in rain are:

- Keep the tyres in good condition.
- Check that the windscreen wipers work well.
- Brake smoothly so that the wheels have more time to stop.
- Check that the brakes work well after driving through a puddle.
- Keep more distance from the vehicle in front.
- Reduce speed.

Watch video

You must be especially careful when the first drops of water start to fall. These drops mix with dust, grease, and other dirt on the road, forming mud that can make the vehicle skid. When the wheels skid you must lift your foot off the accelerator, but without braking.

Motorcycle riders must pay special attention to the road lines when it rains because they can slip on them.

Aquaplaning

What is it?

Losing control of the vehicle because the tyre cannot clear all the water it picks up from the road and it skids. It is easier to aquaplane and lose control when the vehicle is travelling at high speed and when the tyres are very wide or very worn.

Therefore, the best way to prevent aquaplaning is to travel slowly in areas where there is water. After passing through that area, you must check that the brakes work well.

Driving in snow

Driving in snow is more difficult for the following reasons:

- The road, the signs, and the vehicles are seen worse.
- The tyres grip the road less. When this happens, the vehicle needs more space to brake. The precautions you must take to drive in snow are:
- Start the vehicle with the wheels straight.
- Choose a gear in the gearbox that allows the vehicle to start at a higher speed, but with little engine power.

Gearbox. Vehicle mechanism that is responsible for converting the engine power into movement of the wheels. With the gearbox the driver can choose to go faster or slower and with more or less engine power.

- Do not make sudden movements with the steering wheel and do not change gear suddenly. When going downhill you must do it more slowly than normal and choose low gears (first or second) so that they hold the vehicle back.
- Reduce speed little by little.
- Keep a greater safe following distance from the other vehicles.
- Use the brake as little as possible and do it smoothly.
- Drive over the marks made by other vehicles with their wheels. When the sun comes out after it has snowed, it is advisable to use sunglasses so that the light does not dazzle you. The R-412 sign indicates that you are obliged to fit snow chains on the vehicle to continue driving in snow. The chains are fitted, at least, on the driving wheels. You must fit at least one chain on each side of the vehicle.

Driving on ice

Ice makes the wheels unable to grip the road and the vehicle skids a lot. The precautions you must take to drive on ice are the same as for driving on snow. Drive slowly and keep a very large safe following distance from other vehicles because you will need more space to brake. There may be ice on the road, during the night or early morning, in the following places:

- In damp areas.
- In areas where the sun does not reach.
- On bridges and flyovers.
- In dips.
- Where there are P-34 or P319 signs with a panel on which the word Ice is written.

Ice can also remain on the vehicle windows. In those situations, before you start driving, you must:

- Scrape the ice off all the windows, without scratching them.
- Clean the rear-view mirrors.

Driving in fog

Fog is very dangerous for driving for the following reasons:

- The road, the signs and the vehicles are harder to see.
- The tyres grip the road less because the surface is damp. The precautions you must take to drive in fog are:
- Keep the vehicle well ventilated so that the windows do not mist up.
- Switch on dipped headlights and fog lights.
- Drive slowly.
- Keep a very large safe following distance from other vehicles.
- Pay close attention to the signs and road markings.
- Try not to overtake other vehicles if it is not necessary.
- Be more careful when you approach junctions.

Driving in clouds of dust or smoke

They are a danger because they do not allow you to see well the road, the signs, or other vehicles. Also, they can appear suddenly. To drive in clouds of dust and smoke, you must take the same precautions as for driving in fog.

Driving in strong wind

It is dangerous to drive in strong wind, especially when it blows from the side, because it can make the vehicle lose control, roll over, or leave the road.

Wind affects two-wheeled vehicles more and those towing a trailer.

Wind is more dangerous for driving in the following situations:

- When you pass another vehicle.
- When overtaking a vehicle that takes up a lot of space.
- When passing in front of buildings, trees and other objects that can cause the wind to appear suddenly. The precautions you must take to drive in strong side wind are:
- Drive more slowly.
- Hold the steering wheel firmly and in the direction against the wind.
- Keep the windows closed.
- Bear in mind that trees, branches, or stones may fall.

Driving in heat

It is dangerous to drive in heat for the following reasons:

- Sleepiness and fatigue appear sooner, which can cause distractions and mistakes.
- You may need more time to react to unexpected events.
- Aggressiveness towards other drivers may increase. The precautions you must take to drive on a very hot day are:
- Use the vehicle's air conditioning and keep it at a temperature between 20 and 23 degrees.
- Take more breaks during the journey.

- Drink plenty of water or juices.
- Pay more attention when you drive after eating because you may become sleepy.
- Wear light-coloured, light, loose clothing.

Topic 15. Vehicle mechanics and maintenance

Contents

Vehicle systems

- Fuel system
- Electrical system
- Lubrication system
- Cooling system
- Transmission system
- Steering system
- Suspension system
- Braking system

Wheel and tyre checks

Vehicle systems

What are they? All the mechanisms and elements of the vehicle that make it work and be safe. Vehicles have eight main systems:

- Fuel system
- Electrical system
- Lubrication system
- Cooling system
- Transmission system
- Steering system
- Suspension system
- Braking system

Fuel system

What is its function? To bring air and fuel to the engine so that the vehicle can work.

The air can carry dirt from the street. That is why the system has a filter that is responsible for cleaning that air so that it reaches the engine in good condition. You must clean this filter from time to time because, when it is dirty, the fuel passes through it worse. This makes you use more fuel than when it is clean. Also, when the filter is very dirty or broken, black smoke may come out from the fuel through the exhaust pipe.

Exhaust pipe. A pipe that cars have at the back to expel the gases that are created in the engine. You must check the filter more often in summer than in winter and when you drive on very dusty tracks.

Electrical system

What is its function? To provide energy to the car so that it switches on, the engine can start and work, and other elements such as the lights or the horn work. This system has several parts:

Battery

Ignition circuit

Charging circuit

Lighting circuit

Battery

Provides the energy needed to start the engine. It also supplies energy to the rest of the vehicle when necessary. The battery must be kept clean, dry, and well fitted in its place.

Ignition circuit

It is responsible for producing the spark of electricity needed so that the air and fuel turn into energy and the vehicle starts to work. This spark of electricity is produced in an engine part called a spark plug. To keep the ignition circuit in good condition, you must check the condition of the spark plugs and change them when necessary. You must also check that the engine is in good condition.

Charging circuit

It is responsible for supplying electric current to the battery and to all the systems that need it while the vehicle engine is running.

Watch video

The part that allows the vehicle to produce electricity and store it in the battery is called the alternator. The alternator is driven by a belt.

You must check from time to time that this belt is in good condition and change it when necessary.

Lighting circuit

It is made up of all the elements that allow the vehicle lights to be switched on.

Watch video

It is important to check the lighting system. You must change the bulbs when they give less light.

Lubrication system

What is its function? To distribute oil to all the parts of the vehicle engine to create a layer that covers the engine parts to protect them and prevent them from rubbing against each other. The oils used for vehicles are special substances that prevent the vehicle parts from wearing out more. The instruments that monitor the oil in the lubrication system are:

- Dipstick. It indicates how much oil there is in the engine.
- Pressure gauge or warning light. It monitors the oil pressure in the engine. The oil must have the correct pressure to be distributed well throughout the engine and cover all its parts.

Pressure gauge

Warning light

Watch video

When the warning light comes on it means there is little oil or that it does not have the pressure it needs. In that case you must stop the vehicle and not continue until you know what the problem is and fix it. To check the oil level you must look at the dipstick when the engine is stopped and cold. The oil level must be between the minimum and maximum marks.

Types of oils

Depending on where they come from	Depending on their thickness
Depending on where they come from Type of oil Mineral Synthetic Semi-synthetic	How is it obtained? From petroleum. Nowadays it is used very little. It is made in a laboratory by mixing substances from petroleum and from other materials. A mixture of mineral and synthetic oils.

Why are synthetic oils used more? Because they have the following advantages:

- They resist cold and heat better.
- They allow the vehicle to start better when it is very cold.
- They protect the engine better because they are thicker and less liquid than mineral oils.
- Less oil is needed.
- It lasts longer.

Therefore, you must change the vehicle oil less often.

Depending on their thickness

Oil is more liquid when it heats up and thicker when it is cold. That is why it is better to use an oil that is more liquid when the temperature is low. This way it will circulate better through the engine when starting the vehicle, even if it is very cold.

However, when the engine is hot it is better to use a thicker oil. If it is too liquid it will not cover well the engine parts. To avoid these problems there are oils that adapt well to cold and heat. They are called multigrade oils and they have this label.

10W/40

10W indicates that this oil is distributed well through the engine, even if the temperature is low. 40 indicates that it can withstand very high temperatures without being damaged.

To top up the vehicle with oil you must lift the engine cover and pour the oil through the cap that is in that place. From time to time it is necessary to change the oil and the filter. To change it, the vehicle must be level and the engine must be stopped and warm. You must avoid the oil falling on the ground because it is very polluting. If oil spills while you top up the vehicle, you must collect it. When a bluish white smoke comes

out through the exhaust pipe it means that the vehicle has too much oil.

Cooling system

What is its function? To ensure that the engine stays at a good temperature so it can keep working and to prevent its parts from wearing out and breaking due to excess heat. Without this system, the engine would heat up very quickly when running and would break down.

A liquid called coolant passes through the engine parts and absorbs the heat so that it stays cool. Then this liquid goes to a part called the radiator. There it cools down again and returns to the engine driven by a water pump. You must check that the belts that drive the water pump are in good condition and that the whole cooling circuit works well.

Watch video

You must also check that the vehicle has enough coolant. It must be kept between the minimum and maximum marks when the engine is cold. White smoke may come out of the exhaust pipe. That smoke is water vapour. This is normal when starting, especially when it is cold.

However, it may indicate that there is a broken part or a fault if the white smoke comes out when the vehicle has been running for a while and the vehicle is hot. At the beginning of winter it is important to check that the coolant is in good condition and is not going to freeze. If the vehicle is in a very cold place during winter, the liquid can freeze and break the engine or the cooling system.

Transmission system

What is its function? To transmit the engine power to the wheels so that the vehicle can move. A vehicle engine can drive some wheels or others. Depending on which wheels are driven, the vehicle can be:

Which wheels does the engine drive? Front-wheel drive To the front ones.

Rear-wheel drive To the rear ones.

Four-wheel drive To all the wheels. First gear is the one that sends the most force to the wheels, but it is the slowest.

Steering system

What is its function? To transmit the movement of the steering wheel to the front wheels so that the driver can turn the vehicle without problems.

Signs that the steering system is failing:

- It takes a lot of effort to move the steering wheel.
- The tyres may have low pressure and may need inflating more.
- The steering wheel is too loose.
- The vehicle pulls to one side when you let go of the steering wheel on a straight road.
- Some tyres may be more inflated than others.
- The tyres wear out very quickly.
- The steering wheel vibrates while you drive. The wheels may not be properly balanced.

Power steering

A system that helps the driver to use less force to turn the steering wheel and control the vehicle. There is also a type of power steering called progressive.

Progressive power steering helps you turn the steering wheel very easily when the vehicle is going slowly and makes it take a little more effort when the vehicle is travelling at a higher speed. This will make it easier to control the vehicle.

Suspension system

What is its function? To keep contact between the tyres and the road at all times.

Thanks to this system, the vehicle does not lose stability and allows passengers to travel more comfortably.

Without the suspension system the driver would have more difficulty getting over small obstacles on the road and driving without jolts.

What happens when the suspension system fails?

- The vehicle dips forwards or the rear lifts a lot when braking.
- The vehicle sways from side to side or leans a lot when taking a bend.
- You feel bumps or the wind too much.
- The tyres wear out very quickly.
- The vehicle lights move up and down when they are on.

What consequences does a suspension system in poor condition have?

- The vehicle loses stability.

Especially on bends and when it is windy.

- There is a higher risk of aquaplaning.
- The vehicle needs more space to brake. Especially when the road surface is wet.
- The system that prevents the brakes from locking works worse.
- Other parts of the vehicle wear out and break more easily.
- The lights move up and down and can dazzle other drivers.
- The driver can feel tired more easily.

Braking system

What is its function? To reduce speed or stop the vehicle and prevent it from moving again when it should not. When you press the brake pedal, a force is transmitted to the vehicle's wheels so that they stop and stop working. This force passes through a fluid called brake fluid and reaches the parts that brake the wheels, which are the pads or the shoes. The pads and the shoes rub against other parts of the wheels and make them stop. What must you check?

- That there is brake fluid in the vehicle's reservoir. And that it is between the minimum and maximum values and that it is in good condition.
- That the pads or the shoes are not worn and are properly secured.
- That the tyres are properly inflated, with the pressure they need.

Watch video

Wheel and tyre check

To keep the vehicle's wheels and tyres in good condition, you must:

- Check that the tyres are properly inflated, at least once a month. Especially if you are going to make a long journey, or load the vehicle a lot.

You must do this check before you start driving.

Watch video

- Look at the tyre tread to make sure it is not very worn.

You must always carry a spare wheel in the vehicle and other systems. This wheel will be slightly more inflated than the others.

Topic 16. Road traffic accidents

Contents

Avoiding road traffic accidents

- Risk factors
- Consequences of road traffic accidents for society

People with a higher risk of having road traffic accidents

- Pedestrians
- Drivers What to do in case of an accident
- Steps to follow
- PAS rule

First aid

- Vital signs
- Knowing the condition of the injured
- Moving the injured

Avoiding road traffic accidents

Risk factors

Most road traffic accidents can be avoided. Some circumstances related to the driver, the vehicle, the road and the environment mean that there is a higher risk of having an accident. The circumstances that make a person more likely to have an accident are called risk factors.

What are the main risk factors?

Risk factor Some causes

Number of accidents

Human

Speeding. Consuming alcohol. Distractions.

Between 70 and 90 out of every 100.

Road condition The road is wet.

Losing control on a bend.

Between 10 and 35 out of every 100.

Vehicle failure The brakes fail. A tyre punctures.

Between 4 and 13 out of every 100.

Watch video

Many of these risk factors can be avoided if the driver behaves responsibly. For example, by checking the tyres before starting a journey to check that they are in good condition and will not have a blowout.

Accidents can also be avoided by reducing speed and leaving a greater safe following distance from the vehicle in front on a road in the rain. Where and when are there more road traffic accidents?

- Most fatal accidents happen on roads outside towns or cities.
- Most accidents happen on straight stretches and not on bends.
- There are fewer accidents on motorways and dual carriageways than on the rest of the roads.
- The times of year when there are more accidents are: Easter, December and summer.
- There are more accidents at weekends and on public holidays. Especially in the early hours of the morning.
- During the day, most accidents happen at the times people go to and leave work. Especially at the end of the working day.
- The accidents that happen most within cities are pedestrians being run over.

Consequences of road traffic accidents for society Road traffic accidents are one of the main causes of injury and death. Especially in young people.

Also, each road traffic accident has very negative consequences for the whole of society.

These consequences are:

Human

Material

Medical and healthcare costs

Costs in security resources

Human

The death of a person means physical and psychological suffering for the people around them.

Material

Road traffic accidents cause damage to vehicles, the road and the environment.

Spending on medical and healthcare resources

A lot of staff and money must be invested to provide first aid to the injured, treatment, rehabilitation and the adaptations that each injured person may need.

Spending on security resources

A lot of staff and money must also be invested in the work done by police officers and firefighters at accidents.

Also, each accident involves extra costs that have to be paid by insurers and the organisations that provide services to the injured. For all these reasons, the World Health

Organization considers that road traffic accidents are a health problem that affects the whole of society. And it asks all of us to work together so that there are fewer deaths and injuries in road traffic accidents.

People with a higher risk of having road traffic accidents There are studies that show that some groups of people are more likely to have an accident. These groups of people are called risk groups.

Within these risk groups, we must distinguish between the risk faced by pedestrians and the risk faced by drivers.

Pedestrians

When a vehicle runs over a pedestrian, the pedestrian usually suffers very serious injuries. They are more likely to die when the accident is on a road outside the city because vehicles travel faster. Pedestrians being run over happens more in areas where drivers cannot see them well because there is some kind of obstacle. To avoid being run over, pedestrians must take the following precautions:

- Cross at pedestrian crossings.
- Cross when the traffic lights are green.
- Walk in the permitted areas on roads and hard shoulders.
- Wait on the pavement to cross. Not on the road.
- Always look before crossing.
- Check that no other vehicle is coming before getting out of yours.

Drivers must also take measures not to run over pedestrians.

- Drive at a moderate speed. Especially within the city.
- Drive more carefully when you reach places where there are pedestrians and other vehicles parked with people inside. Some of these people may get out of the vehicle without looking.
- Do not go through when the traffic light is red or amber.
- Let pedestrians cross when they have right of way.
- Take special care when a pedestrian crosses while talking on their mobile. They may be distracted.
- Pay attention to pedestrians who walk in places close to areas for parties and entertainment.
- Take special care at the exit of garages.
- Do not park the vehicle on the pavement.

Vehicles on the pavement force pedestrians to walk on the road.

BEEEP

- Do not park on pedestrian crossings.
- Do not make changes to the outside of vehicles. The outside of today's vehicles is designed to cause less damage when pedestrians are run over.
- Drive more slowly when passing near a bus that is stopped.

Especially if it is a bus that takes children to school.

- Drive more carefully on rainy days.
- Take many precautions when reversing. There is a system called a pedestrian detector that can detect when there is a pedestrian in front of a vehicle. It warns the driver and, if they do not respond, the system activates the brakes and stops the vehicle.

Children and older people

The pedestrians who are most at risk of having a road traffic accident are children and older people.

Children

- They are shorter than adults. This means they can see less of the road and drivers can see them worse.
- They pay less attention to the road because they have less sense of danger.
- They do not know the traffic rules.

Most children being run over happens when they leave school.

Older people

The main problems that many older people have as pedestrians are:

- The noise in the environment does not let them hear a vehicle when it is approaching.
- They may have difficulty knowing the speed at which a vehicle is coming.
- They may have difficulty distinguishing the colour of traffic lights.
- There are few pedestrian crossings in some areas.
- Some streets are poorly lit and it is harder to cross them.
- Some kerbs are too high.
- There are obstacles on pavements such as plant pots or badly parked cars.
- They may have orientation problems when they do not know the streets well.

Older people have more accidents when they go alone than when they are accompanied by children. When they go with children they are more careful to set a good example and to protect them.

Drivers

The drivers with the highest risk of having accidents are:

Young people

Over

65 years old

Two-wheeled vehicles

Bicycles

Motorcycles and mopeds

Young people

In Spain, road traffic accidents are the leading cause of death for young people between 15 and 29 years old. These accidents happen, above all, in big cities, on the routes to and from work, and in places where people go out partying. The times when more young people die in road traffic accidents are: the summer months, Christmas, and weekends. The hours when there are most accidents of this type are at night and in the early hours of the morning.

Main causes of road traffic accidents among young people:

- Driving at excessive speed.
- Consuming alcohol and drugs.
- Not respecting traffic rules.
- Taking many risks when driving.
- Little experience.
- Competing with other drivers.

Km/h

20

40

60

80

100 120 140 160

180

200

220

240

STOP 80

- Not wearing the seat belt or the helmet because they think they do not need it.
- Following the advice of some advertising that encourages drivers to take risks.

People over 65 years old

These people are usually very careful when driving and comply with traffic rules. But they can have more accidents for the following reasons:

- They may have less ability to pay attention and need more time to react to an unexpected event.

- Many people over 65 years old see and hear worse.
- Some older people take medicines that affect driving. In the event of an accident, older people have more serious injuries and less chance of surviving.

Drivers of two-wheeled vehicles

Two-wheeled vehicles are more fragile and are seen worse than the rest of the vehicles. For this reason, their drivers must drive with special care.

Bicycles

Cyclist accidents happen more at weekends. Especially when the good weather arrives. There are more accidents within cities, but more cyclists die on roads outside cities. Most of these accidents happen on straight sections or at junctions due to a collision with another vehicle. The causes of these accidents may be due to offences committed by cyclists or offences committed by drivers of the other vehicles.

Offences committed by cyclists	Offences committed by drivers of other vehicles
Distractions. Riding in the opposite or prohibited direction. Making prohibited turns. Not respecting the STOP sign or the sign giving priority to other vehicles. Entering the road without caution.	Distractions. Driving very fast. Overtaking when they must not. Making prohibited turns.

To avoid running over cyclists, drivers of other vehicles must:

- Leave space in front until you have overtaken them,

as well as leave enough lateral separation when we are overtaking them.

- Not overtake other vehicles if cyclists are coming in the opposite direction.
- Be more careful when there is rain, snow, fog, or visibility is poor.
- Not sound the horn near cyclists so as not to frighten them.
- Be careful when there are parked vehicles.

Cyclists may appear between these vehicles.

- Drive with more caution near buildings and leisure areas.

Motorcycles and mopeds

The most common accidents in this type of vehicle when overtaking are head-on and side collisions.

Most of these accidents occur within the city. They have more serious consequences when the people travelling on the motorcycle or moped are not wearing a helmet.

The causes of these accidents may be due to offences committed by moped or motorcycle riders or offences committed by drivers of the other vehicles.

Offences committed by motorcycle riders	Offences committed by drivers of other vehicles
Using the lane of the road that goes in the opposite direction. Distractions.	Not respecting the motorcycles' right of way. Using the lane of the road that goes in the opposite direction. Distractions.

To avoid accidents with motorcycles and mopeds, drivers of other vehicles must:

- Check that there are no two-wheeled vehicles travelling before changing lane.
- Not stay alongside a two-wheeled vehicle. Travel in front of it or behind it.
- Keep the safe following distance from these vehicles.
- Respect situations in which they have right of way.
- Be more careful when there is rain, snow, fog, or visibility is poor.

Act in case of an accident

Steps to follow

All people who witness a road traffic accident must:

- Attend to the victims or ask for help so that someone helps them.
- Cooperate so that there is no more damage or injuries.
- Help traffic to become safe again when possible.
- Explain what happened to the authorities and emergency services.
- Give your name and contact details to the authorities and to other people who are part of the accident.

Drivers must also give their vehicle details. When you witness an accident, but you cannot do anything to help or the traffic officers have already arrived, it is best to keep driving so as not to obstruct the accident site.

Failure to assist

You must always try to help the person who has had an accident. Anyone who does not do so is committing an offence called failure to assist, that is, not helping, and may go to prison. This offence is committed by the person who:

- Does not help the victims of an accident when they can do so without putting themselves in danger.
- Does not report the accident so that someone comes to help.
- Causes an accident and flees instead of helping.

Important moments after an accident

After an accident there are three moments when people are at risk of dying:

First moment

First seconds after the accident. It is called immediate death.

Second moment One or two hours after the accident. In this period, resuscitation work is very important to try to save life. It is called the golden hour.

Last moment

Days or weeks after the accident. It is called late mortality.

PAS rule

What is it? These are the guidelines that must be followed to help a person who has had an accident. These guidelines are:

Watch video

It is very important to follow the PAS rule in the first 10 minutes after the accident. Acting quickly helps to:

- Save lives.
- Prevent a more serious accident from happening.
- Prevent the person from suffering more harm.
- Prevent material damage. For example, more vehicles and road elements being damaged.

Protect

Alert

Assist

Protect

What must you protect?

- Yourself. Do not put yourself in danger or take risks of having an accident.
- The victims.
- The accident site. Mark the place so that it can be seen well, avoid more accidents, and help the emergency services to find the place more easily.

What steps must you take to protect?

- Keep calm.
- Stop and park the vehicle in a safe place, leaving access for the emergency services.
- Stop the engine and switch on the hazard warning lights.
- Put on the high-visibility vest before getting out of the vehicle. This vest must be approved to ensure that it meets all safety rules. It can be yellow, orange, or red.

Approved. It is officially registered and meets the characteristics set by law.

- Observe the situation to assess the seriousness of the accident and decide how to help.
- Mark the accident area to warn other vehicles.
- Apply the handbrake of all vehicles in the accident to ensure that none of them moves.
- Do not smoke or light a fire at the accident site to prevent a fire from starting. If the fire has already started, try to put it out with a fire extinguisher or by throwing earth, sand, or blankets over the fire. You must not throw water on the fire.

- Help the area to become safe again and traffic to be able to move again.
- Try not to touch or move people or things if there are dead people or people with very serious injuries.

Alert

Before alerting the emergency services, you should try to gather as much information as possible to make it easier for the necessary services to come and to do their job well. For example, report if there is a fire so that the firefighters come. Or explain how many injured people there are so that they know how many ambulances they must send. To ask for help you must call the emergency number 112 from your mobile. If you do not have a mobile, you can go to the nearest SOS post to alert them.

You must stay at the accident site or nearby until the authorities and emergency services arrive. In some vehicles there is a system called eCall. This system sends a message automatically to 112 in the event of an accident so that the emergency services go to the place, even if nobody alerts them.

Assist

You must help the injured quickly but keeping calm. You must stay with the injured and not leave them alone. General rules for assisting.

- Do not move the injured unless it is necessary to put them in a safe place.
- Keep the neck protected from sudden movements. You can place clothing, sand, or your hands at the sides of the injured person's neck. Never under the neck.
- Do not remove the injured person's helmet.
- Loosen any clothing they are wearing that is tight on the body.
- Do not give medicines, food, or drinks to the injured.
- Do not put ointments or alcohol on their wounds.
- Cover the injured with clothing or thin blankets so that they maintain their body temperature.
- If in doubt about how to act, talk to the injured and give them emotional support. It is better not to touch or move them if you are not sure how to do it because you could make their injuries worse. How must you move an injured person? When it is necessary to move an injured person because in that place their life is in danger, it must be done by three people to move them as if they were a single block. The three people must be coordinated so as not to move their neck and to ensure that the head, neck, and body remain aligned.

To remove an injured person from their vehicle, you must follow these steps:

- Put your arms under their armpits, carefully.
- Hold their arm with one hand and support the chin with the other.
- Remove the person from the vehicle slowly and keep their head, neck, and trunk in the same line, as if it were a block.
- Place them in a safe place.

First aid

Vital signs

In road traffic accidents, the parts of the body that usually suffer the most damage are the head and the legs. To know how serious the injured people are, the first step is to check their vital signs, always following the same order.

Consciousness
Breathing
Consciousness

To know if a person is conscious, you can ask them: Are you OK? If they do not answer, you can give them a small pinch to see if they react with any movement or sound.

How to act depending on their reaction? The injured person responds by answering or moving The injured person does not respond and does not move Do not move the injured person.

Check whether they have bleeding, burns or fractures.

Watch them every few minutes to see if there are changes. They are unconscious. Check whether they are breathing.

Breathing

Steps to follow to check whether a person is breathing:

- Look at their chest to see if it moves.
- Bring your face close to their mouth to hear their breathing and feel on your cheek the air coming out of their mouth. When the person is face down, turn them to put them on their back and lift their chin to open their airways.
- Check whether there is normal breathing for 10 seconds.

Normal breathing for an adult is between 15 and 20 breaths per minute.

Normal breathing for a baby is between 30 and 40 breaths per minute.

Normal breathing for a child is between 26 and 30 breaths per minute. How to act depending on their reaction?

Breathes normally	Does not breathe normally
It is a sign that the heart is working. Check whether they have bleeding, burns or fractures. Stay with them until the emergency services arrive. Place them on their side if you have to leave them alone to help other injured people. Press 30 times Continue like this until medical help arrives.	Start cardiopulmonary resuscitation, until the emergency services arrive, as follows: - Place both hands with fingers interlocked on top of the chest of the person who is not breathing. - With your elbows straight, push your hands up and down about 100 times per minute. - Give 2 mouth-to-mouth breaths after every 30 chest compressions so that air goes into the injured person's mouth. Blow 2 times

Knowing the condition of the injured people

When the injured person is conscious and breathing you must check whether they have other types of injuries or wounds. To assess their condition you must take into account whether there are:

Bleeding

Burns

Fractures

External

Internal

Bleeding

What is it?

Losing a lot of blood because blood vessels have been broken. A person can go into shock or die if they lose a lot of blood.

Shock. Danger of death because the body does not receive the blood it needs. When the bleeding happens because an object has become stuck in the person's body, you must not try to remove the object.

Bleeding can be external or internal.

External bleeding

The blood comes out of the person's body. To control external bleeding three methods can be used:

Direct pressure

- Press on the wound with your hand or with your fist using a gauze pad or a handkerchief that is as clean as possible.
- Do not remove the gauze pad or the handkerchief, even if they become soaked with blood.
- If the wound is on an arm or a leg, keep the arm or the leg raised.

Arterial pressure

Artery. Tube that carries blood from the heart to all parts of the body.

- It is used when direct pressure fails.
- Press the main artery of the arm or the leg to stop blood circulation.
- Keep the area pressed until the emergency services arrive.

Tourniquet

- It is only applied for bleeding in arms and legs when direct pressure and arterial pressure have not worked.
- Tourniquets are usually made with a folded handkerchief. Do not use very thin objects.
- It is done above the place

where the wound is. Never below.

- You must write on a piece of paper the time when the tourniquet was applied and the place where it has been placed.
- Only medical staff can remove the tourniquet.

Internal bleeding

It happens inside the body and cannot be seen. But you can know that there is this type of bleeding because the injured person goes into shock. What symptoms does a person in shock have?

- They breathe very fast and very shallowly.
- The pulse is fast and weak.
- Their skin is pale and their sweat is cold and sticky.
- They say things that do not make sense. In these cases you must place the injured person lying down with their feet higher than their head. If they are vomiting, place them on their side and keep their feet higher than their head.

There is a type of internal bleeding in which the wound cannot be seen, but the blood does come out of the body through an opening such as the nose, the ear or the mouth. This type of bleeding is called externalised. In these cases you must:

- Treat the injured person as very serious.
- Do not plug or stop this bleeding.
- Put the injured person on their side if there is a risk of them choking.
- When blood from the bleeding comes out through the ear you must cover it with gauze, without pressing.

Burns

What to do in case of burns?

- Do not touch the burned area.
- Do not remove clothing that has become stuck to that area.
- Pour cold water over the burned area.
- Do not cut or burst blisters.
- Apply a disinfected and damp dressing.
- Do not bandage two burned areas together.
- When the burned area is on an arm or a leg, keep that arm or leg raised so that it swells less.
- You can give the injured person water to drink as long as they have no other injuries, are conscious and do not vomit.

Fracture

A fracture is a broken bone. What are the symptoms of a fracture?

- Severe pain in the area.
- The area swells or becomes deformed.
- The person cannot move that area or has difficulty doing so.
- When the fracture is in the spine the person cannot move and loses sensation. What to do in case of a fracture?

- Keep the fracture area immobile.
- Prevent the injured person from moving that area.
- Cover the wound if there is one.

Transporting injured people

You must always wait for the ambulance to arrive to transport injured people to hospital. Injured people should only be taken to hospital without waiting for the ambulance in these cases:

- The accident has happened in an isolated place where you cannot ask the emergency services for help.
- The emergency services take more than 30 minutes to arrive and the injured person is unconscious or has bleeding that cannot be controlled. Transport should be done, if possible, in a van or lorry so that the injured person can be laid down. You must drive slowly and smoothly.

Topic 17. Preventive and efficient driving

Contents

Prepare the journey

- Safety checks
- Choose the route

Preventive driving

- What is it?
- Vision
- Anticipation
- Control the space

Efficient driving

- What is it?
- Pollution
- Techniques to drive efficiently
- What makes more fuel be used?
- Environmental labels

Prepare the journey

Safety checks

Before starting a journey in a car or in a vehicle that transports goods, you must check that you have:

- The warning light or the warning triangles that you must place on the road in case of breakdown or accident.
- A high-visibility vest.
- A spare wheel. It is also advisable to carry a first aid kit.

People who wear glasses are recommended to carry a spare pair.

Choose the route

Before starting the journey, you must plan the route you are going to follow by looking at a map. It is important to choose the safest and most comfortable route taking into account which roads have roadworks, the weather and the number of vehicles that are using each road that day. You must also plan the breaks you are going to take during the journey. When you travel in winter, you must take the following precautions:

- Carry snow chains in the vehicle.
- Avoid driving at night.
- Travel with the fuel tank full.
- Carry warm clothes, water, food and a torch in the vehicle.

Preventive driving

What is it? A way of driving that allows you to anticipate unexpected events that may happen during the journey. In preventive driving, the driver:

- Collects the necessary information before starting the journey to drive safely.
- Anticipates unexpected events that may happen and reacts well to them.
- Adapts to the circumstances at each moment of the journey. To drive preventively, you must follow a technique based on three principles:

What does it consist of?

Vision

Look in all directions to gather information and do not focus on things that could be a distraction.

Anticipation Anticipate the movements and reactions of other drivers so you can react in time.

Space

Keep a safe following distance from the rest of the vehicles at all times.

Vision

Look far ahead

Look to the sides

Look far ahead

You must look far ahead to control what happens in the space that your vehicle will cover in the next 20 seconds.

This way you can anticipate dangerous situations and avoid harsh braking and sudden acceleration. The faster you drive, the further ahead you must look.

Look to the sides

It takes us longer to see what happens to the sides than what happens in front. For this reason, it is necessary to check all the time what is happening to the left and to the right during the journey. This is done through the rear-view mirrors because they allow you to see what happens behind and to the sides of the vehicle. You must look at them during the journey, even if you are not going to do any manoeuvre.

How often you look at them will depend on the type of road you are on. You look quickly, briefly so you keep paying attention to the rest of the road.

You must bear in mind that there is always a space on both sides of the vehicle that you cannot see, even if your rear-view mirrors are well adjusted. These spaces are called blind spots and they are dangerous because you cannot control what is happening in that place.

Blind spot

Blind spot

Watch video

To make sure there is no other vehicle in the blind spot area, you can turn your head and look through the window. It is important to check that there is no other vehicle in the blind spot before turning the vehicle towards that side to change lane.

Blind spot detector (BSM)

There is a system that warns you when there is another vehicle in the blind spot area that is moving towards your vehicle. In those cases a light comes on that is placed in the rear-view mirror. Some blind spot detectors light up whenever another vehicle is in the blind spot area. Other detectors only light up if you activate the direction indicator to change lane and there is another vehicle in the blind spot.

Anticipation

Watch video

To avoid risks and react quickly to unexpected events on the road, you must:

- Keep a speed suitable for your circumstances, the vehicle's circumstances and the road's circumstances at all times.
- Switch on the lights that are necessary at each moment so that other vehicles can see you and know what movement you are going to make.
- Do not spend a long time in the blind spot of another vehicle's rear-view mirrors. Especially if you are travelling alongside or behind a very large vehicle.
- Always signal the manoeuvres you are going to do for as long as necessary.

STOP

?

Control the space

Front space

Rear space

Space to the sides

Space when stopping the vehicle

Front space

It is advisable that between your vehicle and the one in front there are two or three seconds of separation. That is, that it takes you two or three seconds to reach it at the speed you are travelling at. However, there are circumstances in which you must leave at least one more second of space. These circumstances are:

- There is rain, snow, fog or it is night.
- The vehicle behind is getting too close.

Rear space

Sometimes it is the vehicle behind that comes too close. What you must do in those situations is:

- Leave more safe following distance from the vehicle in front so you do not brake sharply if something unexpected happens.
- Signal earlier the manoeuvres you are going to do.
- Brake smoothly and with enough time so that the driver of the vehicle behind can react.

Space at the sides

You must drive at a suitable speed and keep the necessary separation distance when overtaking other vehicles or passing them in the opposite direction. For example, when changing lane or opening the vehicle door to get out. You must try to stay out of the blind spot area of other vehicles.

Space when stopping the vehicle

When you stop the vehicle for some reason, such as traffic lights or a traffic jam, you must keep a distance of at least two or three metres from the vehicle in front. This way:

- You will have time to warn the driver in front if they roll their vehicle backwards.
- You will be able to overtake the vehicle in front if it stays parked for some reason.
- You will not hit the vehicle in front if another vehicle hits yours from behind. Also, the driver of the last vehicle in the queue, to avoid rear-end collisions, can:
- Keep a distance of five or six metres from the vehicle in front.
- Keep the brake pressed and switch on the hazard warning lights in a traffic jam so that they can be seen better.
- Look in the rear-view mirror and watch the vehicles coming from behind.

Efficient driving

What is it? A way of driving in which less fuel is used, it helps to pollute less, and driving is safer. What do you achieve with efficient driving?

- Improve air quality because fewer polluting gases are released into the atmosphere.
- Save money because less is spent on fuel and on vehicle maintenance.
- Increase safety because techniques similar to preventive driving are used.
- Make passengers travel more comfortably because sharp braking is avoided and there are no big

changes in speed.

- Reduce the noise that vehicles make when driving.

Pollution

Vehicles pollute because they produce toxic gases that reach the atmosphere. The measures you can take to pollute less with your vehicle are:

- Check the engine frequently so that it does not give off clouds of smoke.
- Do not accelerate sharply when the vehicle is stopped and starts moving again.
- Use the horn only when necessary.
- Secure the load properly so that it does not move and make noise when it hits the vehicle.
- Prevent oil or other vehicle fluids from falling onto the road.
- Do not wash the vehicle on the road. It must be washed in places prepared to do so.
- Do not throw objects onto the road that can dirty it, start fires, or cause accidents.
- Do not make unnecessary noise or emit more gases or smoke than allowed. In vehicles there is a device in the exhaust pipe called a catalytic converter. The catalytic converter is responsible for reducing the pollution produced by the gases that come out of the exhaust pipe. When the vehicle battery runs out you must not try to start the engine by pushing the vehicle.

Unburnt fuel can reach the catalytic converter and destroy it.

Techniques for driving efficiently

When starting and beginning to drive When accelerating and changing gear When choosing the speed When reducing speed and stopping When starting and beginning to drive

- Start the engine without pressing the accelerator.
- Start driving with the vehicle just after starting the engine.
- Start driving at low speed and increase it when the engine warms up.

Driving at high speed with a cold engine makes more fuel be used and makes the engine wear out sooner.

See video

When accelerating and changing gear

- Accelerate little by little pressing the pedal smoothly.
- Start driving in first gear. In this gear the engine has a lot of power, but it goes at very low speed.
- Change to second gear after a few seconds. In this gear the vehicle has a little less power and goes a bit faster.
- After that you can increase speed smoothly and little by little. The vehicle uses less fuel in fourth and fifth gear, which is when the engine runs at higher speed and has less power. In the city you should try to use the highest gear possible, always respecting the speed limits.

When choosing the speed you drive at From 80 or 90 kilometres per hour, the vehicle uses much more fuel and pollutes more.

Changing gear many times also makes more fuel be used. Also, it makes the driver get more tired. You should travel at a constant speed, that does not change all the time, and avoid sharp braking and sudden acceleration. When reducing speed and stopping the vehicle To reduce the vehicle's speed you have to:

- Lift your foot off the accelerator and let the vehicle roll.
- Use the brake smoothly to reduce speed little by little.
- Switch off the engine when the vehicle is going to be stopped for more than one minute.

What makes more fuel be used?

- The air outside
- Not doing vehicle checks
- Driving in towns or cities
- Using air conditioning
- Carrying a lot of weight

The air outside
Most of the fuel that is used is used to fight against the air that does not let the vehicle move forward. Some of the elements fitted to the vehicle make it harder to fight against the air and, therefore, it uses more fuel. For example, a roof rack or deflectors.

Deflectors. An accessory fitted to the vehicle to divert the air and prevent it from entering the interior.

So that the vehicle does not use so much fuel because of the air, the following measures can be taken:

- Do not fit a roof rack to the vehicle. It is better for luggage and other loads to go in the boot.
- Do not lower the vehicle windows while it is driving.
- Do not tow a trailer or caravan.

Especially if the trailer or caravan is wider or taller than the vehicle.

Not doing vehicle checks

Having the vehicle checked and checking that it is in good condition helps to use less fuel. To save fuel you must check:

- The ignition system. You must check the spark plugs following the instructions given by the manufacturer.
- The cooling and lubrication systems.
- Tyre pressure. They must be properly inflated.
- The engine fuel system. You must change the air filter when necessary and keep the idle speed at the correct value.

Idle speed. The ability of the engine to run without the driver pressing the accelerator. An idle speed that is too high will make more fuel be used.

Driving in towns or cities

In towns and cities more fuel is used even though vehicles drive more slowly than on other roads.

See video

The reasons are that there are more traffic jams, the vehicle has to stop more times, you have to change gear more, and the engine is strained more. To save fuel in towns and cities and not suffer traffic jams, it is advisable to:

- Not use the vehicle for short distances.
- Use public transport.
- Avoid driving at the busiest traffic times.
- Choose the route where there are fewer vehicles, even if it is the longer route.

See video

Using air conditioning

When using air conditioning in the vehicle more fuel is used. It is advisable to have a temperature inside the vehicle of 23 or 24 degrees.

Carrying a lot of weight

A vehicle uses more fuel when it carries a lot of weight. That is why you should try to carry only the amount of luggage and loads that are necessary. It is also important to distribute the load well in the vehicle to use less fuel.

Measures to save fuel that are false Some measures used to save fuel are false and must not be done because they damage the vehicle. These measures are:

- Putting fuel in the vehicle different from what the manufacturer specifies.

Unsuitable fuel will break the vehicle's engine.

- Going downhill in neutral. The vehicle can lose stability and you will have to use the brakes too much.
- Not spending money on the necessary vehicle maintenance.

Worn parts and faults will make more fuel be used.

Neutral. Position of the gearbox in which the engine's movement is not transmitted to the wheels.

New techniques to use less fuel

Start-Stop function

This system starts and stops the engine automatically when the vehicle stops for a moment at traffic lights, in a traffic jam, or for other reasons. This way, when the vehicle is stopped it does not use fuel and does not release gases that pollute the environment.

Eco mode

This system gives the option to set the engine to less power and speed to save fuel. It takes into account all the elements that make a vehicle use fuel to know how it can work without using all its power and speed.

By limiting the vehicle's speed and temperature less fuel is used. Eco mode is switched on and off manually in each vehicle.

Environmental labels

What are they? A way of classifying vehicles depending on the pollution caused by their gases and the damage they do to the environment. There are four different types of environmental labels.

Zero emissions

Eco

B

C

The information shown on each label is:

The aim of this classification is to give preference to vehicles that damage the environment less. City authorities can:

- Create low-emission zones in the city. These are zones that only vehicles that damage the environment less can enter. This is done to improve air quality.

Low-emission zones are indicated with a vertical sign.

Only vehicles that have the environmental label shown on the lower part of the sign will be able to enter that zone.

- Decide which vehicles can enter the city centre on days when there is a lot of air pollution.
- Give economic or traffic benefits to vehicles that help to protect the environment. For example, they can allow vehicles with the “zero emissions” label to use the VAO lane, even if only the driver is travelling in the vehicle.

Environmental labels must be placed on the windscreen, on the right-hand side of the vehicle's front glass. In vehicles that do not have a windscreen, it will be placed in a visible place.

Shared vehicles

Another measure to protect the environment is the use of shared vehicles. This initiative consists of a person being able to rent a vehicle through an application (App) for hours or minutes and share it with other people.

Shared-use vehicles carry this label.

Annex points. The points-based driving licence

Contents

How many points does each driver have?

Losing points

Which offences take points away from the driving licence?

The aim of the points-based driving licence is to reduce the number of road traffic accidents. How many points does each driver have?

Drivers who start driving have 8 points during the first two years. If during that time they do not lose any points, they will then have 12 points.

Drivers who keep the 12 points for three more years will get another two points. Therefore, they will have 14.

Drivers who keep the 14 points for another three more years will get another point. The maximum number of points a driver can get is 15.

Type of driver	Points
New drivers	8
Drivers with more than two years of driving licence	12
Drivers with more than five years of driving licence	14
Drivers with more than eight years of driving licence	15

Losing points

A driver loses points when they commit some offences or penalties while driving. The person who loses all the points will not be able to drive again until six months have passed. After those six months, they will have to do a 24-hour course and a test at the Traffic Headquarters to recover the points and the driving licence. If, in the following three years, they lose all the points again, they will have to wait 12 months more to recover them again.

Drivers who only lose some points can recover them again if they do not lose any more in two years. Also, they can do a road safety education course lasting 12 hours to recover up to 6 points.

Which offences take points away from the driving licence? Offences that take away from 2 to 6 points

- Driving faster than the permitted speed. The number of points lost for this offence will depend on the speed at which the vehicle is travelling in each case.

Km/h

20

40

60

80

100 120 140 160

180

200

220

240

Offences that take away 3 points

- Making a U-turn where it is not permitted.
- Driving with helmets or headphones connected to the mobile phone or another electronic device.
- Having in the vehicle devices to detect speed cameras.

Radar. A system used to detect an object, know how far away it is and at what speed it is travelling.

Offences that take away 4 points

- Driving with a blood alcohol limit between 0.25 and 0.50 milligrams of alcohol per litre of air.

Professional drivers and those who have had the driving licence for less than two years will lose the 4 points with a blood alcohol limit between 0.15 and 0.30 milligrams of alcohol per litre of air.

- Driving a vehicle with a licence that does not authorise you to drive it. For example, driving a car when the driver's licence only authorises them to drive motorcycles.
- Not complying with the right of way rules for other drivers or pedestrians and not stopping at STOP signs, give way signs and at red traffic lights.
- Overtaking other drivers in places where it is prohibited.
- Reversing on motorways or dual carriageways.
- Not respecting or complying with the signs and orders of traffic officers.
- Not keeping the safe following distance from the vehicle in front.
- Driving a vehicle when the driver has lost the driving licence for committing offences.
- Not wearing the seat belt, the helmet, child restraint systems for children and other safety systems that are compulsory.

Offences that take away 6 points

- Driving with a blood alcohol limit higher than 0.50 milligrams of alcohol per litre of air.

Professional drivers and those who have had the driving licence for less than two years will lose the 6 points with a blood alcohol limit higher than 0.30 milligrams of alcohol per litre of air.

- Driving after taking drugs or when some of these drugs are still in the body.
- Refusing to take the tests to detect alcohol or drugs in the body.
- Driving in the opposite direction to that permitted, taking part in illegal races or endangering other people's lives while driving.
- Driving vehicles that have installed mechanisms so that speed cameras do not detect them and do not know what speed they are travelling at.
- Driving for more than half of the time that is permitted or taking breaks shorter than those permitted (for professional drivers). For example, driving for 6 hours in a row if the time limit is 4 hours or resting for 10 minutes when you must take breaks of at least 20 minutes.
- Fitting in the vehicle elements that change how the devices that limit speed work, or those that count the time that the professional driver has been driving.
- Driving with the mobile phone in your hand.
- Throwing objects onto the road that can cause fires or accidents.
- Overtaking while endangering cyclists or without leaving a separation from them of at least 1.5 metres.

Collaborate: